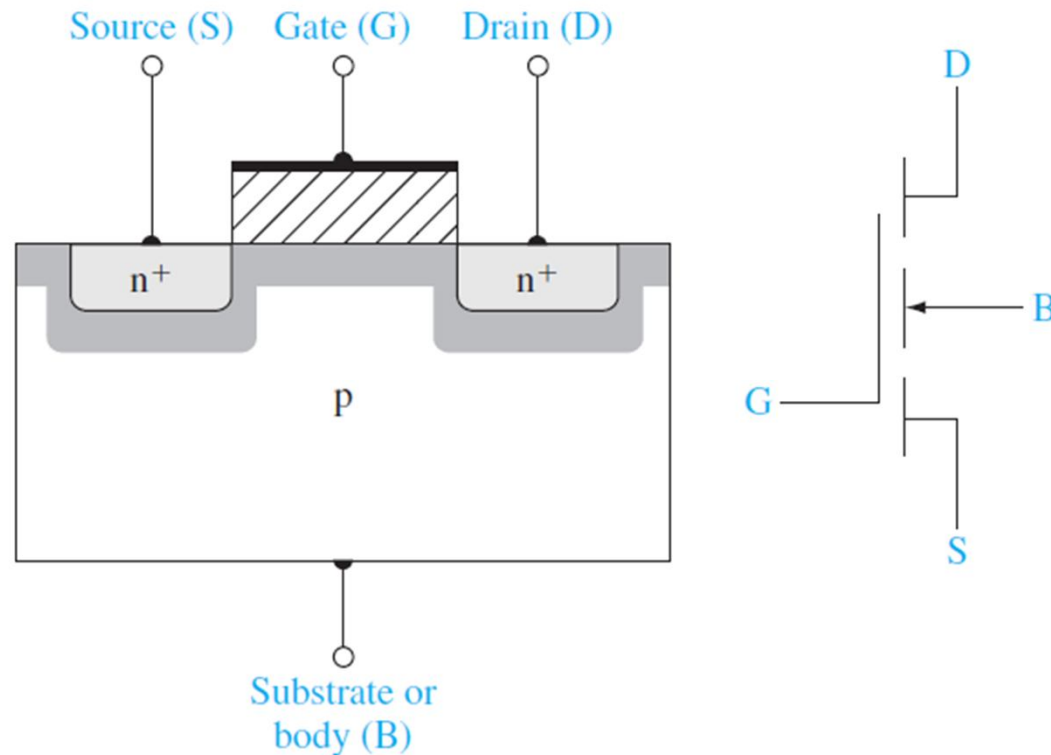


L10: Fundamentals of the Metal–Oxide–Semiconductor Field-Effect Transistor



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Topics

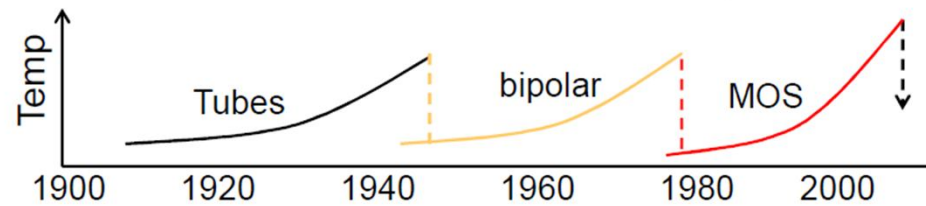
- Study the characteristics of energy bands as a function of applied voltage in the metal–oxide–semiconductor structure known as the MOS capacitor. The **MOS capacitor is the heart** of the MOSFET.
- Discuss the concept of **surface inversion** in the semiconductor of the MOS capacitor.
- Define and derive **the expression for the threshold voltage**, which is a basic parameter of the MOSFET.
- Discuss various **physical structures** of MOSFETs, including enhancement and depletion mode devices.
- Derive the **ideal current–voltage** relationship of the MOSFET.

History of Transistor

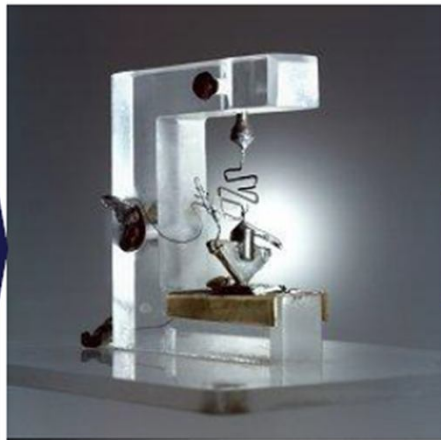
Vacuum tubes, invented by Fleming in 1905 and further developed by Lieben (1906) and by De Forest (1907) used for rectification, amplification, switching and otherwise modification and creation of electrical signal for many years.

Invention of bipolar transistor in 1947 started the solid state electronics era. Demonstration of MOSFET in 1960 revolutionized electronics industry.

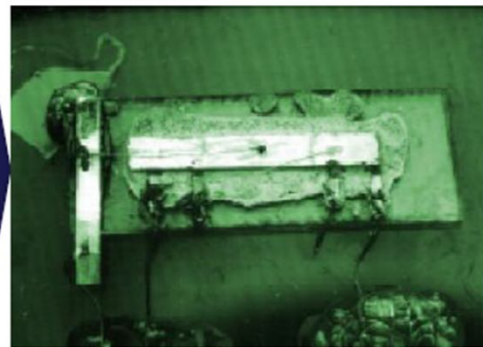
Today MOSFET is the most abundant man made device on Earth. ($10^7 - 10^8$ MOSFETs per person in Western World.)



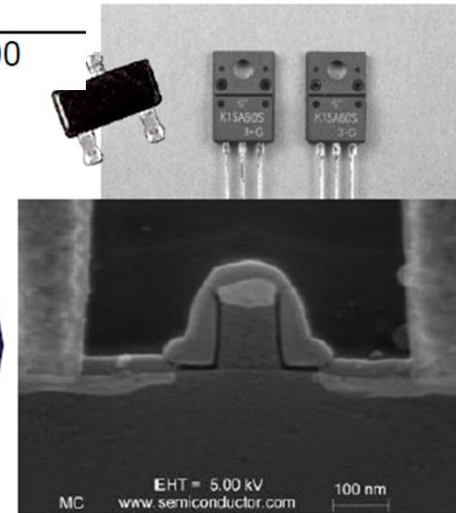
RCA Vacuum tube
1940s



First transistor
(Bardeen et al. 1947)



First MOSFET
(Kahng and Attala, 1960)



Transistors 2000s

1930: Patent on the Field-Effect Transistor

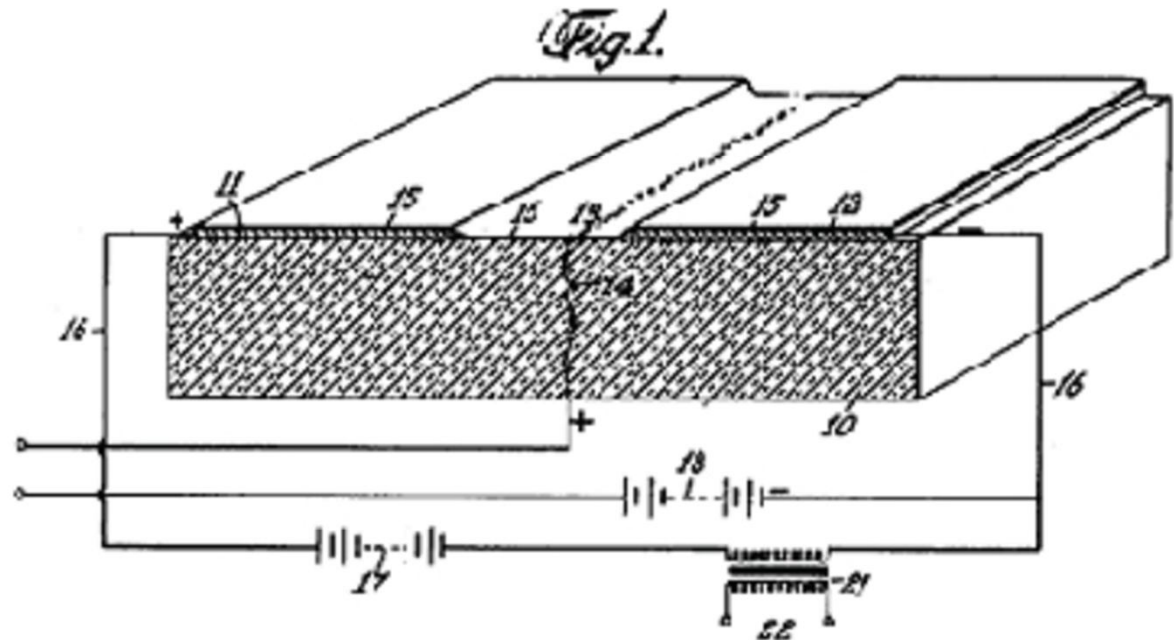
Jan. 28, 1930.

J. E. LILIENFELD

1,745,175

METHOD AND APPARATUS FOR CONTROLLING ELECTRIC CURRENTS

Filed Oct. 8, 1926



Julius Lilienfeld filed a patent describing a three-electrode amplifying device based on the semiconducting properties of copper sulfide. He did not demonstrate the device experimentally

1960: MOS Transistor Demonstrated



Dawon Kahng



John Atalla

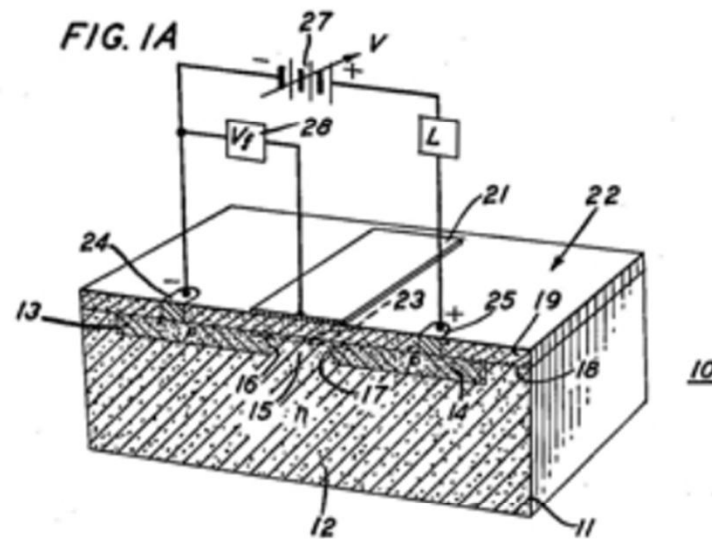
Aug. 27, 1963

DAWON KAHNG

3,102,230

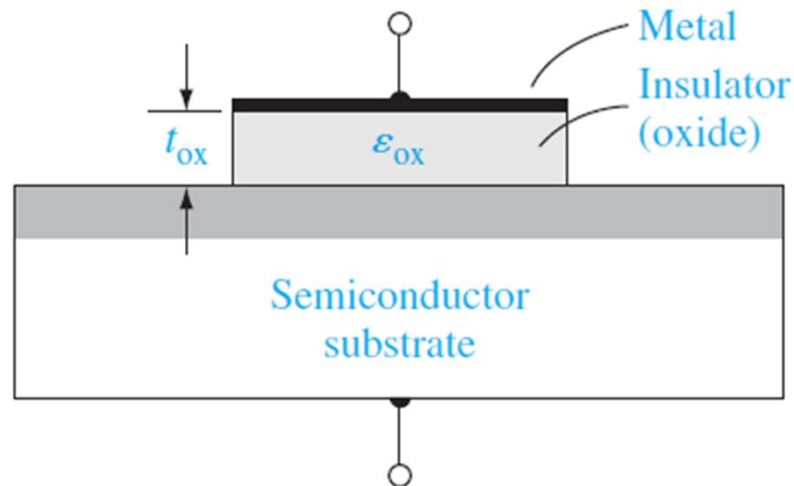
ELECTRIC FIELD CONTROLLED SEMICONDUCTOR DEVICE

Filed May 31, 1960



MOS Cap

Metal Oxide Semiconductor Capacitors



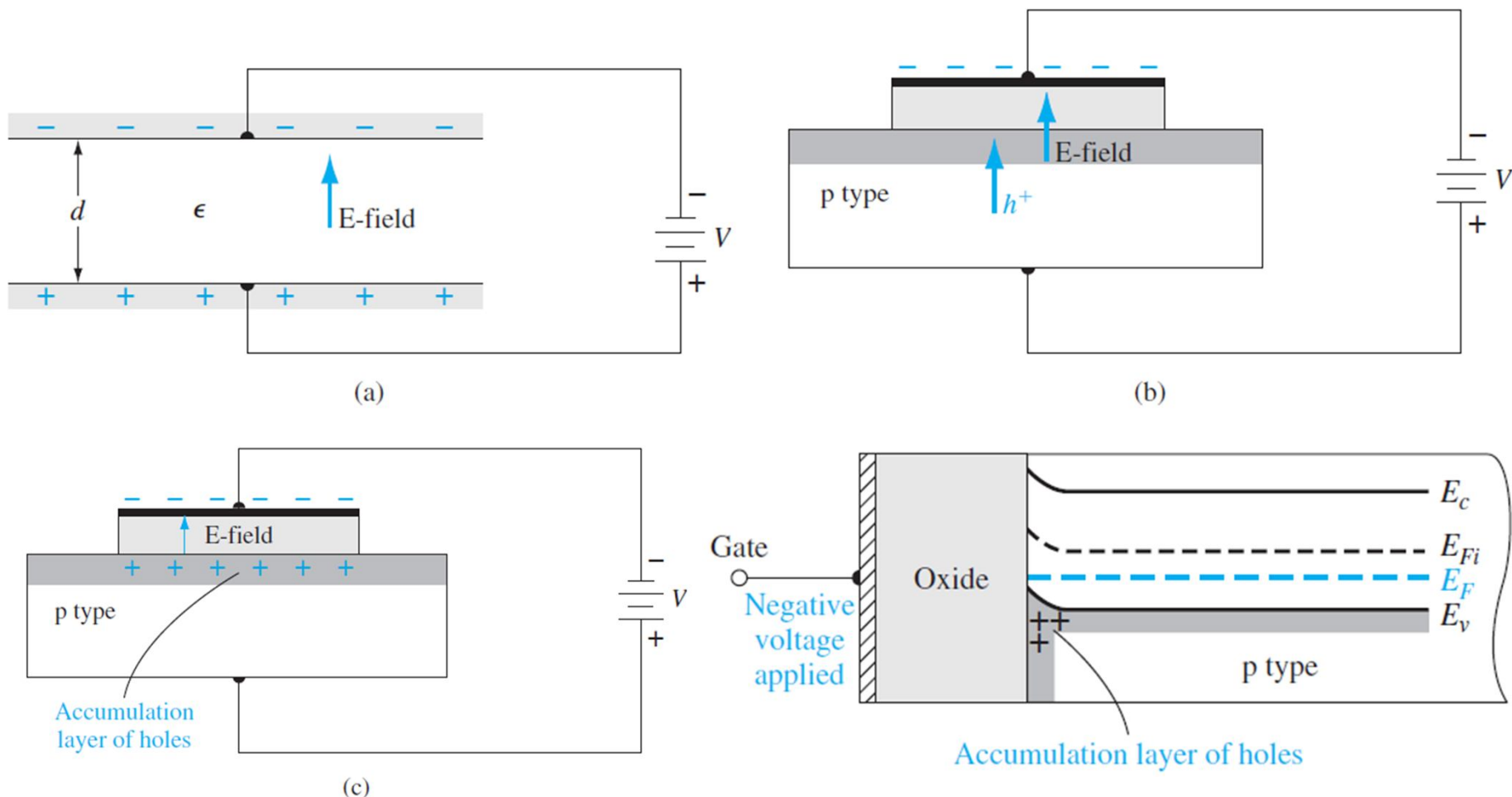
$$C' = \frac{\epsilon}{d}$$

$$Q' = C'V$$

MOS Cap

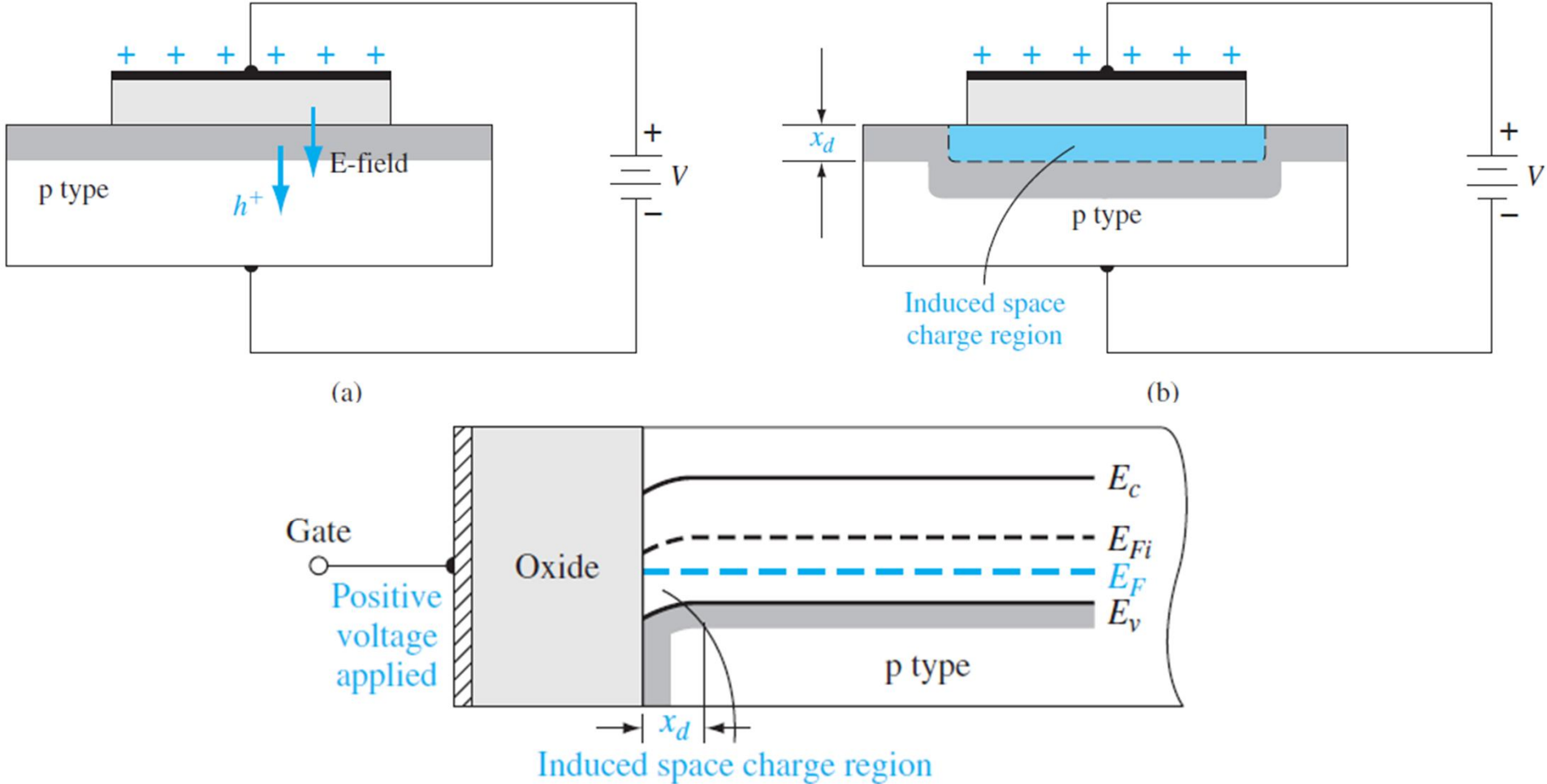
- Used as capacitors: DRAM
- Main ingredient in a MOS transistor (besides p-n junctions)
- Used in imagers (i.e. cameras) → Charge-Coupled Devices (CCD)

MOS Cap Accumulation



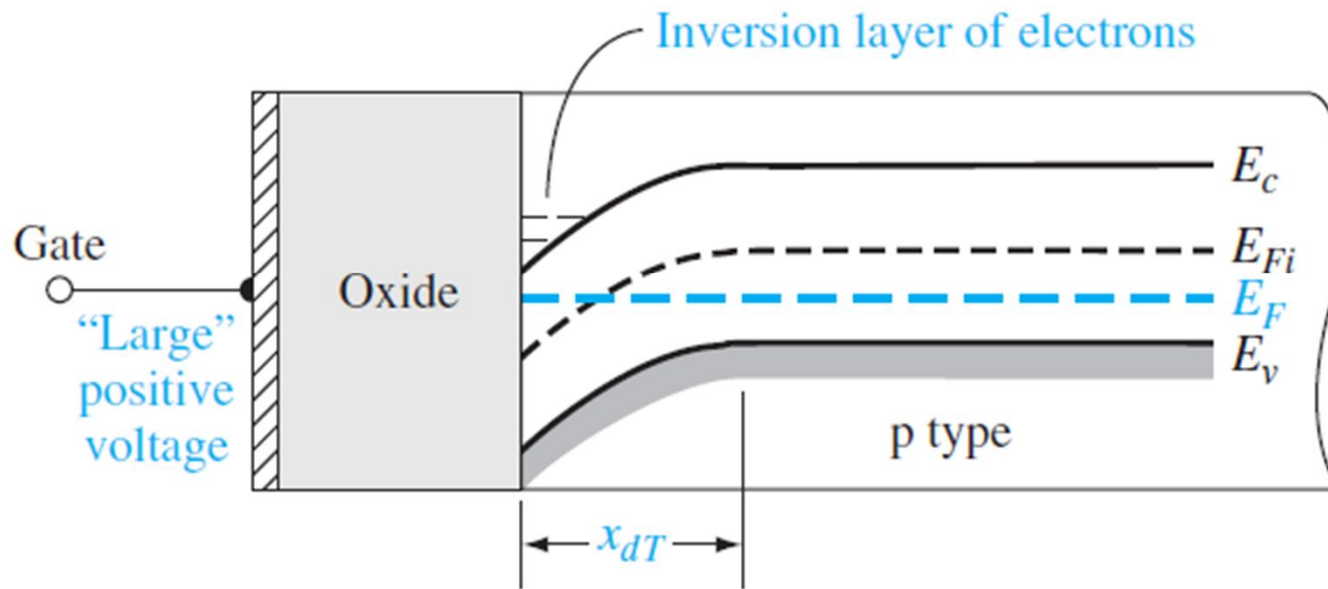
With a negative gate bias, if the electric field were to penetrate into the semiconductor, the majority carrier holes would experience a force toward the oxide– semiconductor interface. An **accumulation layer of holes** at the oxide– semiconductor junction corresponds to the positive charge on the bottom “plate” of the MOS capacitor.

MOS Cap depletion



With a moderate positive gate bias, the holes are pushed away from the oxide/semiconductor interface, a negative **space charge region** is created because of the fixed ionized acceptor atoms.

MOS Cap inversion

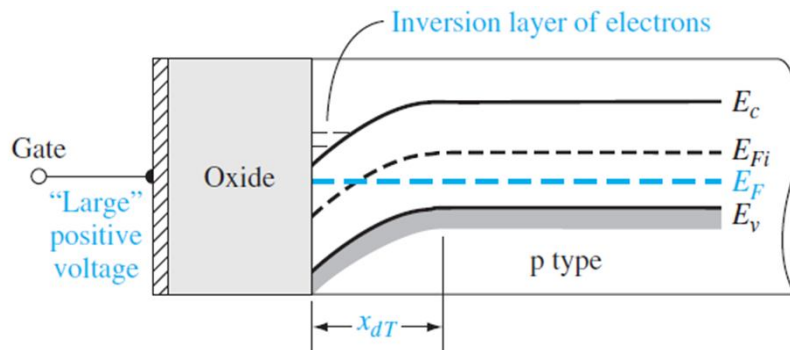
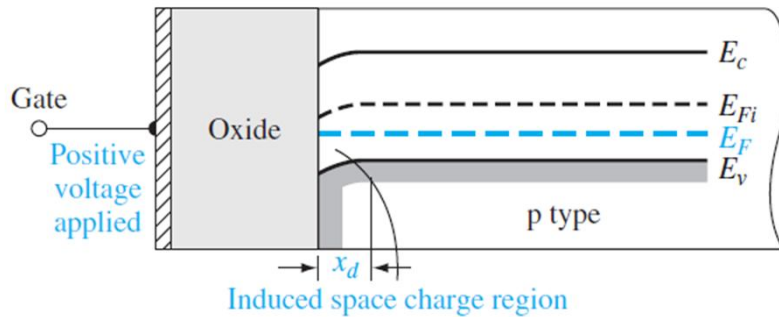
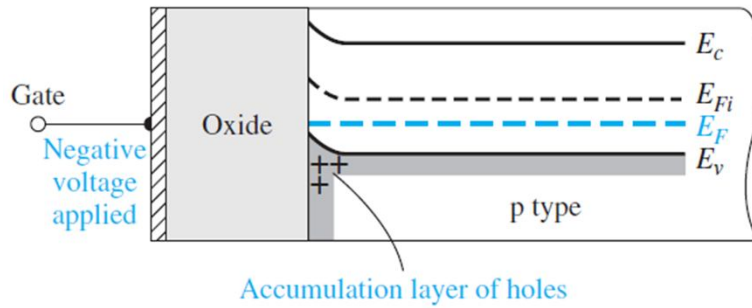


By applying a sufficiently large positive gate voltage, we have inverted the surface of the semiconductor from a p-type to an n-type semiconductor. We have created an *inversion* layer of electrons at the oxide–semiconductor interface.

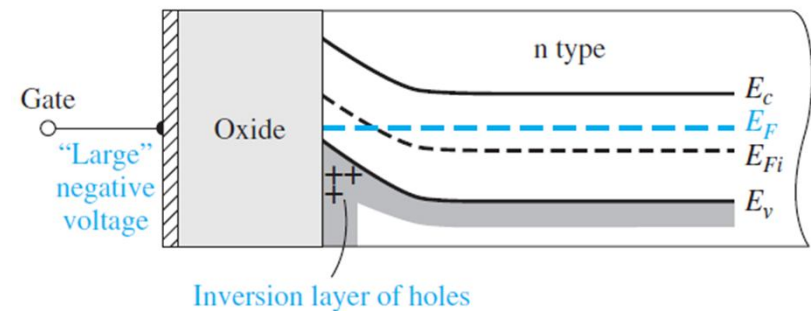
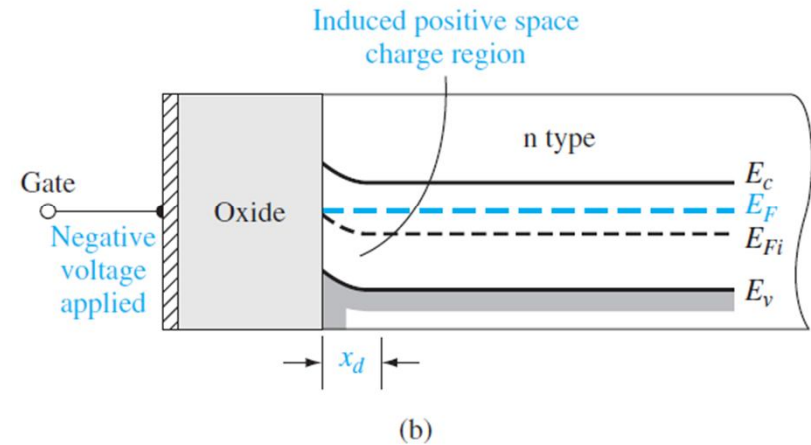
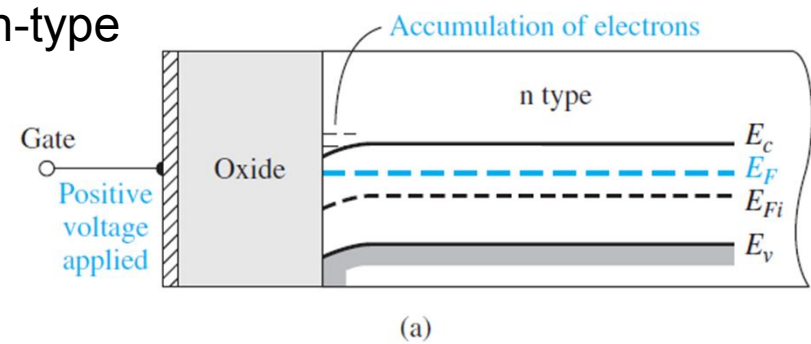
This n-type surface layer is formed not by doping, but instead by *inversion* of the originally p-type semiconductor due to the applied voltage. This inverted layer, separated from the underlying p-type material by a depletion region, is the key to MOS transistor operation.

MOS Cap Accumulation, depletion and inversion

p-type



n-type



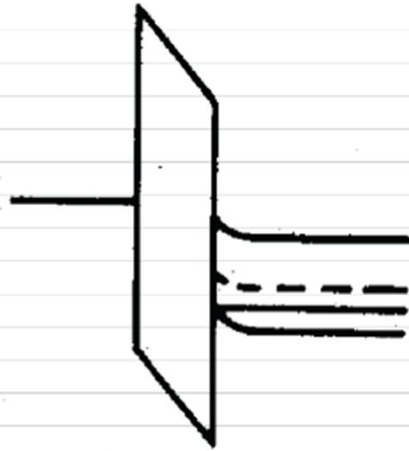
MOS Cap Accumulation, depletion and inversion

$$\Phi_M = \Phi_s$$

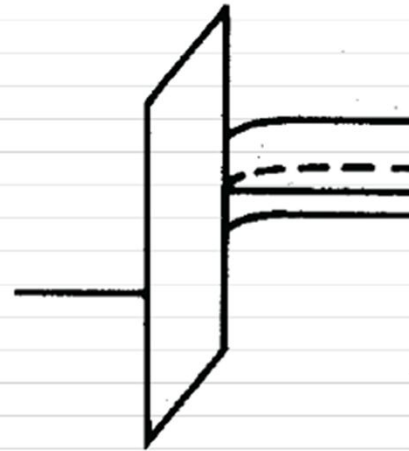
Energy
band
diagram



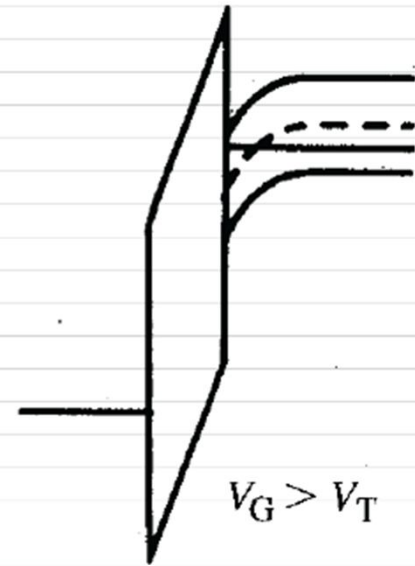
$$V_G = 0$$



$$V_G < 0$$

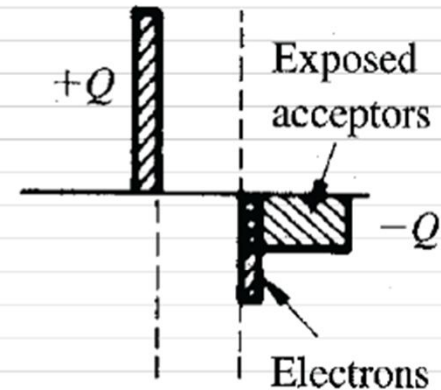
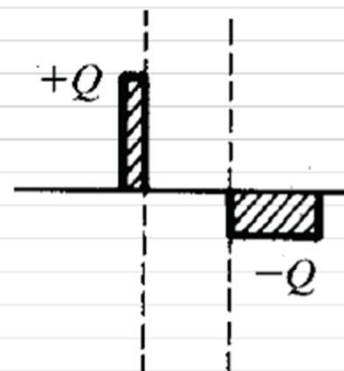
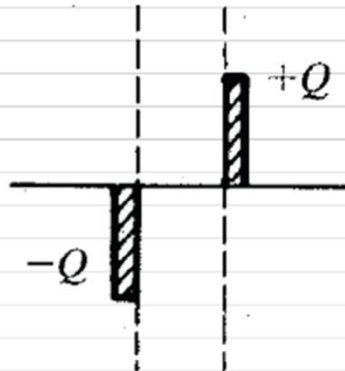


$$V_T > V_G > 0$$



$$V_G > V_T$$

Charge
diagram



Exposed
acceptors

Electrons

Name Flat band

Accumulation

Depletion

Inversion

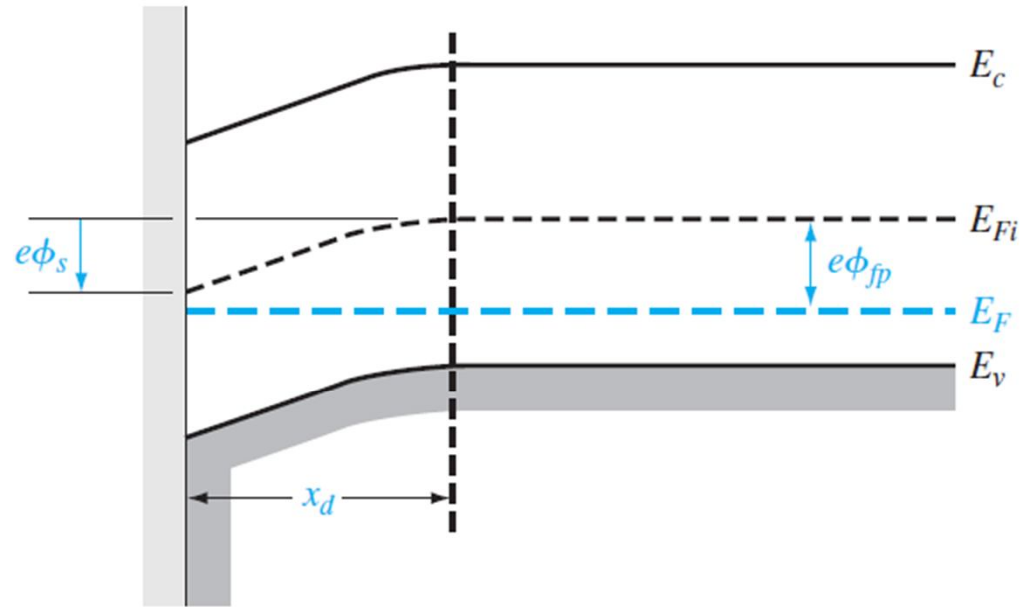
Potential is dropped across SiO_2 and Si.

Depletion Layer Thickness

$$\phi_{fp} = V_t \ln \left(\frac{N_a}{n_i} \right)$$

The potential ϕ_s is called the surface potential; it is the difference (in V) between E_{Fi} measured in the bulk semiconductor and E_{Fi} measured at the surface.

$$\phi_s = \frac{1}{q} (E_{i-BULK} - E_{i-SURFACE})$$



The space charge width can now be written in a form similar to that of a one-sided pn junction

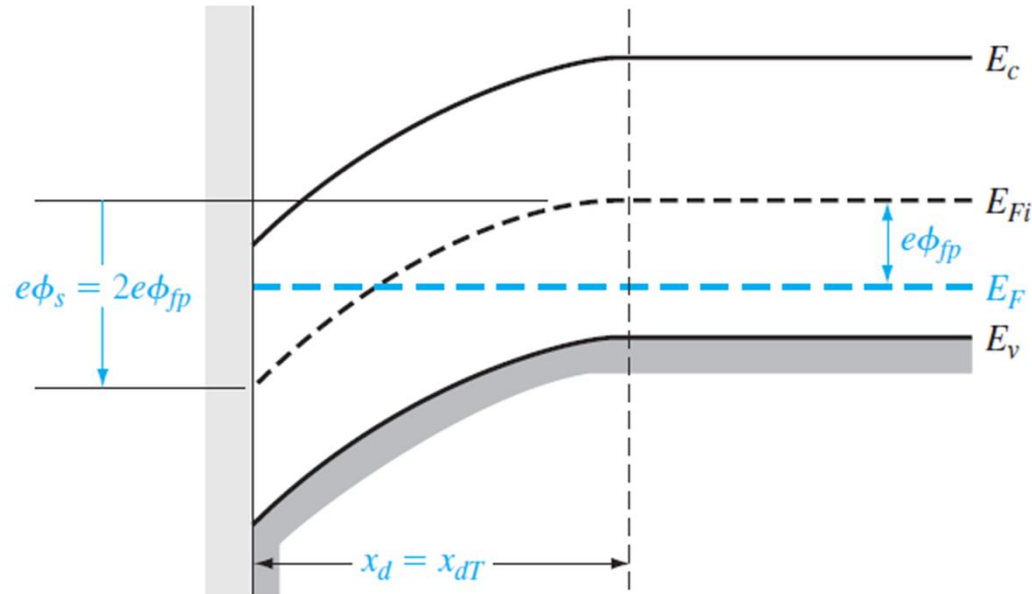
$$x_d = \left(\frac{2\epsilon_s \phi_s}{eN_a} \right)^{1/2}$$

n-channel threshold inversion

$$\phi_s = 2\phi_{fp}$$

The Fermi level at the surface is as far above the intrinsic level as the Fermi level is below the intrinsic level in the bulk semiconductor.

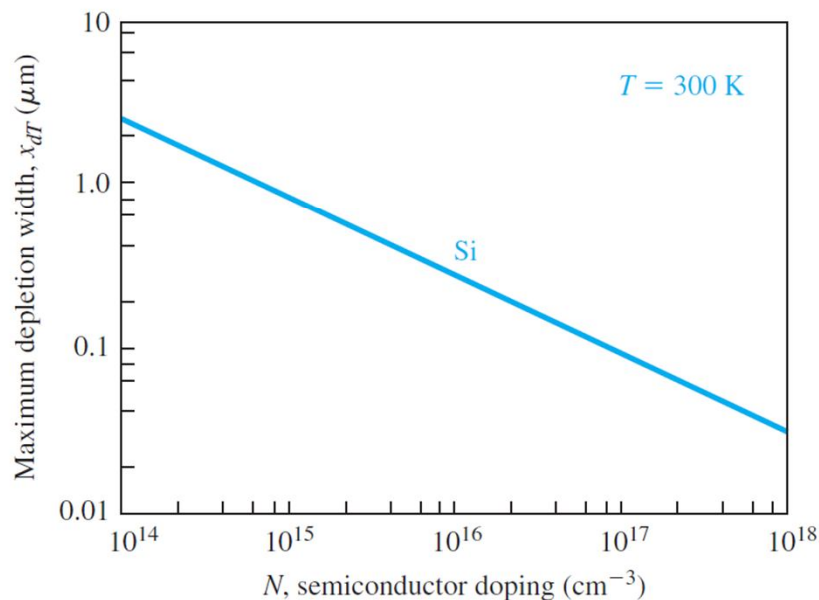
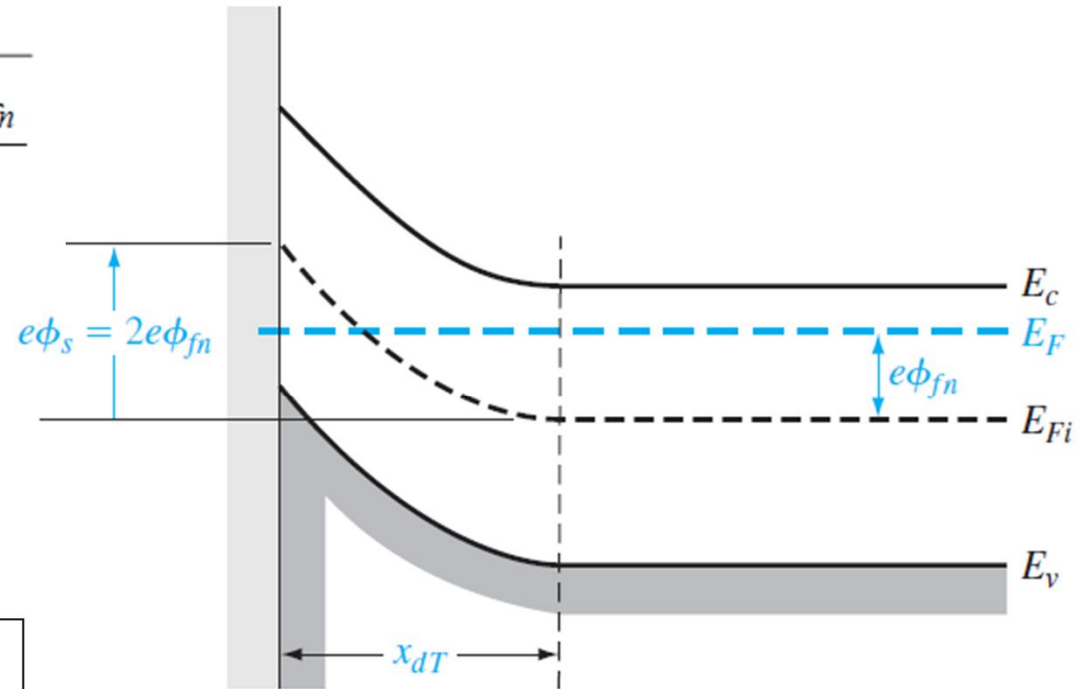
$$x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a} \right)^{1/2}$$



- The electron concentration at the surface is the same as the hole concentration in the bulk material. This condition is known as the threshold inversion point. The applied gate voltage creating this condition is known as the **threshold voltage**.
- If the gate voltage increases above this threshold value, the conduction band will **bend slightly** closer to the Fermi level, but the change in the conduction band at the surface is now only a **slight function** of gate voltage.
- The electron concentration at the surface is an **exponential function** of the surface potential. The surface potential may increase by a few (kT/e) volts, which will change the electron concentration by orders of magnitude, **but the space charge width changes only slightly**. In this case, then, the space charge region has essentially reached a maximum width.

p-channel threshold inversion

$$\phi_{fn} = \frac{kT}{e} \ln \frac{N_d}{n_i} \quad x_{dT} = \sqrt{\frac{4\epsilon_s \phi_{fn}}{eN_d}}$$



The maximum induced space charge width is on the same order of magnitude as pn junction space charge widths.

10.1.3 Surface Charge Density

For a p-type semiconductor substrate, at inversion point, the electron charge density can then be written as:

$$n_s = n_i \exp\left(\frac{\phi_{fp}}{V_t}\right) = N_A$$

The thickness of the inversion layer (~ 4 nm) is much smaller than that of the depletion layer:

$$x_{inv} \ll x_d$$

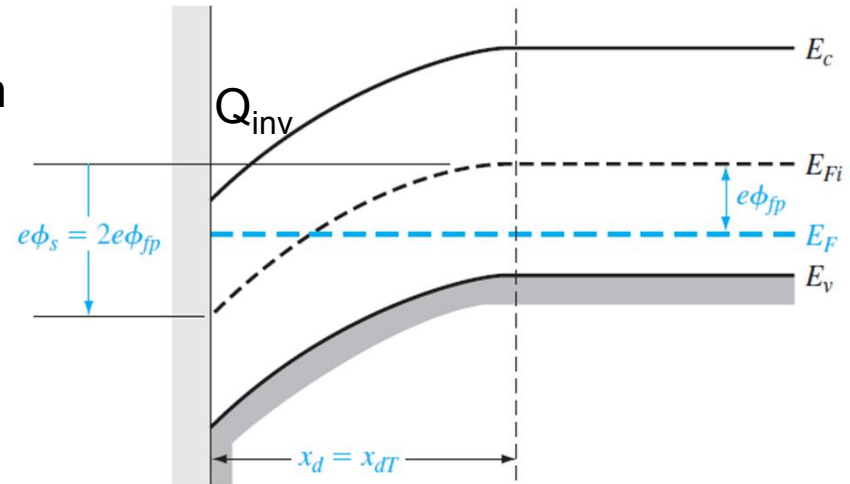
The charge in the inversion layer then is:

$$Q_{inv} = en_s x_{inv} = eN_A x_{inv} \ll eN_A x_d$$

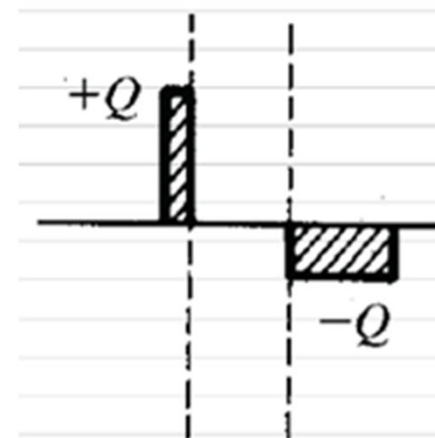
Total charges:

$$Q = Q_{inv} + Q_d \approx Q_d$$

which means that the charges are mainly contributed by the ions of the depletion layer.



Total charges: $Q = Q_{inv} + Q_d \approx Q_d$



10.1.3 Surface Charge Density

Above the inversion point, the electron inversion charge density can be written as

$$n_s = n_i \exp \left[\frac{e(\phi_{fp} + \Delta\phi_s)}{kT} \right] = n_i \exp \left[\frac{\phi_{fp} + \Delta\phi_s}{V_t} \right]$$

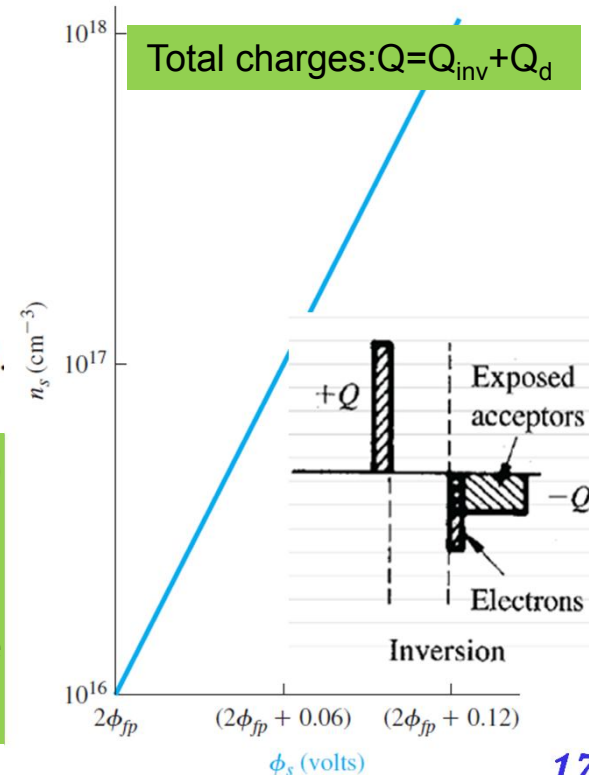
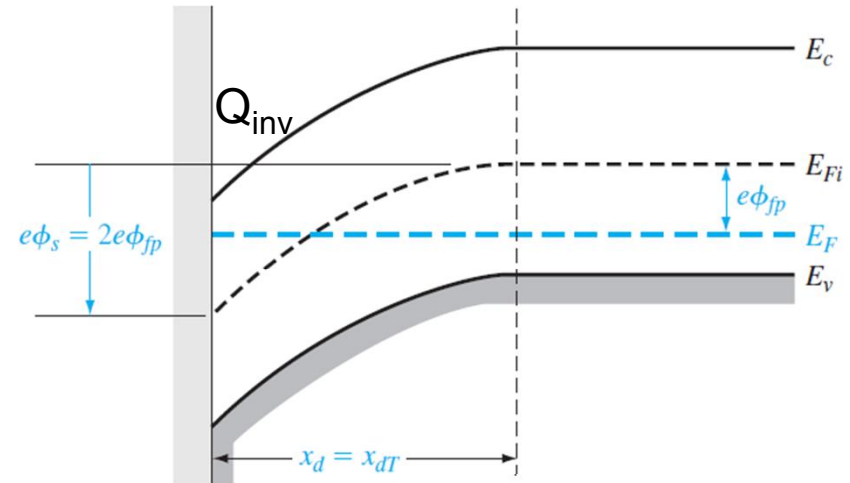
$$n_s = n_i \exp \left(\frac{\phi_{fp}}{V_t} \right) \cdot \exp \left(\frac{\Delta\phi_s}{V_t} \right)$$

where $\Delta\phi_s$ is the surface potential greater than $2\phi_{fp}$.

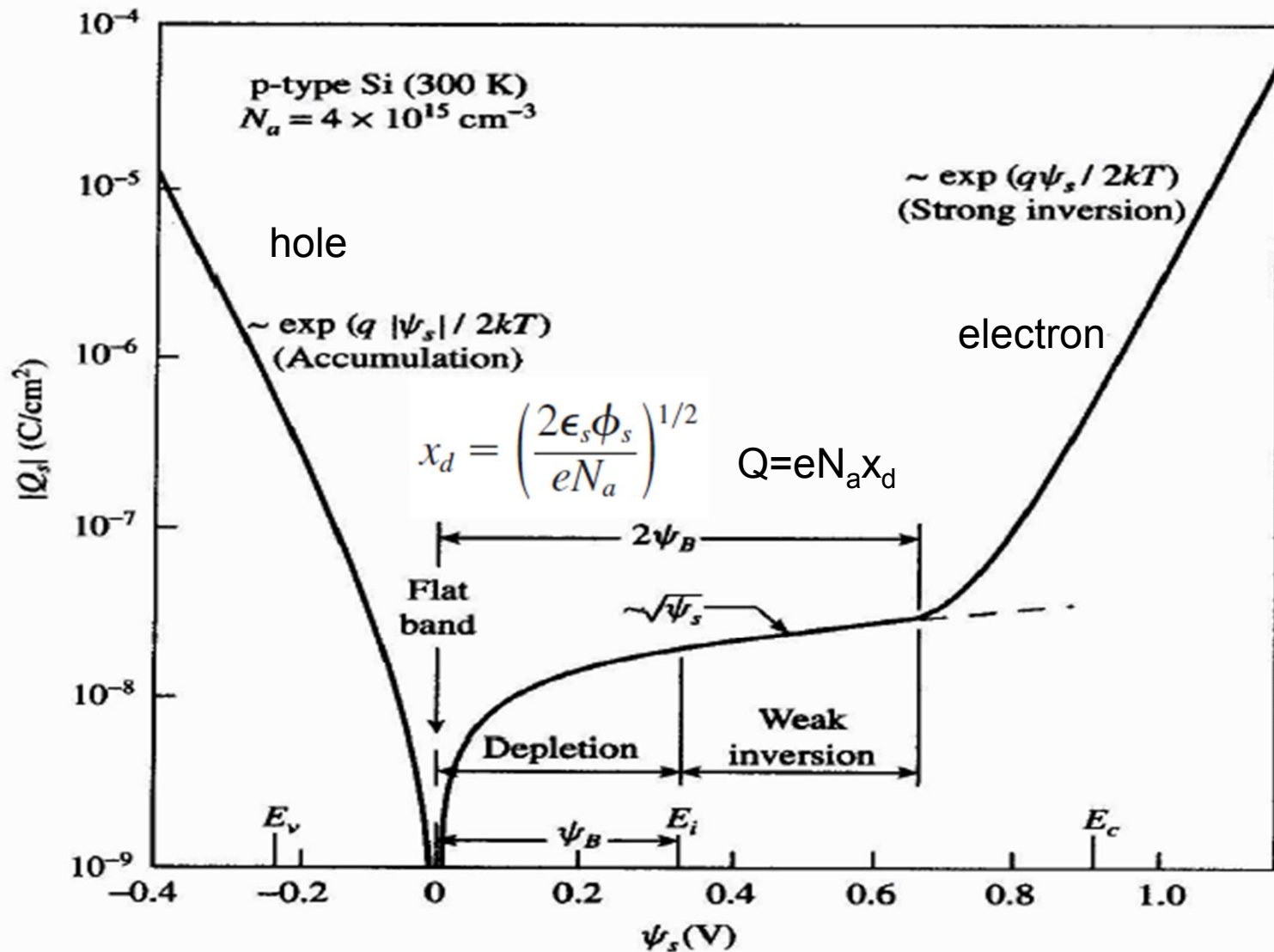
$$n_s = n_{st} \exp \left(\frac{\Delta\phi_s}{V_t} \right)$$

where n_{st} is the surface charge density at the threshold inversion point.

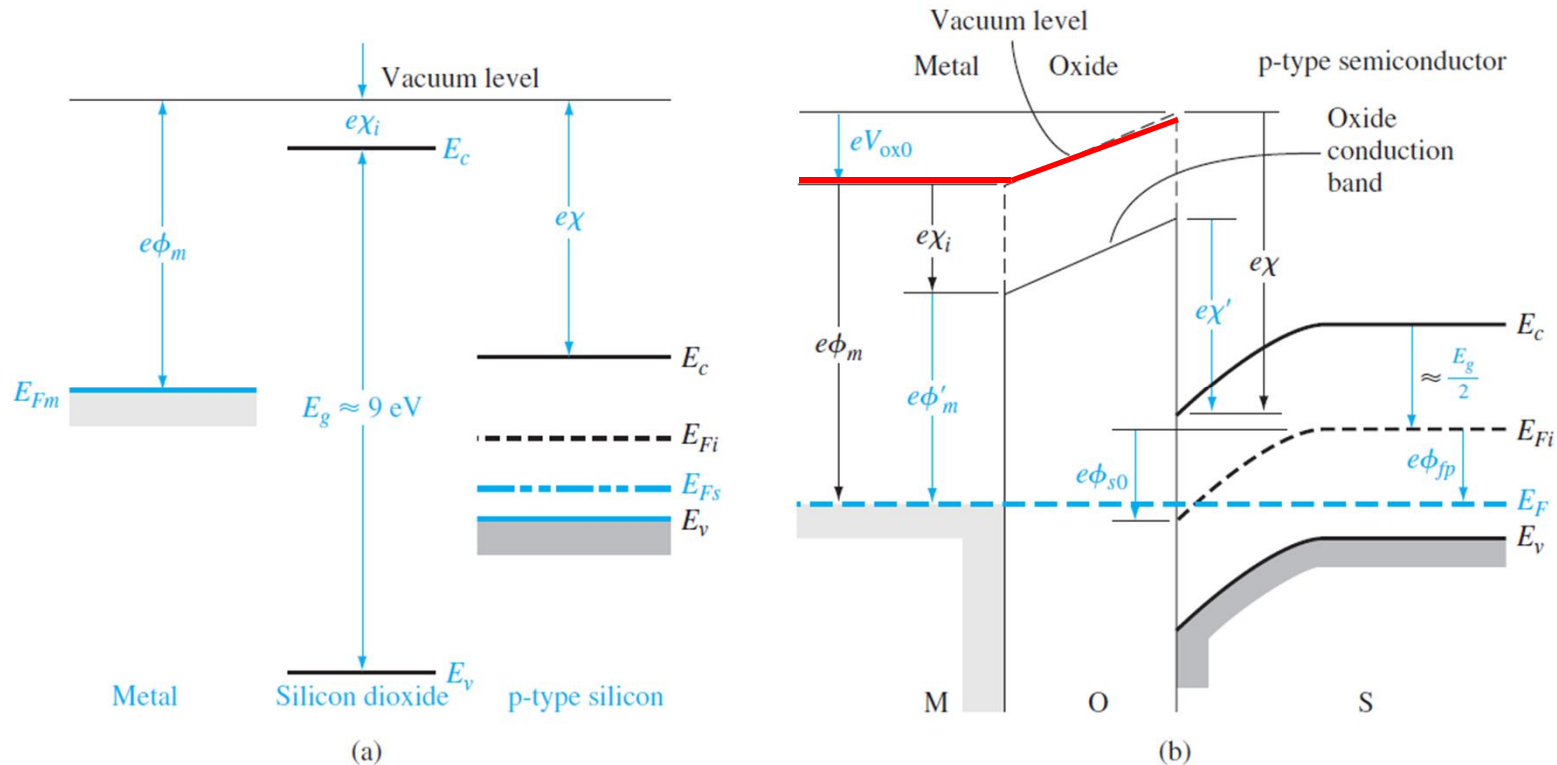
The inversion charge density increases by a factor of 10 with a 60-mV increase in surface potential. The electron inversion charge density increases rapidly with small increases in surface potential, which means that the space charge width essentially reaches a maximum value.



Surface Charge vs. Surface Potential



10.1.4 Work Function Differences



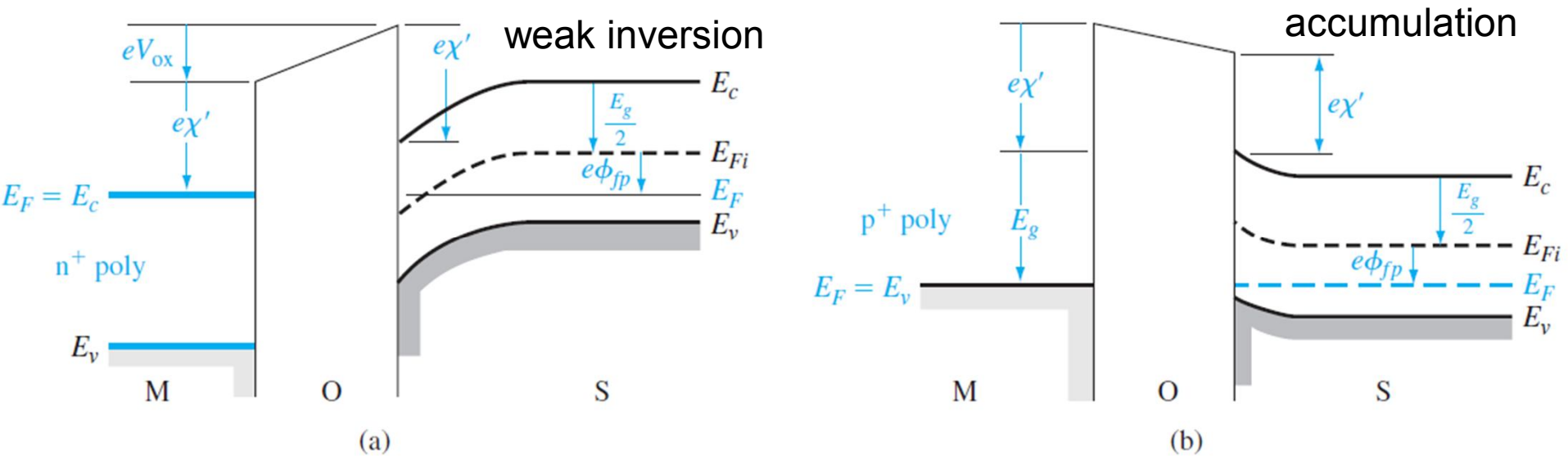
If we sum the energies from the Fermi level on the metal side to the Fermi level on the semiconductor side, we have

$$e\phi'_m + eV_{ox0} = e\chi' + \frac{E_g}{2} - e\phi_{s0} + e\phi_{fp} \quad (10.13)$$

Equation (10.13) can be rewritten as

$$-\phi_{ms} \equiv V_{ox0} + \phi_{s0} = -\left[\phi'_m - \left(\chi' + \frac{E_g}{2e} + \phi_{fp}\right)\right] \quad \text{M-S work function difference}$$

10.1.4 Work Function Differences



For the n⁺ polysilicon gate, the metal–semiconductor work function difference can be written as

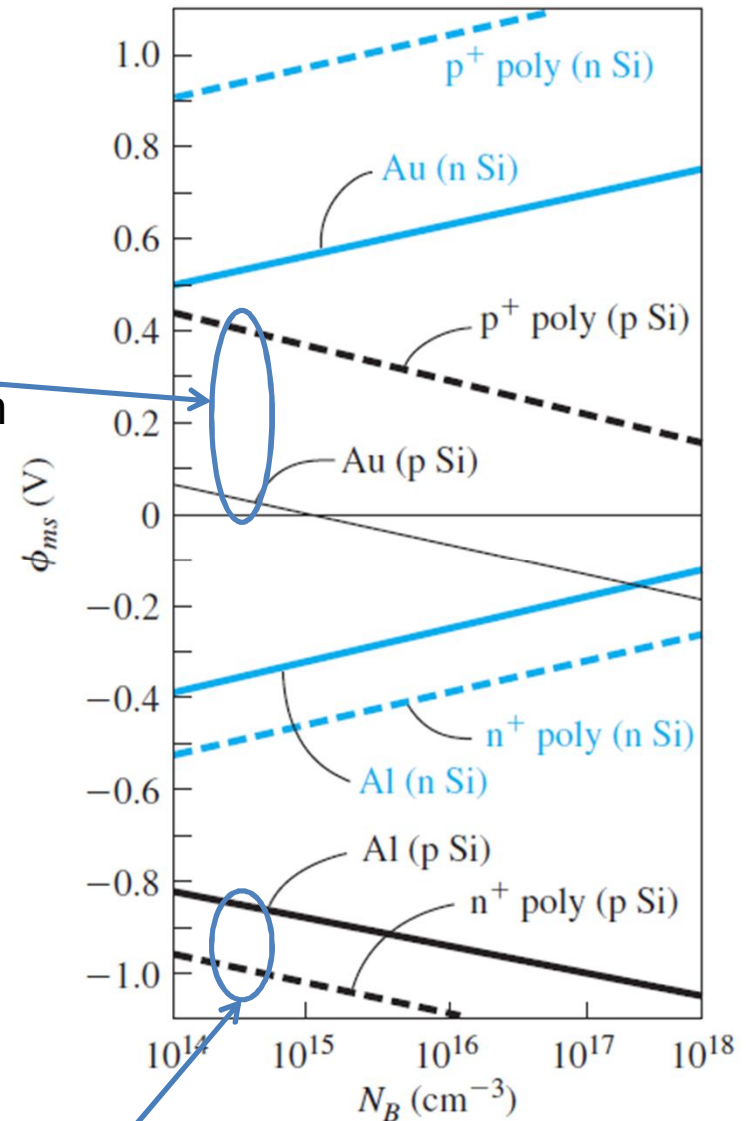
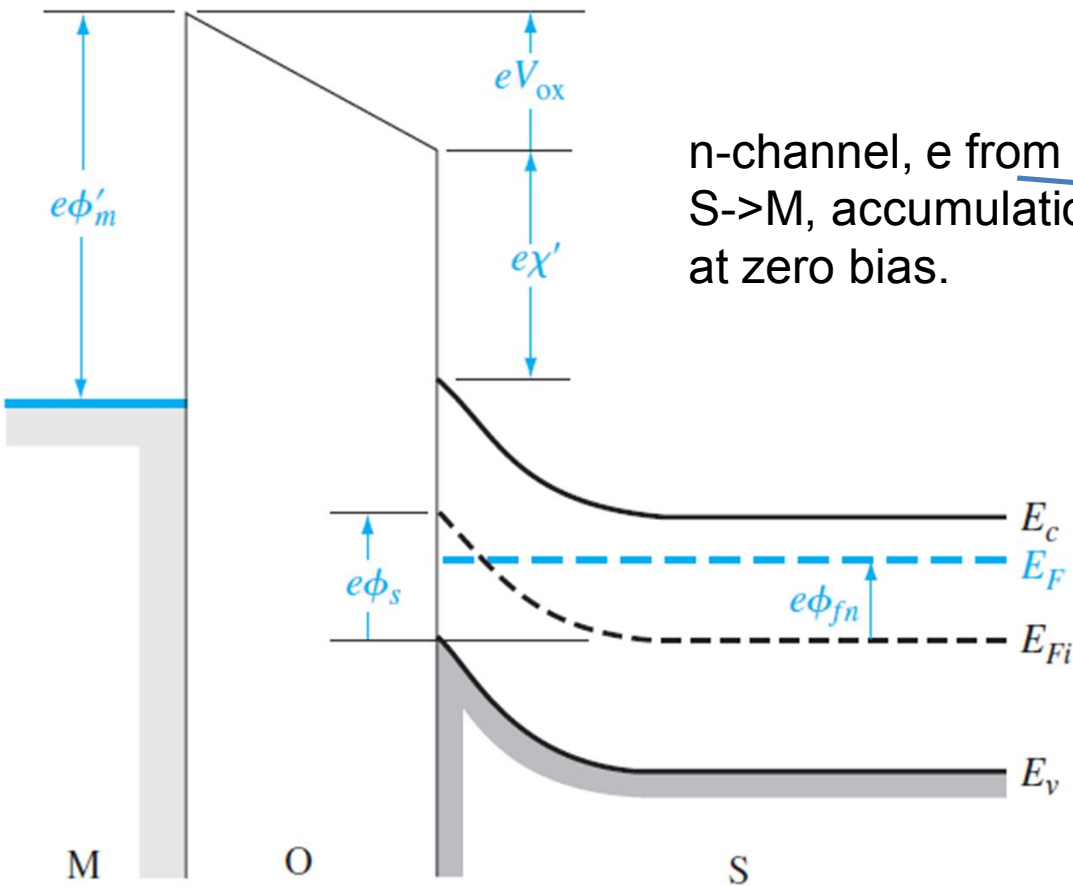
$$\phi_{ms} = \left[\chi' - \left(\chi' + \frac{E_g}{2e} + \phi_{fp} \right) \right] = - \left(\frac{E_g}{2e} + \phi_{fp} \right) \quad (10.16)$$

and for the p⁺ polysilicon gate, we have

$$\phi_{ms} = \left[\left(\chi' + \frac{E_g}{e} \right) - \left(\chi' + \frac{E_g}{2e} + \phi_{fp} \right) \right] = \left(\frac{E_g}{2e} - \phi_{fp} \right) \quad (10.17)$$

10.1.4 Work Function Differences

$$\phi_{ms} = \phi'_m - \left(\chi' + \frac{E_g}{2e} - \phi_{fn} \right)$$

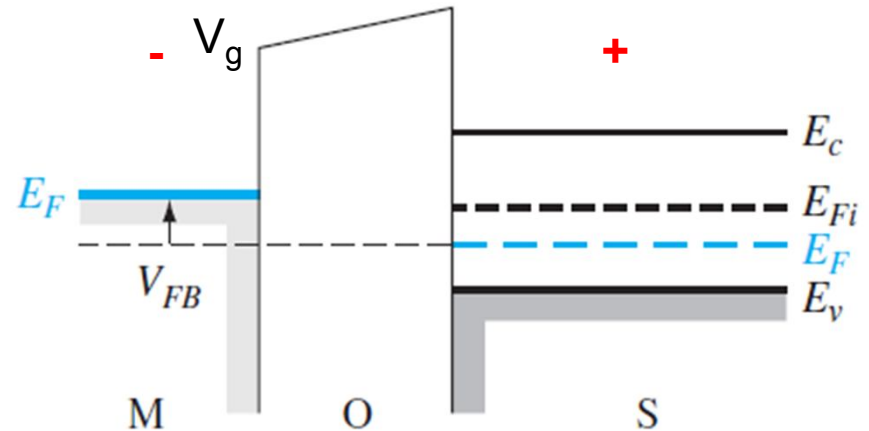


n-channel, e from M→S, weak inversion at zero bias

10.1.5 Flat-Band Voltage

The *flat-band voltage* is defined as the applied gate voltage such that there is no band bending in the semiconductor and, as a result, zero net space charge in this region.

作用：抵消金属与半导体之间的功函数差和氧化层的正电荷对半导体表面的影响



If a gate voltage is applied, the potential drop across the oxide and the surface potential will change. We can then write

$$V_G = \Delta V_{ox} + \Delta \phi_s = (V_{ox} - V_{ox0}) + (\phi_s - \phi_{s0}) \quad (10.20)$$

$$V_{ox0} + \phi_{s0} = -\phi_{ms}$$

$$V_G = V_{ox} + \phi_s + \phi_{ms}$$

$$Q'_m + Q'_{ss} = 0$$

We can relate Q'_m to the voltage across the oxide by

$$V_{ox} = \frac{Q'_m}{C_{ox}} = \frac{-Q'_{ss}}{C_{ox}}$$

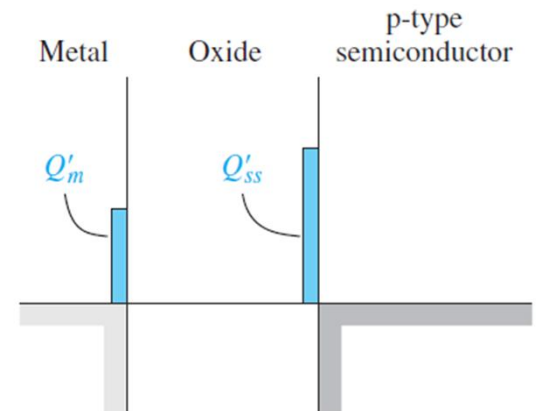


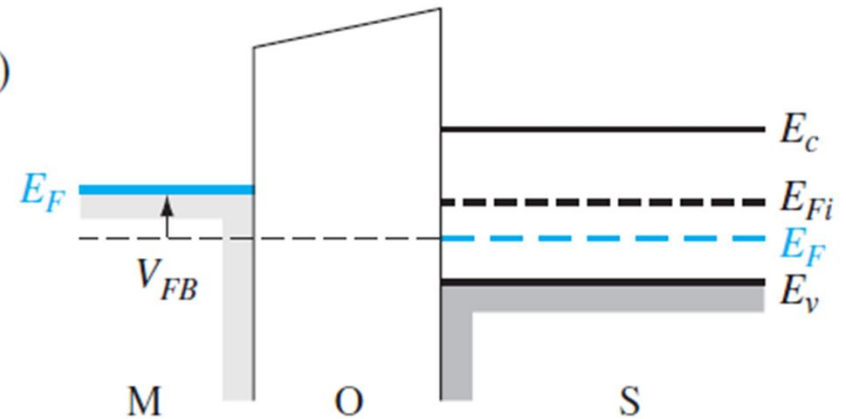
Figure 10.18 | Charge distribution in a MOS capacitor at flat band.

10.1.5 Flat-Band Voltage

$$V_G = \Delta V_{ox} + \Delta \phi_s = (V_{ox} - V_{ox0}) + (\phi_s - \phi_{s0})$$

$$V_{ox0} + \phi_{s0} = -\phi_{ms}$$

$$V_G = V_{ox} + \phi_s + \phi_{ms}$$



$$Q'_m + Q'_{ss} = 0$$

We can relate Q'_m to the voltage across the oxide by

$$V_{ox} = \frac{Q'_m}{C_{ox}} = \frac{-Q'_{ss}}{C_{ox}}$$

where ' indicates charges per unit area

In the flat-band condition, the surface potential is zero, or $\phi_s = 0$. Then from Equation (10.21), we have

$$V_G = \boxed{V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}} \quad (10.25)$$

Equation (10.25) is the flat-band voltage for this MOS device.

Flat-Band Voltage: Example

Consider a MOS capacitor with a p-type silicon substrate doped to $N_a = 10^{16} \text{ cm}^{-3}$, a silicon dioxide insulator with a thickness of $t_{ox} = 20 \text{ nm} = 200 \text{ \AA}$, and an n^+ polysilicon gate. Assume that $Q'_{ss} = 5 \times 10^{10}$ electronic charges per cm^2 .

■ Solution

The work function difference, from Figure 10.16, is $\phi_{ms} \cong -1.1 \text{ V}$. The oxide capacitance is found to be

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{(3.9)(8.85 \times 10^{-14})}{200 \times 10^{-8}} = 1.726 \times 10^{-7} \text{ F/cm}^2$$

The equivalent oxide charge density is

$$Q'_{ss} = (5 \times 10^{10})(1.6 \times 10^{-19}) = 8 \times 10^{-9} \text{ C/cm}^2$$

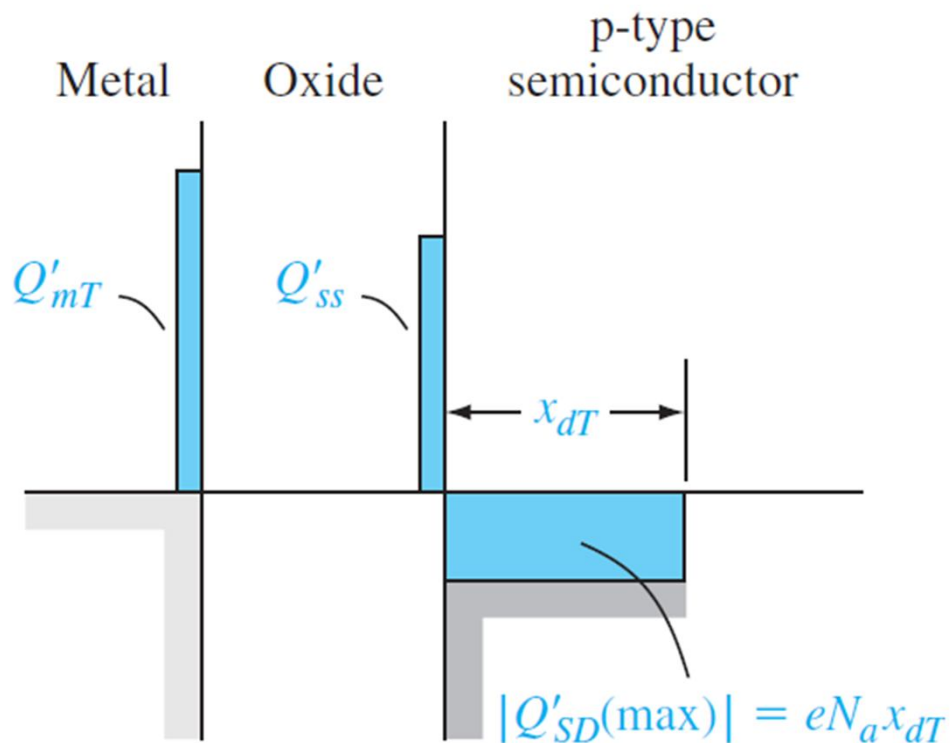
The flat-band voltage is then determined to be

$$V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}} = -1.1 \frac{-8 \times 10^{-9}}{1.726 \times 10^{-7}} = -1.15 \text{ V}$$

The applied gate voltage required to achieve the flat-band condition for this p-type substrate is negative. If the amount of fixed oxide charge increases, the flat-band voltage becomes even more negative.

10.1.6 Threshold Voltage

The threshold voltage is defined as the condition when the surface potential is $\phi_s = 2\phi_{fp}$ for the p - type semiconductor and $\Phi_s = 2\phi_{fn}$ for the n - type semiconductor.



- The space charge width has reached its maximum value.
- There is an equivalent oxide charge Q'_{ss} and the positive charge on the metal gate at threshold is Q'_{mT} .
- neglect the inversion layer charge at this threshold inversion point.

From conservation of charge, we can write

$$Q'_{mT} + Q'_{ss} = |Q'_{SD}(\max)| = eN_a x_{dT}$$

10.1.6 Threshold Voltage

An applied gate voltage will change the voltage across the oxide and will change the surface potential:

$$V_G = \Delta V_{ox} + \Delta \phi_s = V_{ox} + \phi_s + \phi_{ms}$$

At threshold, the surface potential is: $\phi_s = 2\phi_{fp}$

Threshold voltage:

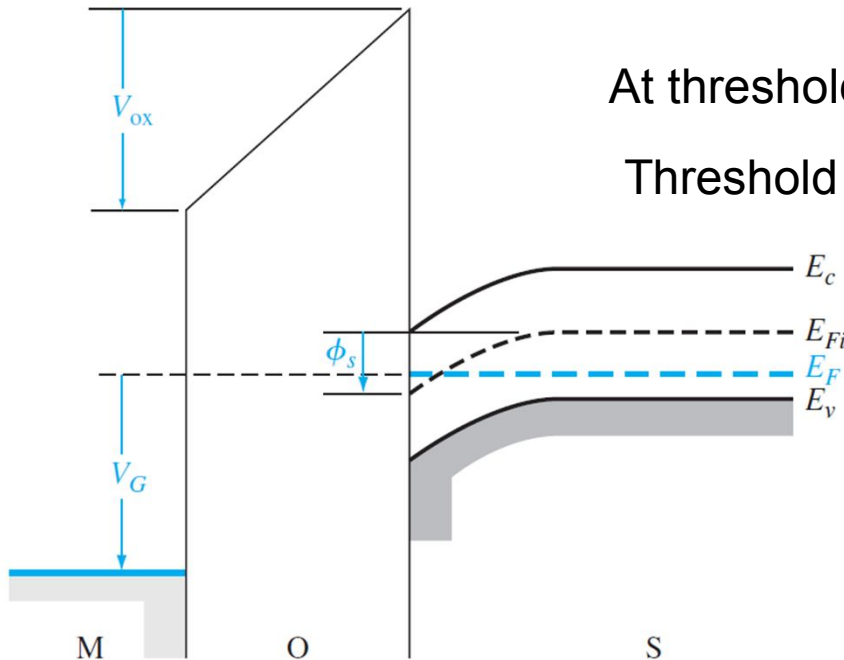
$$V_{TN} = V_{oxT} + 2\phi_{fp} + \phi_{ms}$$

$$V_{oxT} = \frac{Q'_{mT}}{C_{ox}} = \frac{1}{C_{ox}} (|Q'_{SD}(\max)| - Q'_{ss})$$

$$V_{TN} = \frac{|Q'_{SD}(\max)|}{C_{ox}} - \frac{Q'_{ss}}{C_{ox}} + \phi_{ms} + 2\phi_{fp}$$

Since: $V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$

$$V_{TN} = \frac{|Q'_{SD}(\max)|}{C_{ox}} + V_{FB} + 2\phi_{fp}$$



The threshold voltage is a function of semiconductor doping, oxide charge Q'_{ss} , and oxide thickness.

Threshold Voltage: Example

Consider a p-type silicon substrate at $T = 300$ K doped to $N_a = 10^{15} \text{ cm}^{-3}$. Let $Q'_{ss} = 10^{10} \text{ cm}^{-2}$, $t_{ox} = 12 \text{ nm} = 120 \text{ \AA}$, and assume the oxide is silicon dioxide.

■ **Solution** Calculate the threshold voltage with an aluminum gate.

From Figure 10.16, we find $\phi_{ms} \cong -0.88 \text{ V}$. The other parameters are

$$\phi_{fp} = V_t \ln \left(\frac{N_a}{n_i} \right) = (0.0259) \ln \left(\frac{10^{15}}{1.5 \times 10^{10}} \right) = 0.2877 \text{ V}$$

and

$$x_{dT} = \left\{ \frac{4\epsilon_s \phi_{fp}}{eN_a} \right\}^{1/2} = \left\{ \frac{4(11.7)(8.85 \times 10^{-14})(0.2877)}{(1.6 \times 10^{-19})(10^{15})} \right\}^{1/2} = 8.63 \times 10^{-5} \text{ cm}$$

Then

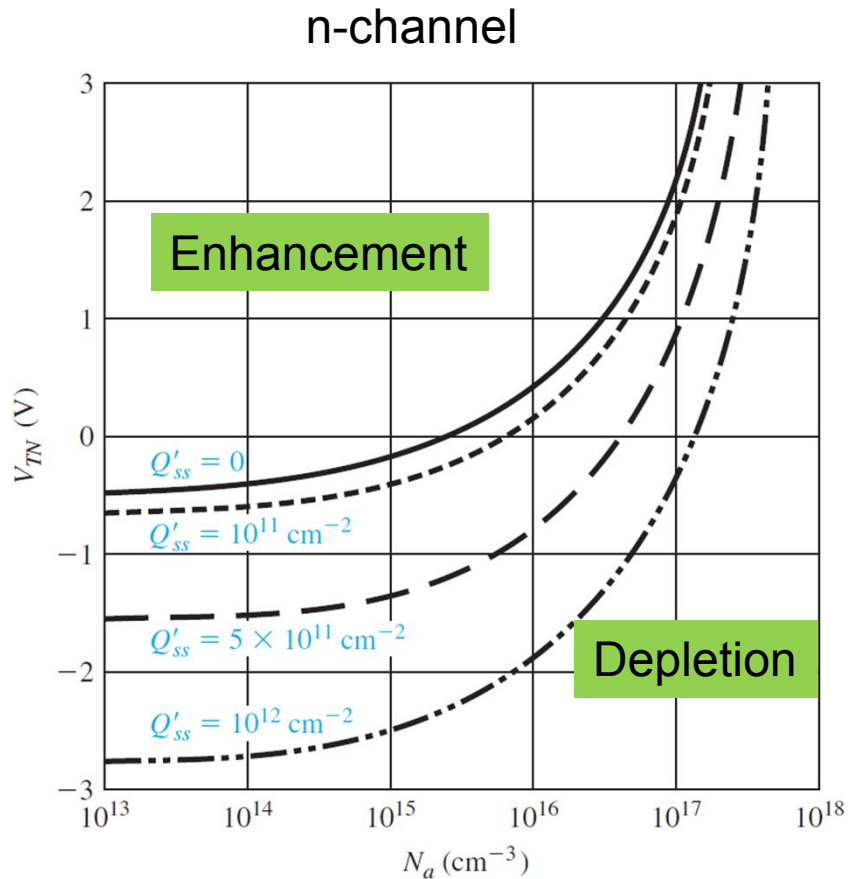
$$|Q'_{SD}(\text{max})| = eN_a x_{dT} = (1.6 \times 10^{-19})(10^{15})(8.63 \times 10^{-5}) = 1.381 \times 10^{-8} \text{ C/cm}^2$$

The threshold voltage is now found to be

$$\begin{aligned} V_{TN} &= (|Q'_{SD}(\text{max})| - Q'_{ss}) \left(\frac{t_{ox}}{\epsilon_{ox}} \right) + \phi_{ms} + 2\phi_{fp} \\ &= [(1.381 \times 10^{-8}) - (10^{10})(1.6 \times 10^{-19})] \cdot \left[\frac{120 \times 10^{-8}}{(3.9)(8.85 \times 10^{-14})} \right] \\ &\quad + (-0.88) + 2(0.2877) = -0.262 \text{ V} \end{aligned}$$

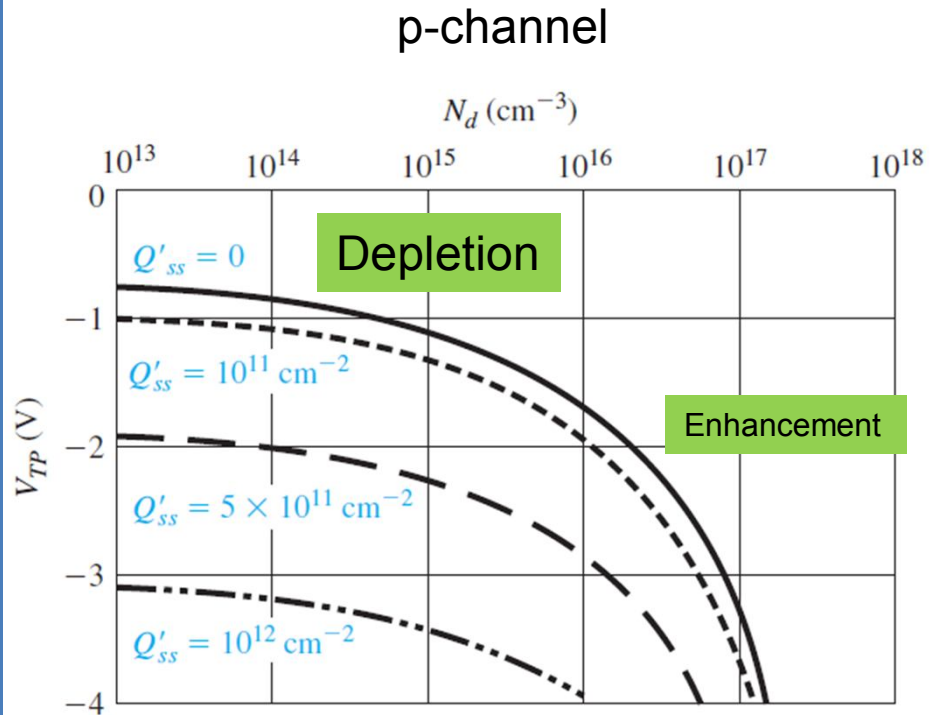
For fairly **lightly doped** semiconductor, in conjunction with the **positive charge** in the oxide and the **work function difference**, is sufficient to induce an electron inversion layer charge even **with zero applied gate voltage**. This condition makes the threshold voltage negative: **depletion mode**

Threshold Voltage



$$V_{TN} = (|Q'_{SD}(\text{max})| - Q'_{ss}) \left(\frac{t_{ox}}{\epsilon_{ox}} \right) + \phi_{ms} + 2\phi_{fp}$$

the p - type semiconductor must be somewhat heavily doped in order to obtain an enhancement mode device.



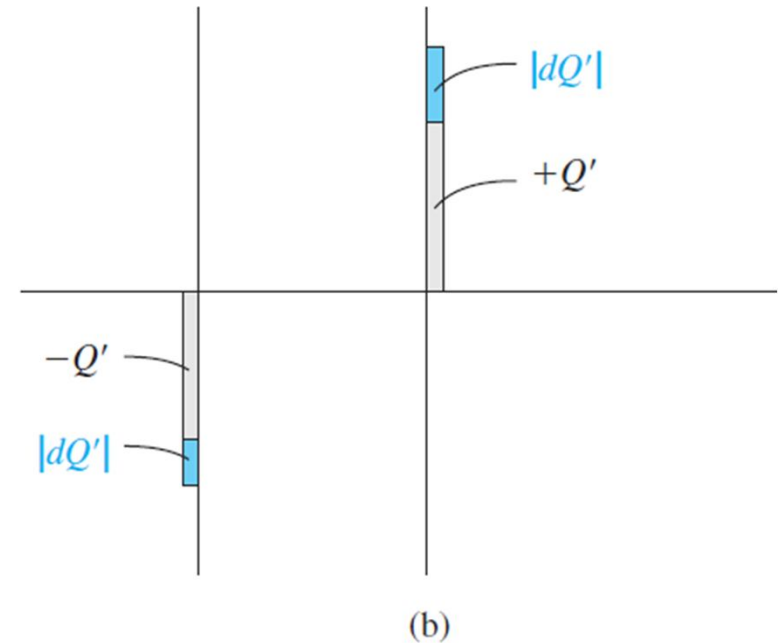
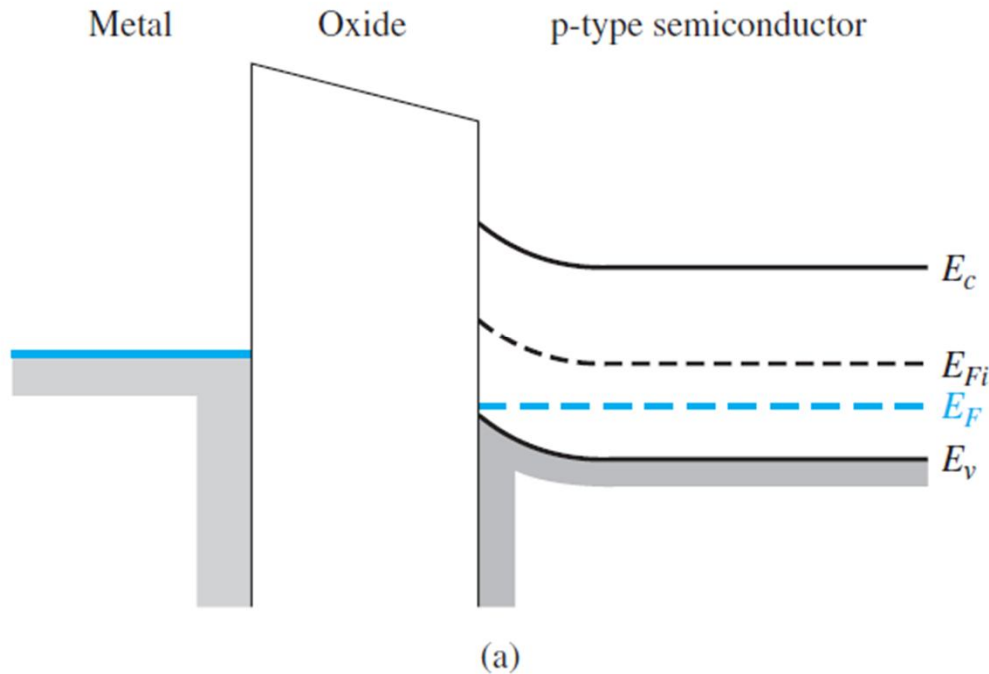
$$V_{TP} = (-|Q'_{SD}(\text{max})| - Q'_{ss}) \left(\frac{t_{ox}}{\epsilon_{ox}} \right) + \phi_{ms} - 2\phi_{fn}$$

for all values of positive oxide charge, this MOS capacitor is always an enhancement mode device

10.2 | CAPACITANCE–VOLTAGE CHARACTERISTICS

There are three operating conditions of interest in the MOS capacitor: accumulation, depletion, and inversion. For ideal C-V characteristics, we will assume that there is zero charge trapped in the oxide and also that there is no charge trapped at the oxide semiconductor interface.

In accumulation mode: $C'(\text{acc}) = C_{\text{ox}} = \frac{\epsilon_{\text{ox}}}{t_{\text{ox}}}$



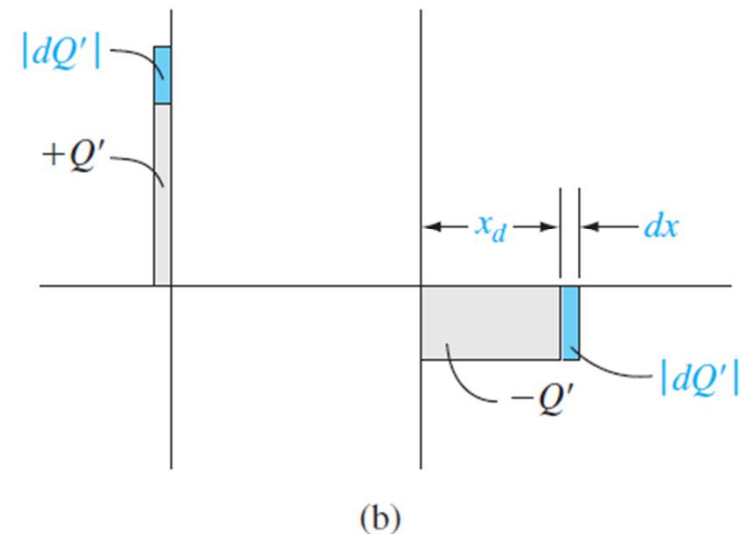
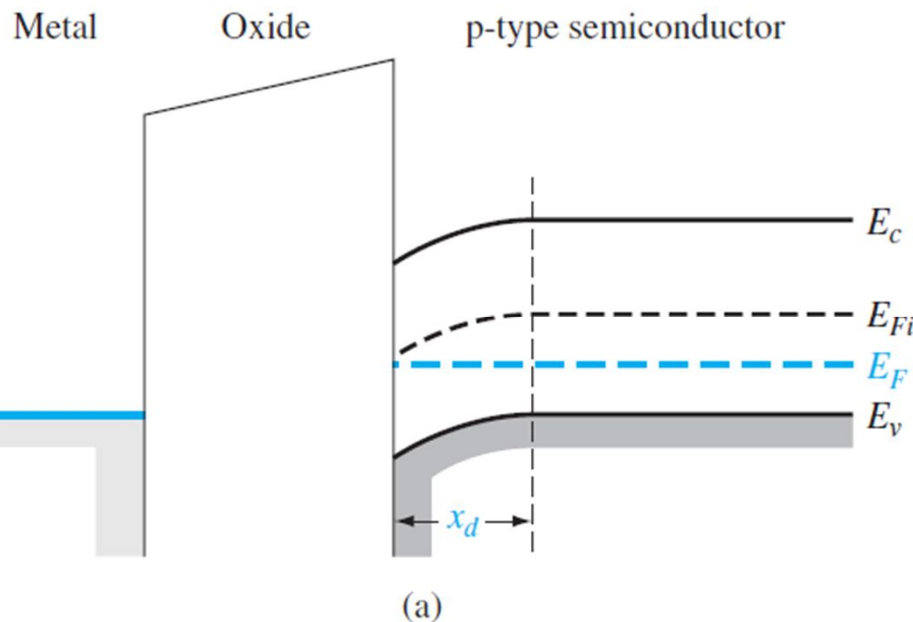
10.2 | CAPACITANCE–VOLTAGE CHARACTERISTICS

In **depletion mode**, the oxide capacitance and the capacitance of the depletion region are in series :

$$\frac{1}{C'(\text{depl})} = \frac{1}{C_{\text{ox}}} + \frac{1}{C'_{SD}} \quad C'(\text{depl}) = \frac{C_{\text{ox}} C'_{SD}}{C_{\text{ox}} + C'_{SD}} = \frac{C_{\text{ox}}}{1 + \frac{C_{\text{ox}}}{C'_{SD}}} = \frac{\epsilon_{\text{ox}}}{t_{\text{ox}} + \left(\frac{\epsilon_{\text{ox}}}{\epsilon_s}\right) x_d}$$

At threshold inversion point, maximum depletion width is reached:

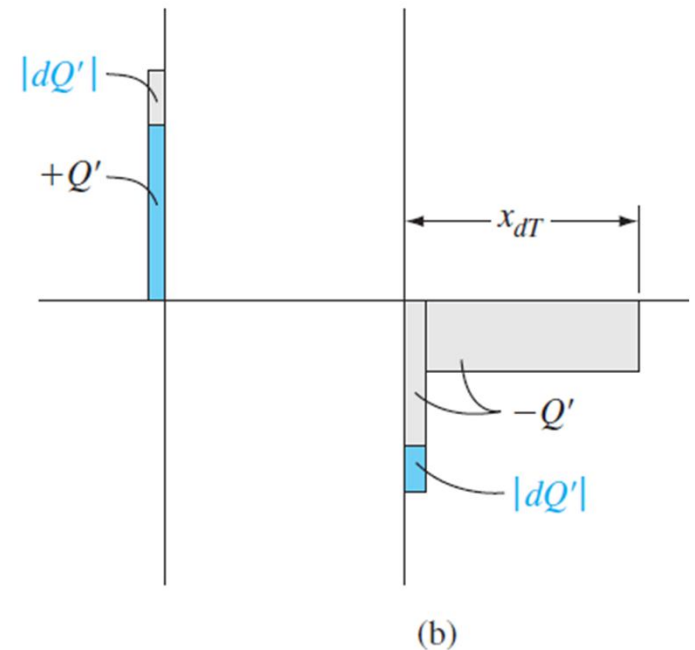
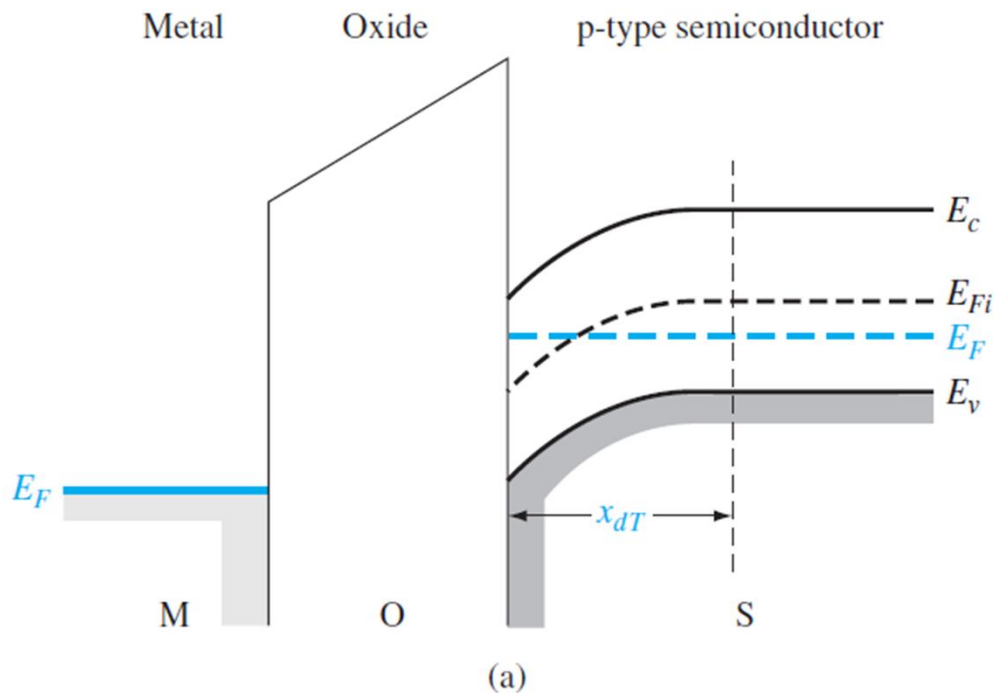
$$C'_{\min} = \frac{\epsilon_{\text{ox}}}{t_{\text{ox}} + \left(\frac{\epsilon_{\text{ox}}}{\epsilon_s}\right) x_{dT}} \quad x_d = \left(\frac{2\epsilon_s \phi_s}{eN_a}\right)^{1/2} \quad x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}$$



10.2 | CAPACITANCE–VOLTAGE CHARACTERISTICS

In **inversion mode**, a small incremental change in the voltage across the MOS capacitor will cause a differential change in the inversion layer charge density. The space charge width does not change. If the inversion charge can respond to the change in capacitor voltage, then the capacitance is again just the oxide capacitance:

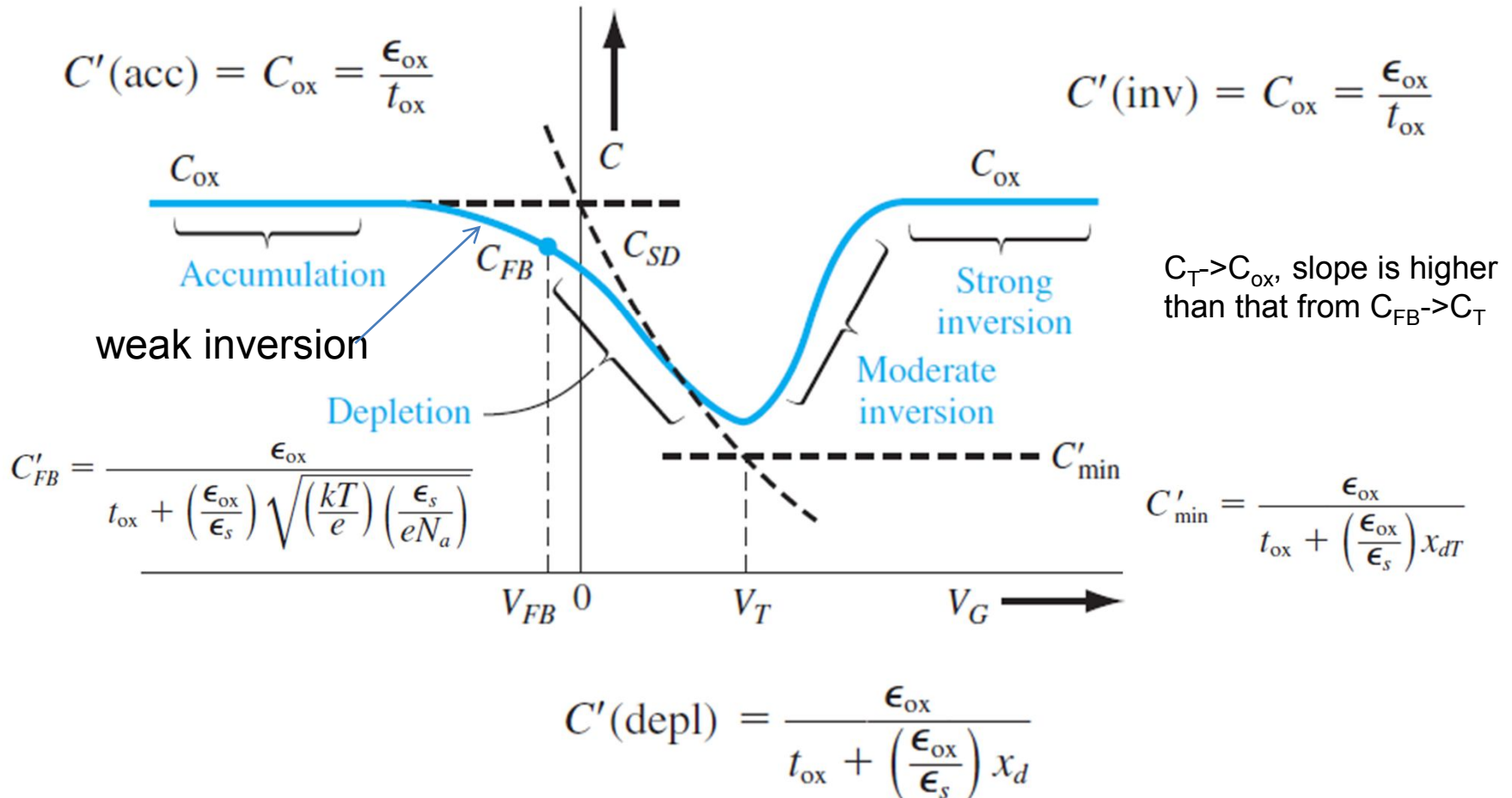
$$C'(\text{inv}) = C_{\text{ox}} = \frac{\epsilon_{\text{ox}}}{t_{\text{ox}}}$$



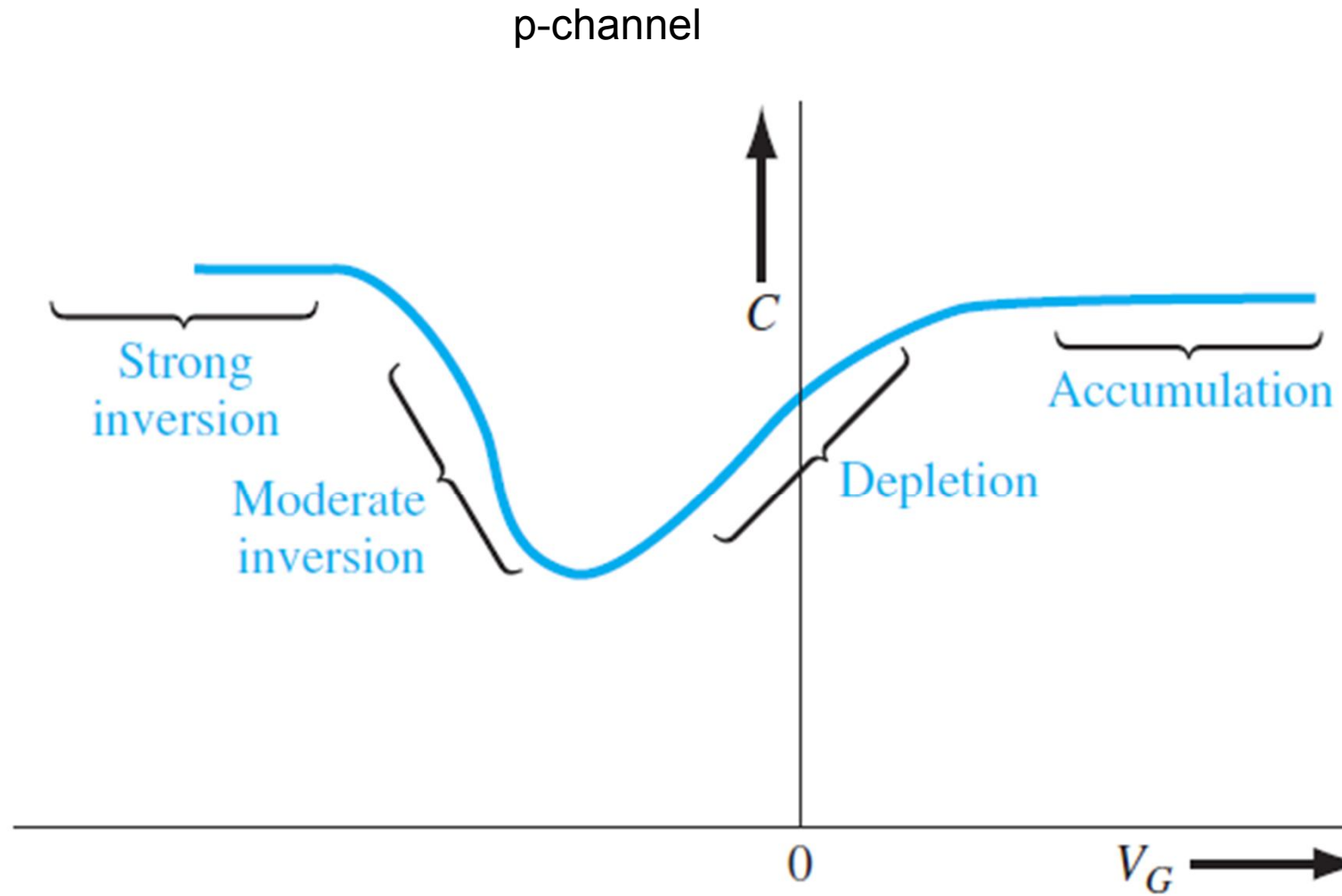
Ideal low frequency $C-V$ Characteristics

n-channel

' indicates capacitance per unit area



Ideal low frequency $C-V$ Characteristics

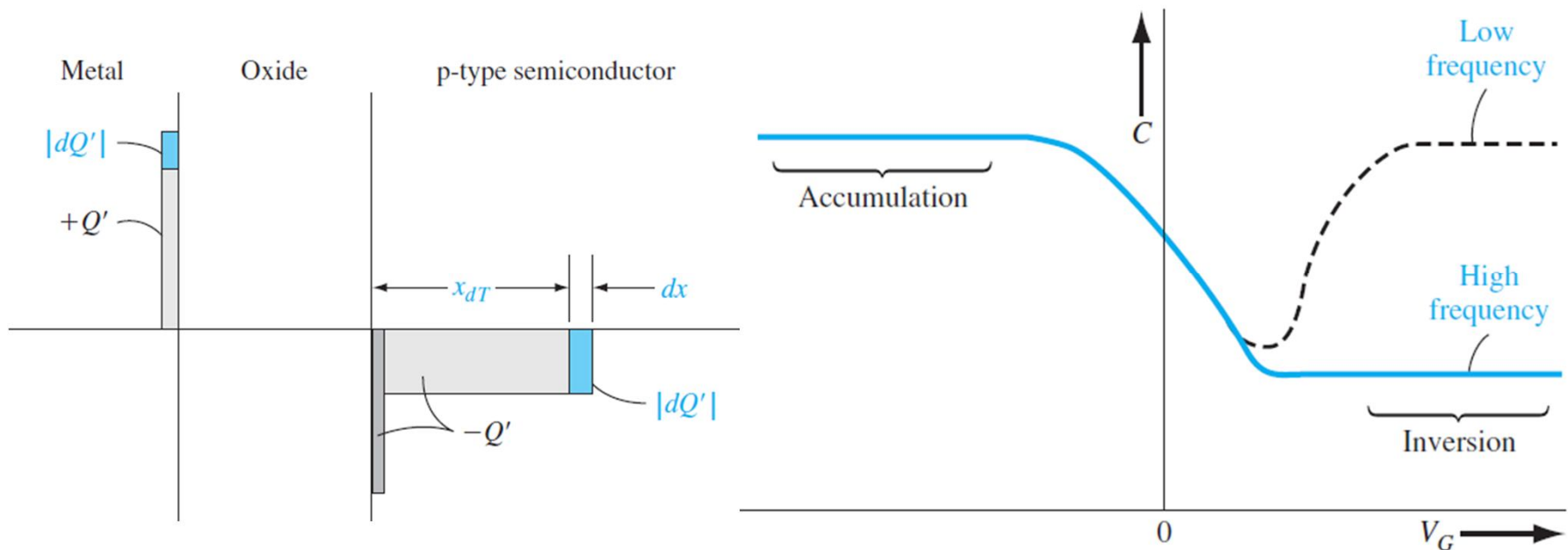


Ideal high frequency $C-V$ Characteristics

Two sources of electrons that can change the charge density of the inversion layer:

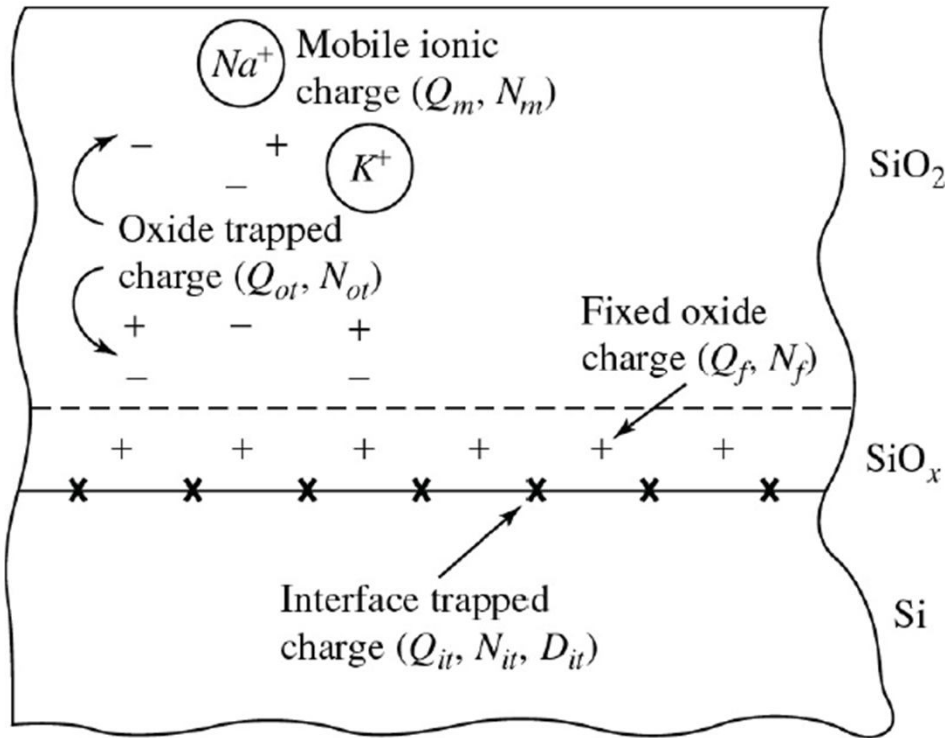
1. by **diffusion of minority carrier** electrons from the p - type substrate across the space charge region.
2. by **thermal generation of electron-hole pairs** within the space charge region.

In the limit of a very high frequency, the inversion layer charge will not respond to a differential change in capacitor voltage.

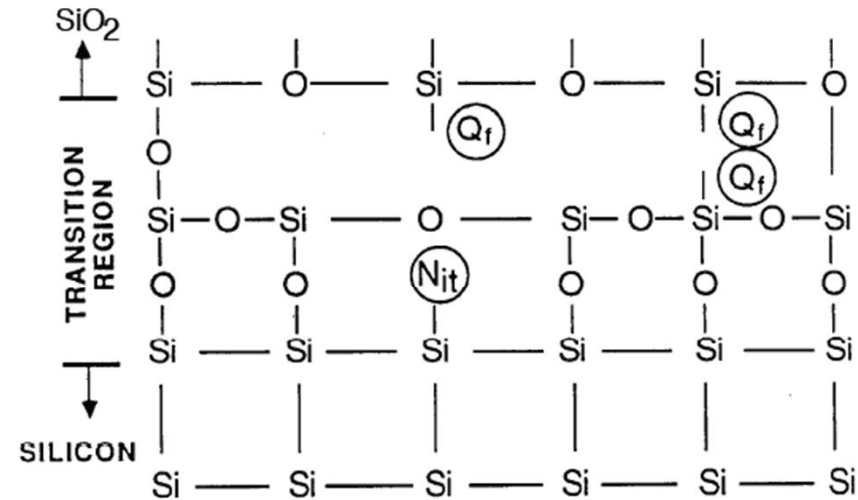


In general, high frequency corresponds to a value on the order of **1 MHz** and low frequency corresponds to values in the range of **5 to 100 Hz**. Typically, the highfrequency characteristics of the MOS capacitor are measured.

Fixed Oxide and Interface Charge Effects



Four categories of oxide charge in the MOS system. The symbols for the charge densities Q (C cm⁻²), and state densities N (states cm⁻²) or D (states cm⁻² eV⁻¹) have been standardized.



Dangling bonds which are not passivated by oxygen give rise to interface energy states and fixed charge

Positive charges due to loss of electrons

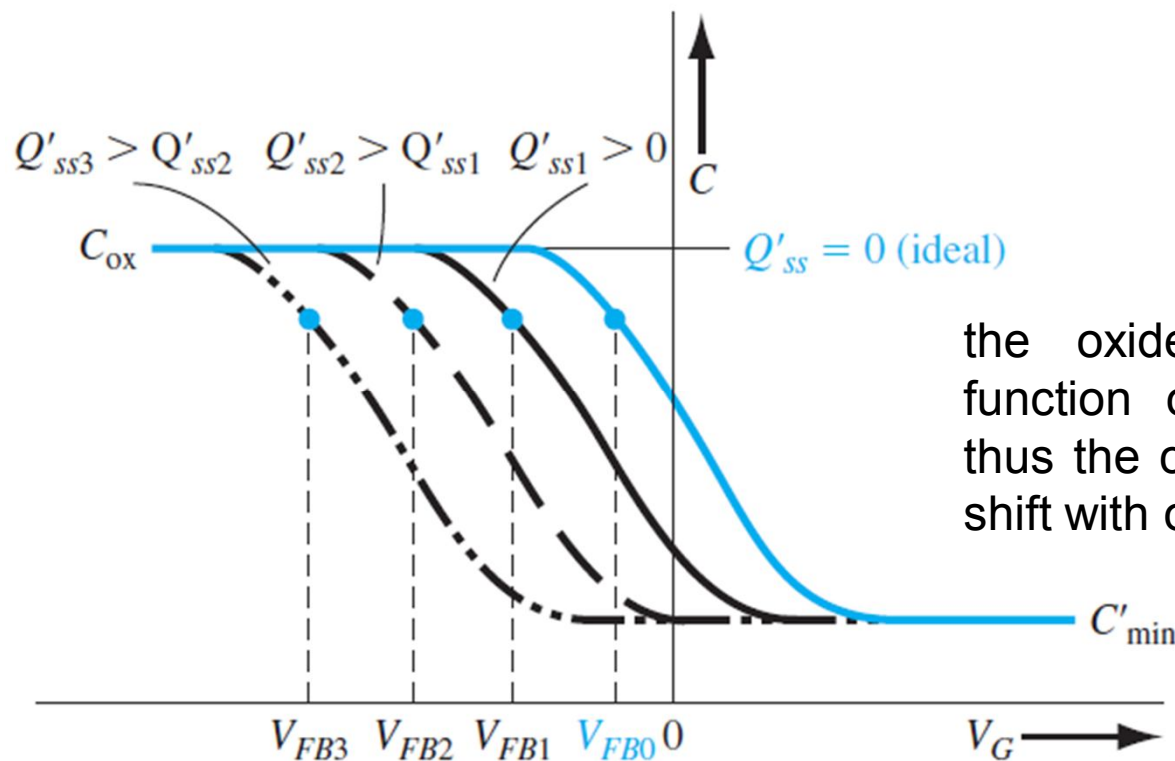
Types and origins of charge:

1. Fixed interface charge, Q_f (Also known as Q_{ss}) Believed to be due to uncompensated silicon-silicon bonds, on the order of 10^{10} cm^{-2} .
2. Interface trapped-charge, Q_{it} Due to unterminated Si bonds, on the order of 10^{10} cm^{-2} .
3. Oxide trapped-charge, Q_{ot} Due to defects in the SiO_2 network, usually negligible in today's MOS devices. Can be generated after a large amount of charges have passed through the SiO_2 and broken up the network.
4. Mobile charge Q_m Due to alkaline ions (e.g., Na^+ , K^+), usually very low concentration ($\approx 10^9 \text{ cm}^{-2}$) in today's technology.

Fixed Oxide Charge Effects

In the presence of fixed oxide or oxide-semiconductor interface charges, the flat-band voltage is modified:

$$V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$$

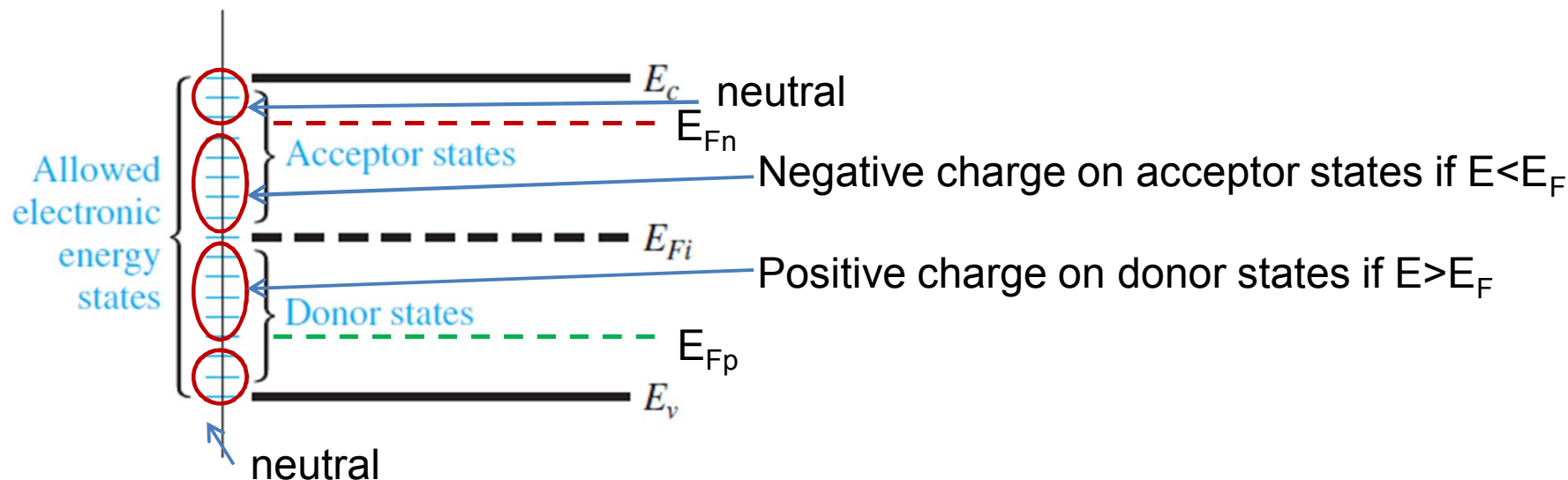


the oxide charge is not a function of gate voltage, and thus the curves show a parallel shift with oxide charge.

can be used to determine the equivalent fixed oxide charge

Interface Charge Effects

- In contrast to the fixed oxide charge, charge can flow between the semiconductor and interface states.
- The net charge in these interface states is a function of the position of the Fermi level in the bandgap.



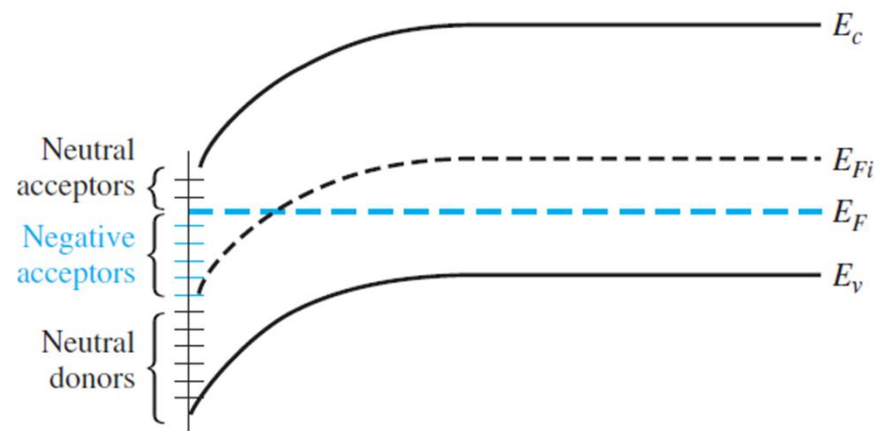
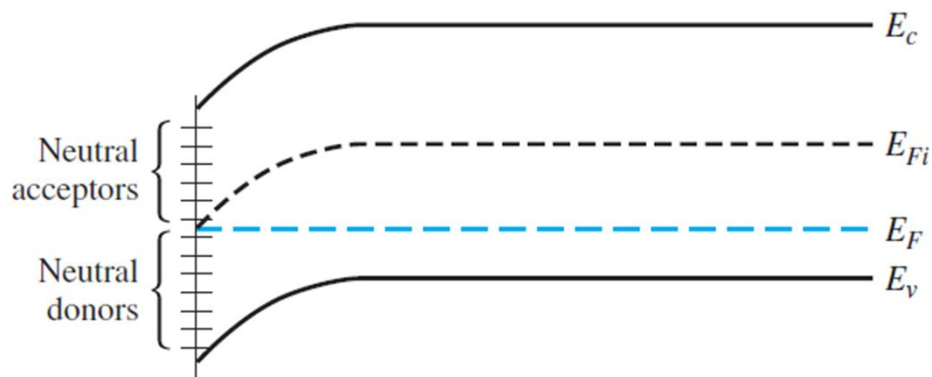
Interface Charge Effects

Accumulation, positive interface charges trapped in the donor states. V_{FB} reduces.

$$V_{FB} = \phi_{ms} - \frac{Q'_{ss}}{C_{ox}}$$

Depletion, all interface states are neutral. *Midgap bias*.

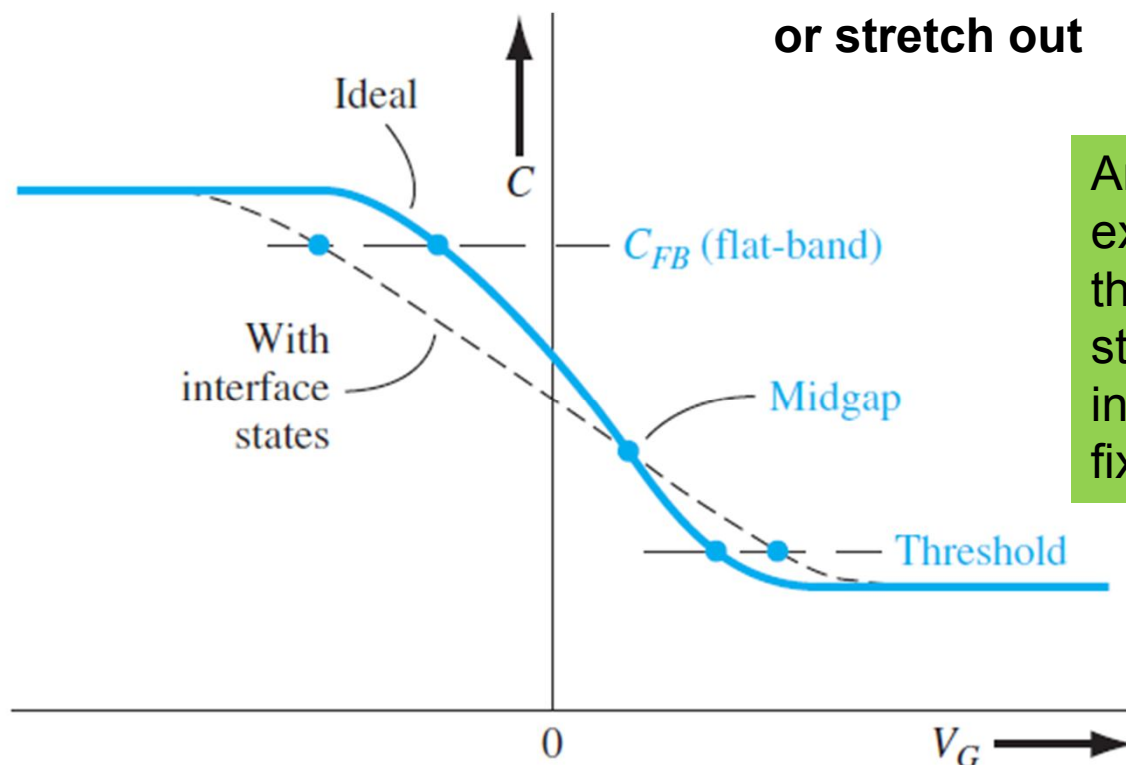
Inversion, negative interface charges trapped in the acceptor states. V_{FB} increases.



Interface Charge Effects

When interface states are present, the amount and direction of the shift change as we sweep through the gate voltage, since the amount and sign of the interface trapped charge change.

C–V curves become “smeared out” or stretch out

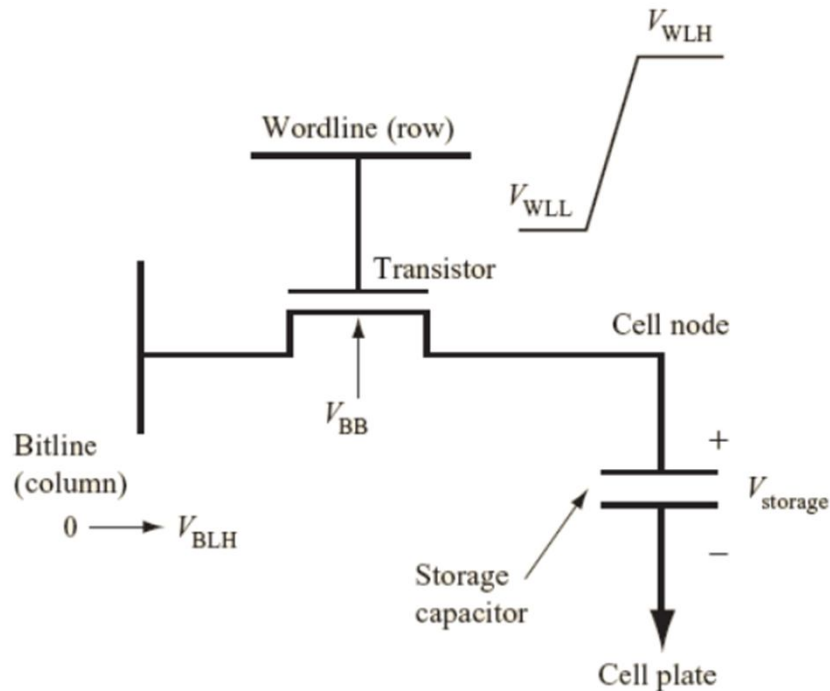


Any “smearing out” in the experimental curve indicates the presence of interface states and any parallel shift indicates the presence of fixed oxide charge.

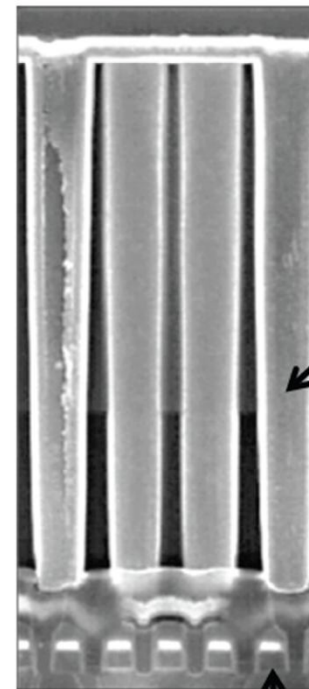
Applications of MOS Capacitors

Dynamic Random Access Memory (DRAM)

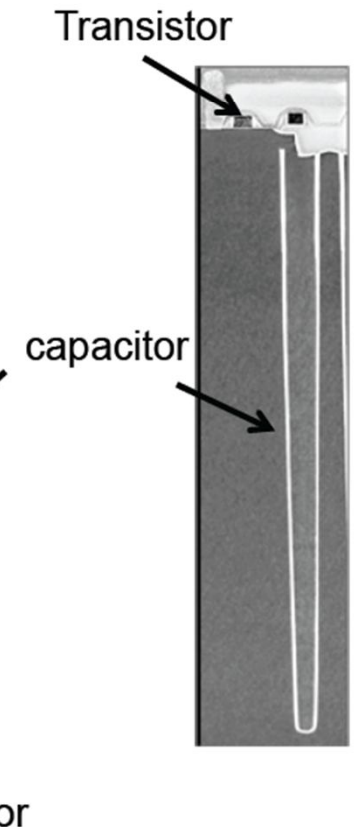
DRAM with one transistor and one capacitor. Charge is stored on the capacitor and sensed by the transistor. Binary data is represented by the presence or absence of the charge.



DRAM:
Stacked
Capacitor



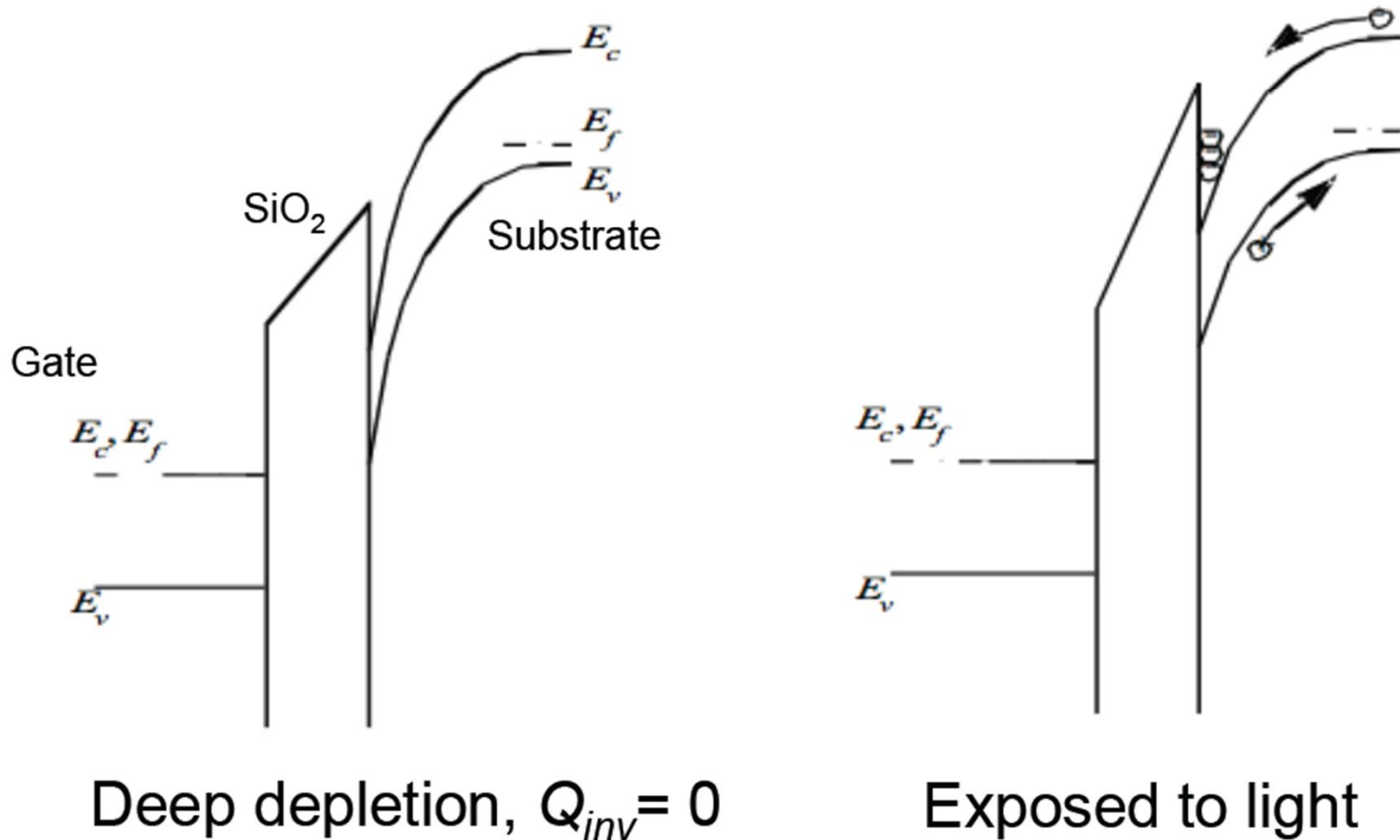
DRAM:
Trenched
Capacitor



Applications of MOS Capacitors

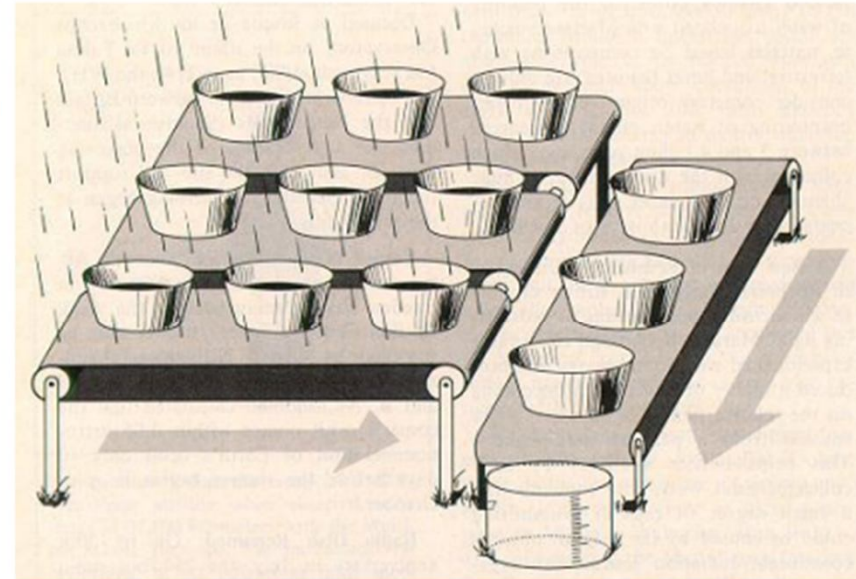
CCD Imager

Extensively used in digital cameras, imagers, etc.

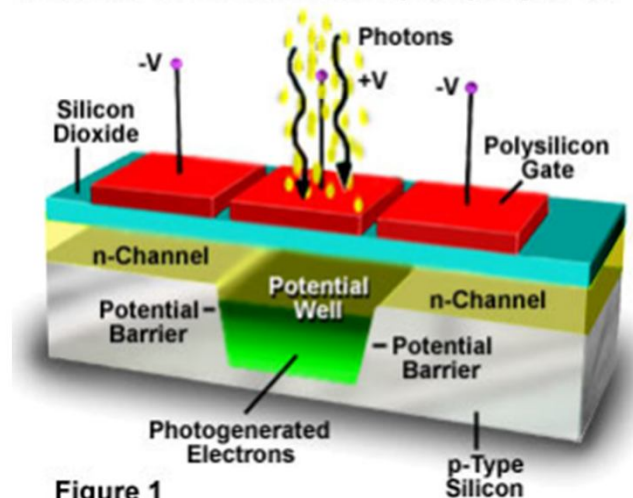


Charge Coupled Device (CCD)

- ❑ **Step 0 - Light into Detector**
- ❑ **Step 1 - Charge Generation**
- ❑ **Step 2 - Charge Collection**
- ❑ **Step 3 - Charges Transfer**
- ❑ **Step 4 - Charge-to-Voltage Conversion**
- ❑ **Step 5 - Digitization**

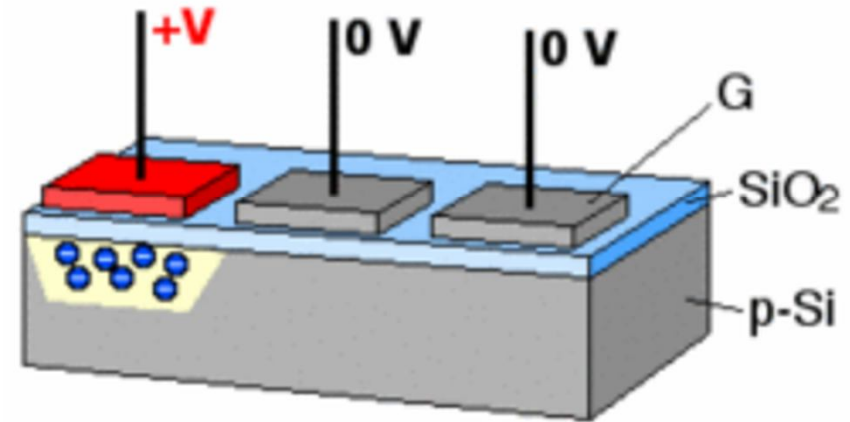
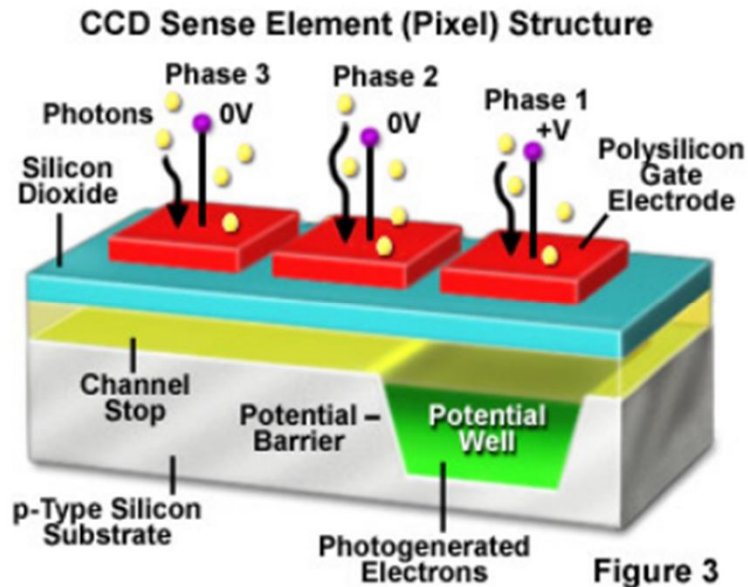


Metal Oxide Semiconductor (MOS) Capacitor



Charge Coupled Device (CCD)

CCD pixel

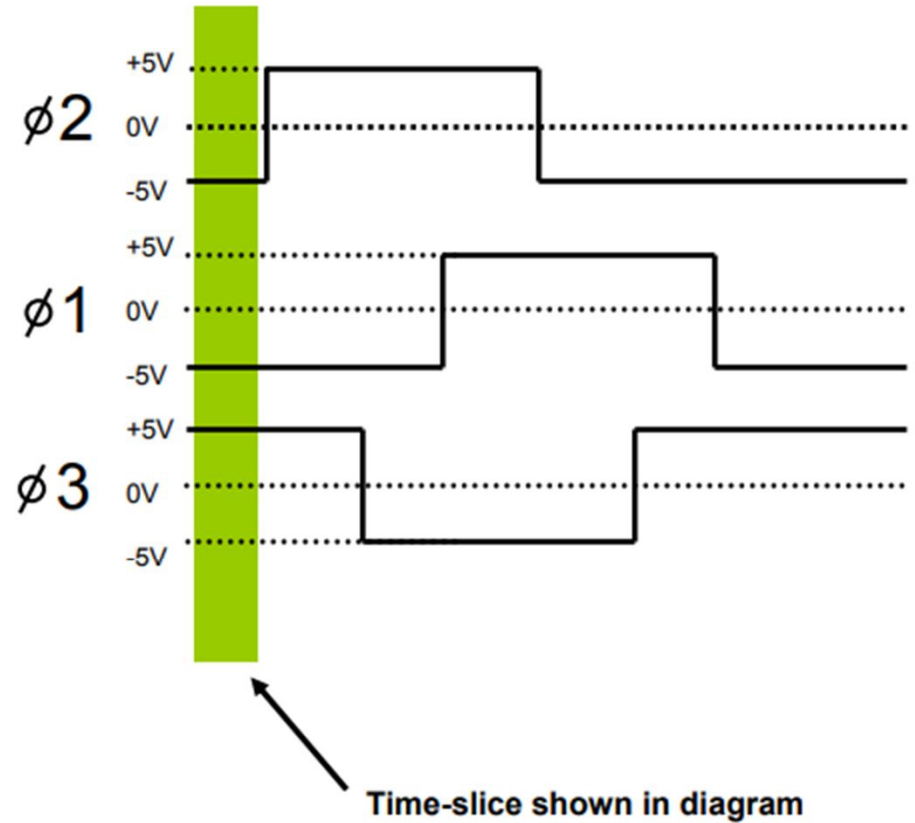
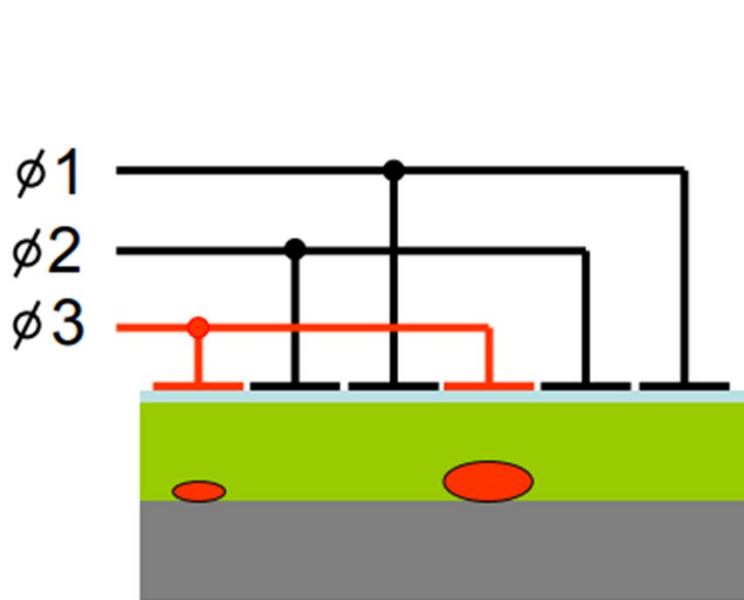


- ❑ *MOS capacitors are the basic building blocks of the CCD*
- ❑ *Each pixel consists of several MOS capacitors*
- ❑ *MOS capacitors follow phase clock to transfer charges*

Charge Coupled Device (CCD)

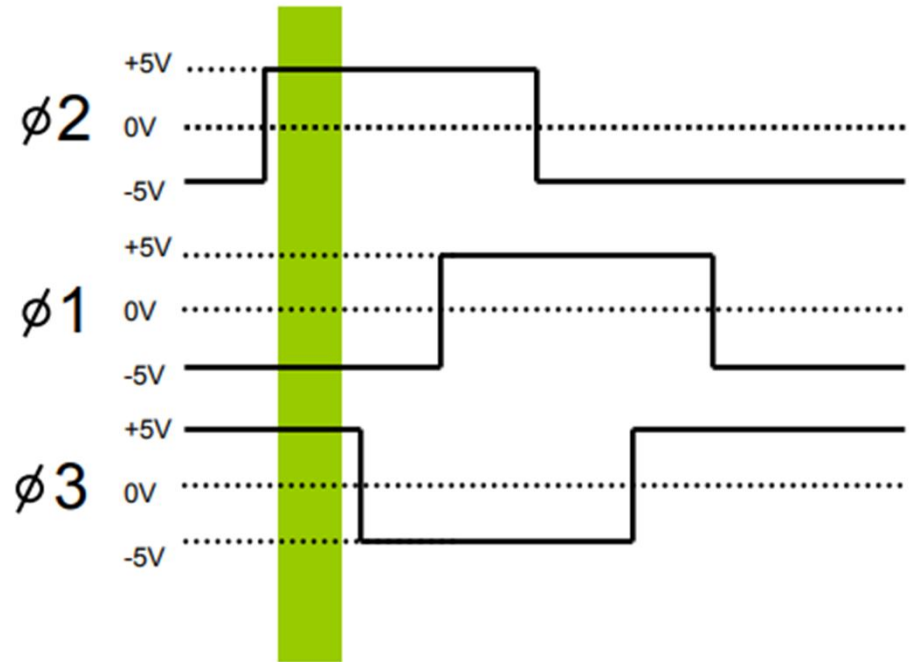
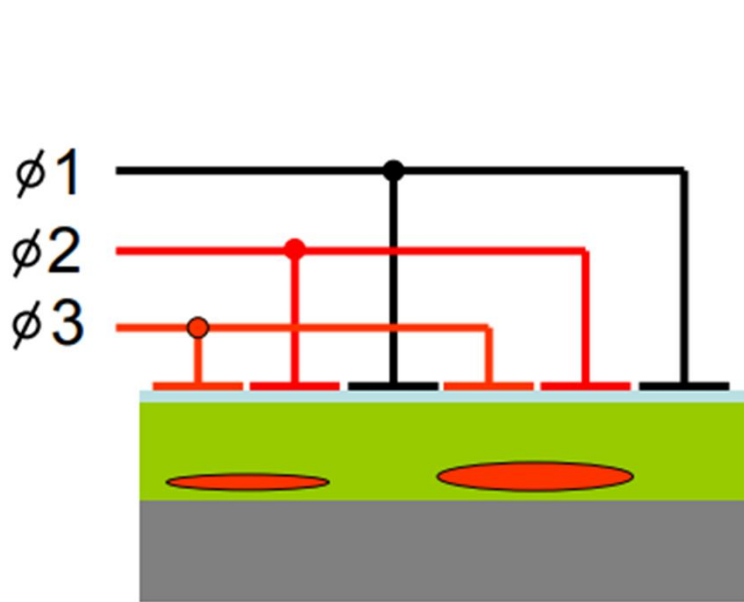
CCD charge transfer

CCD Phased Clocking: Step 1



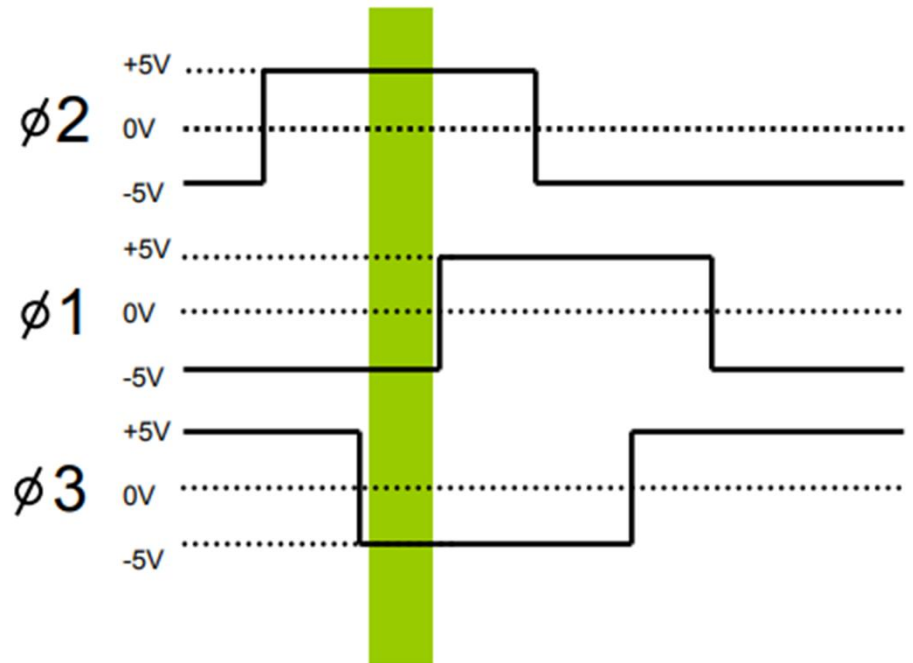
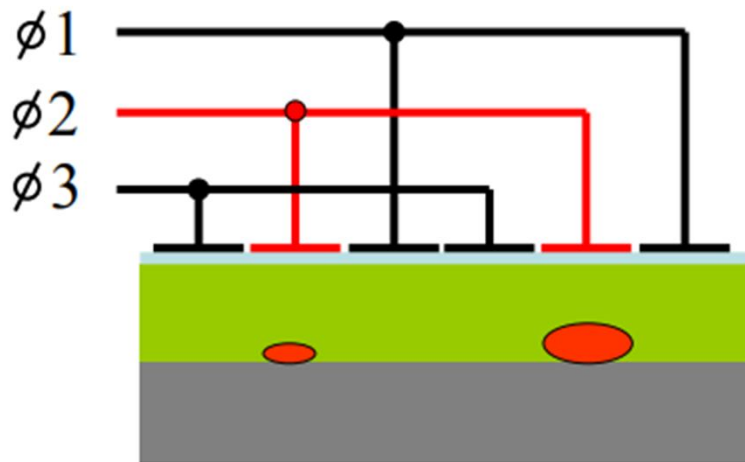
Charge Coupled Device (CCD)

CCD Phased Clocking: Step 2



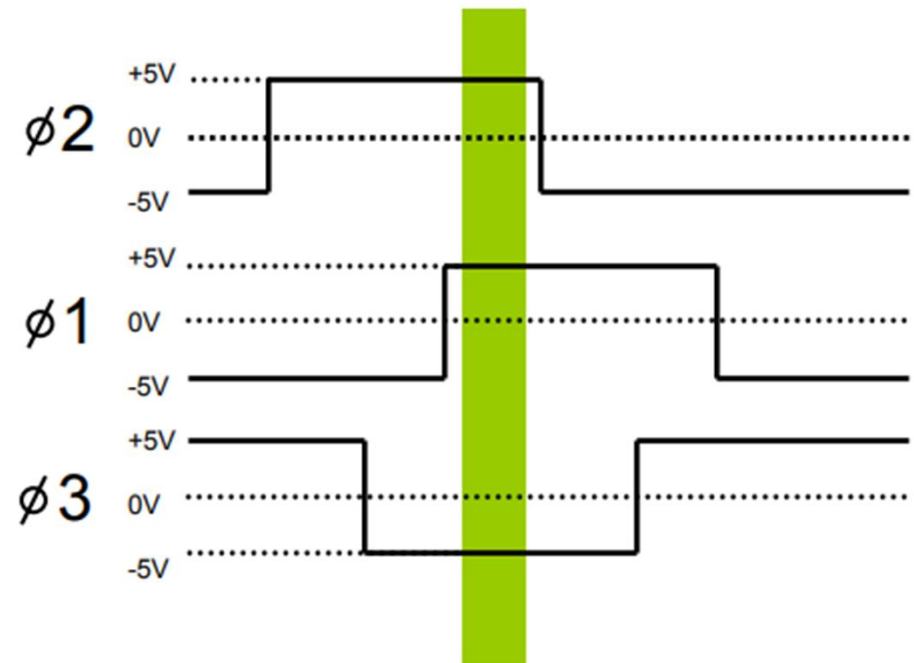
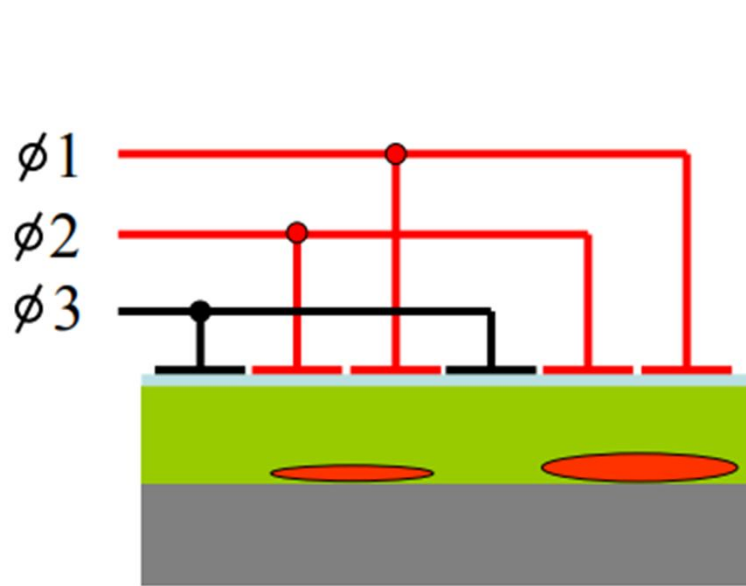
Charge Coupled Device (CCD)

CCD Phased Clocking: Step 3



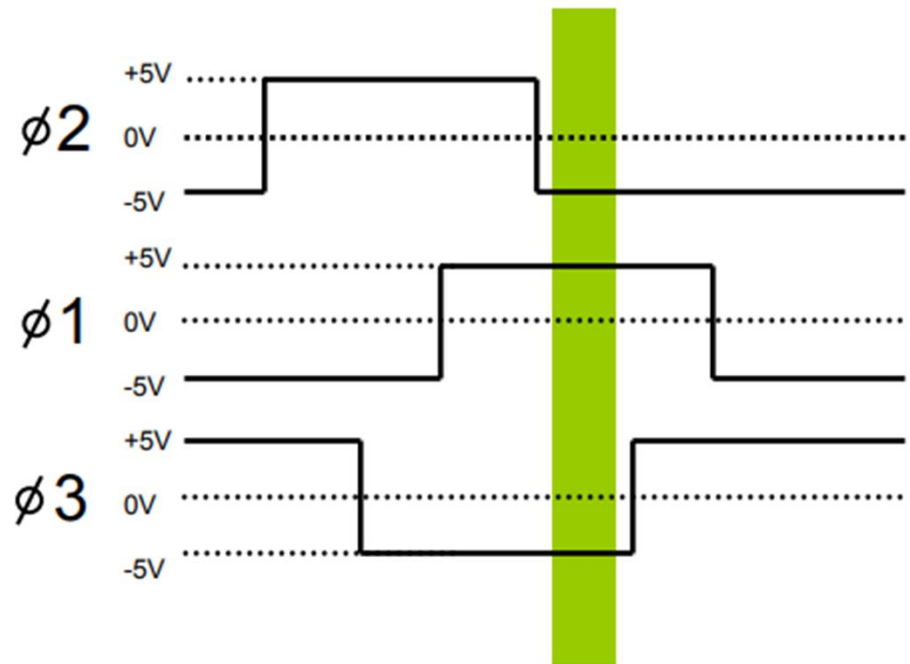
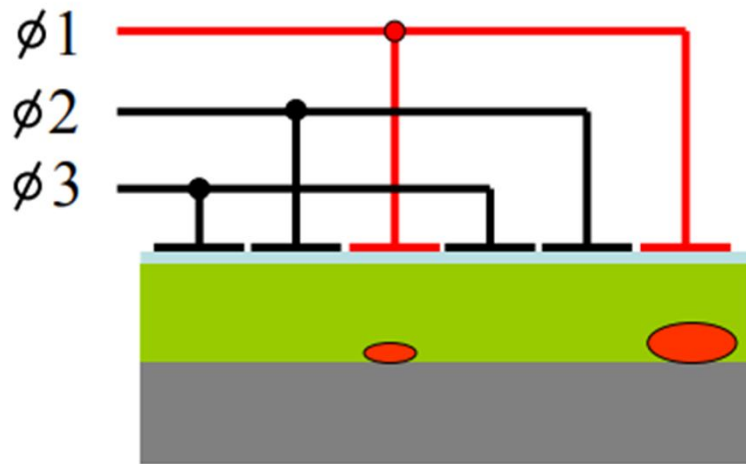
Charge Coupled Device (CCD)

CCD Phased Clocking: Step 4



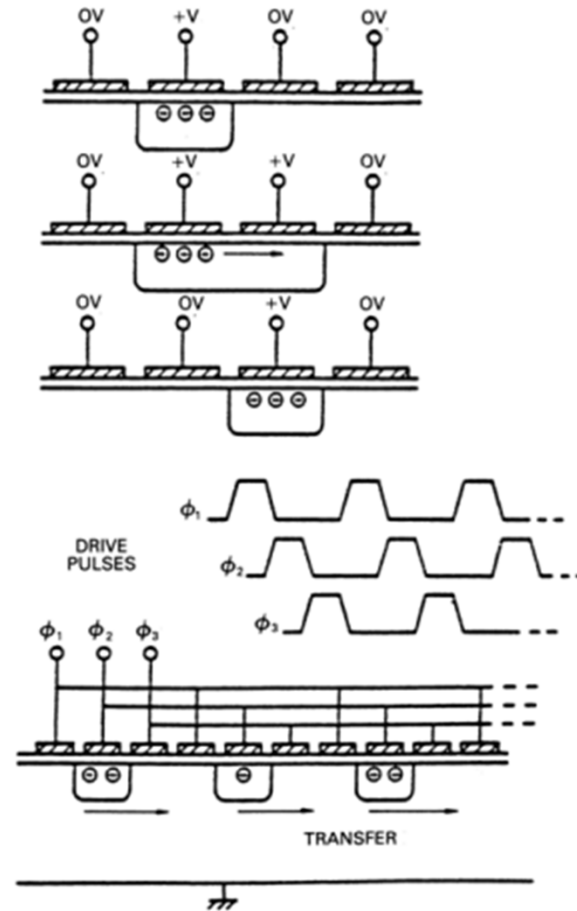
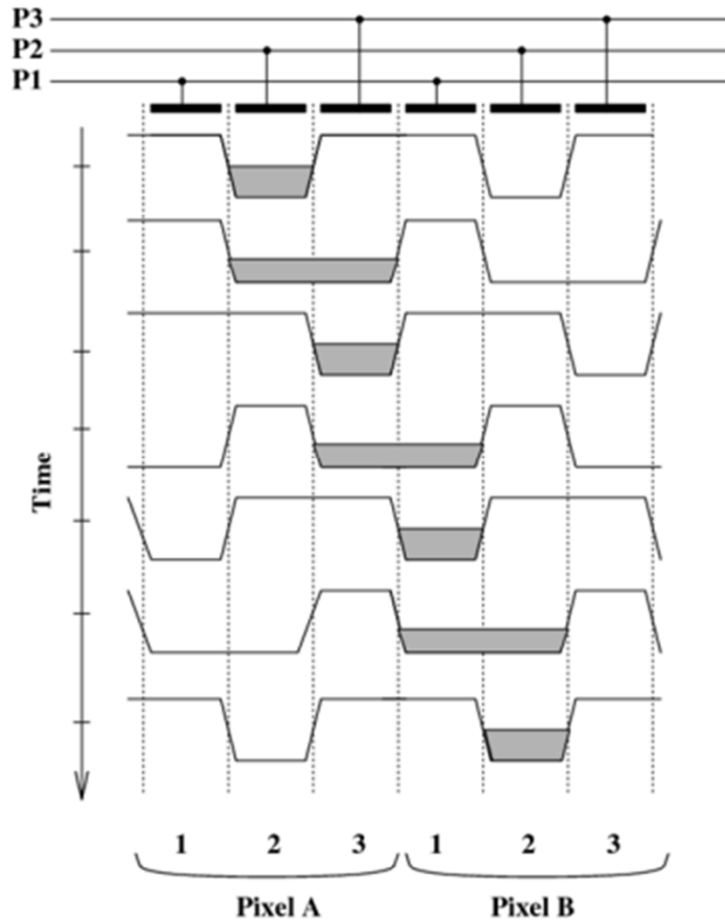
Charge Coupled Device (CCD)

CCD Phased Clocking: Step 5

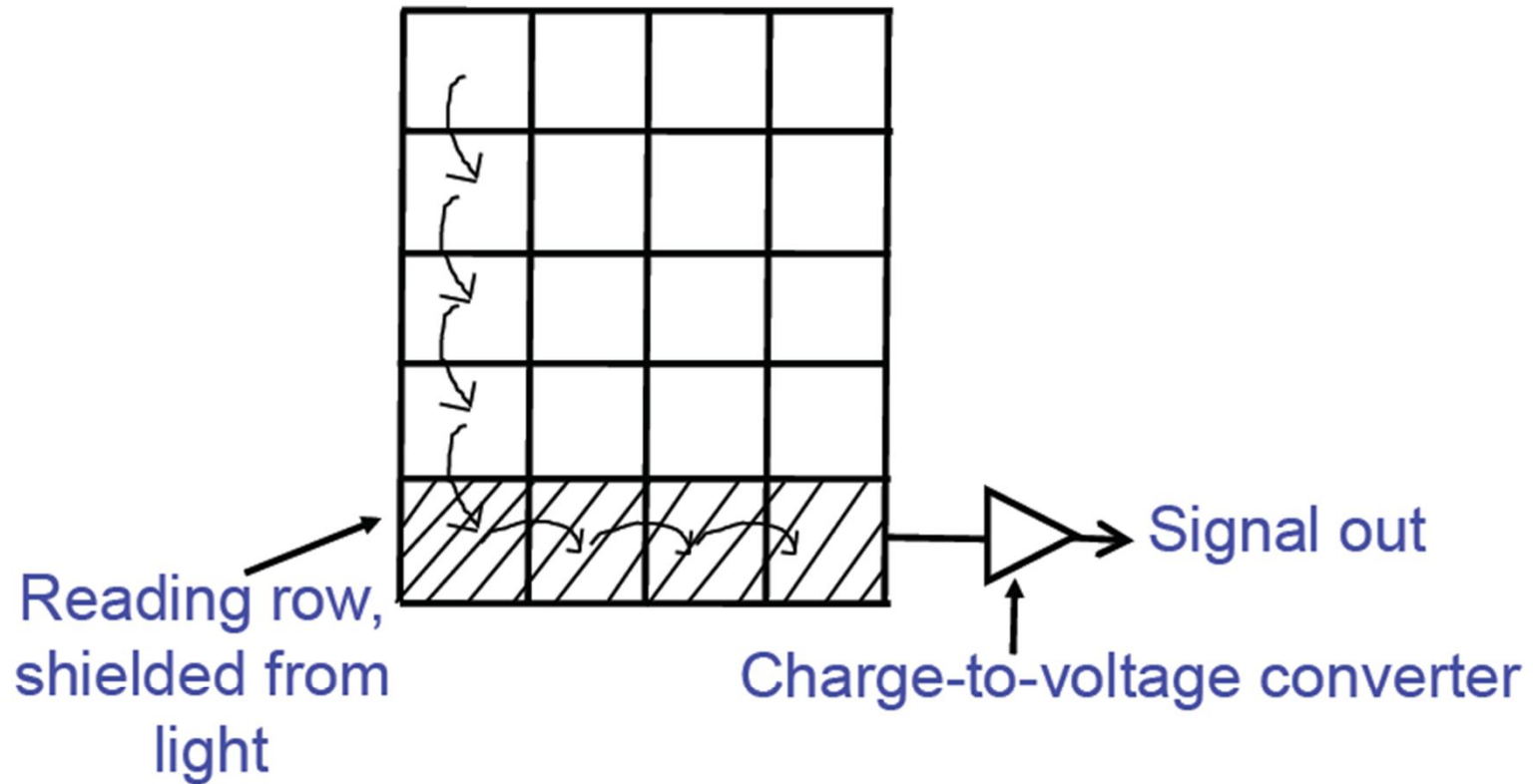


Charge Coupled Device (CCD)

CCD Phased Clocking: Summary

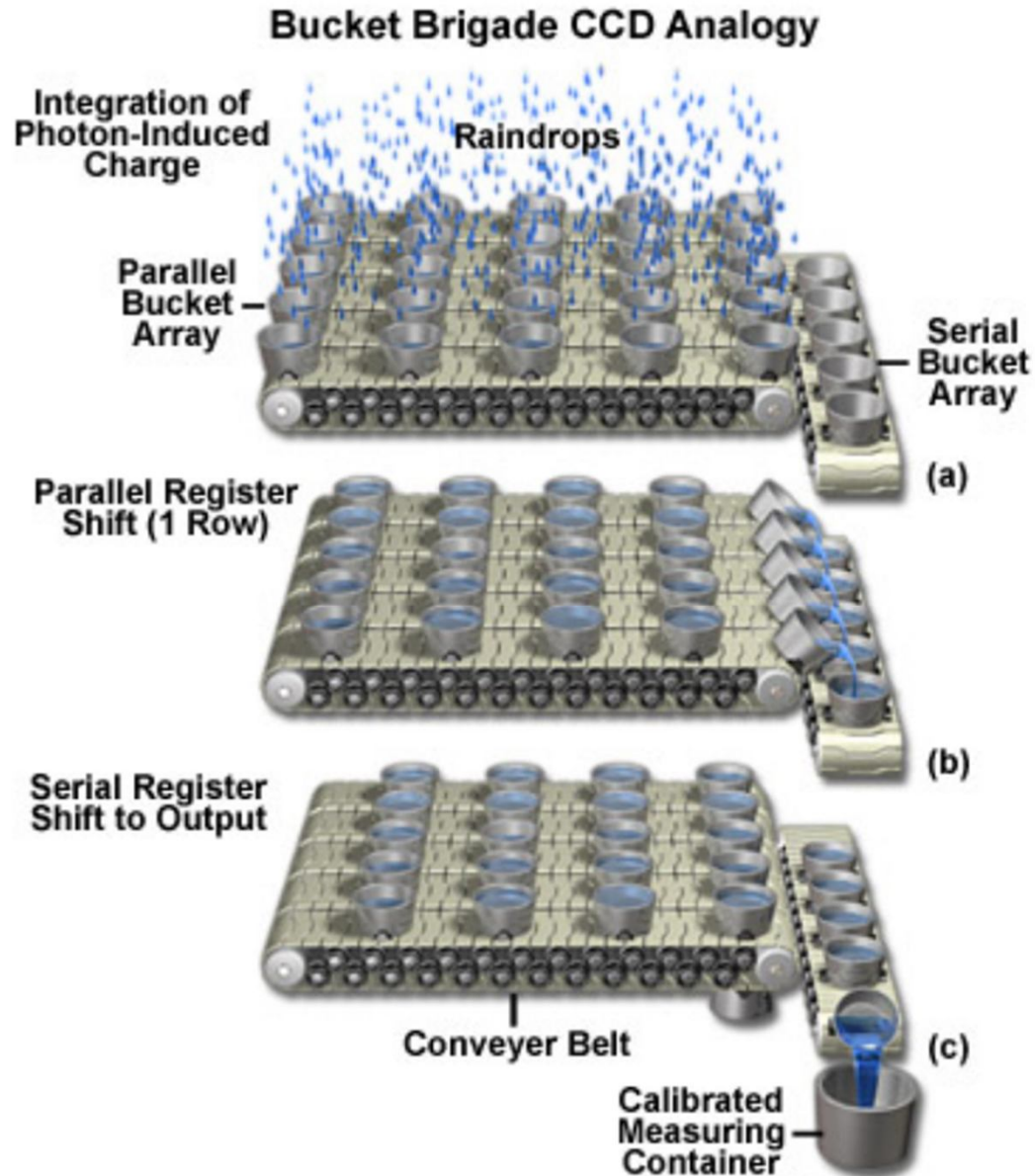


two-dimensional CCD imager



The reading row is shielded from the light by a metal film. The 2-D charge packets are read row by row.

two-dimensional CCD imager



Inventors of CCD Imagers



Willard S. Boyle



George E. Smith

**Both of Bell Laboratories, received Physics Noble Prize
in 2009 *"for the invention of an imaging semiconductor
circuit – the CCD sensor"***

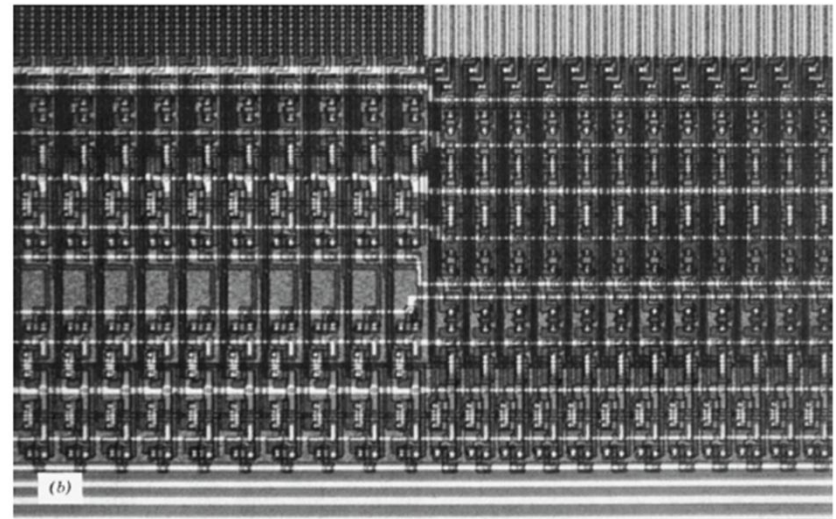
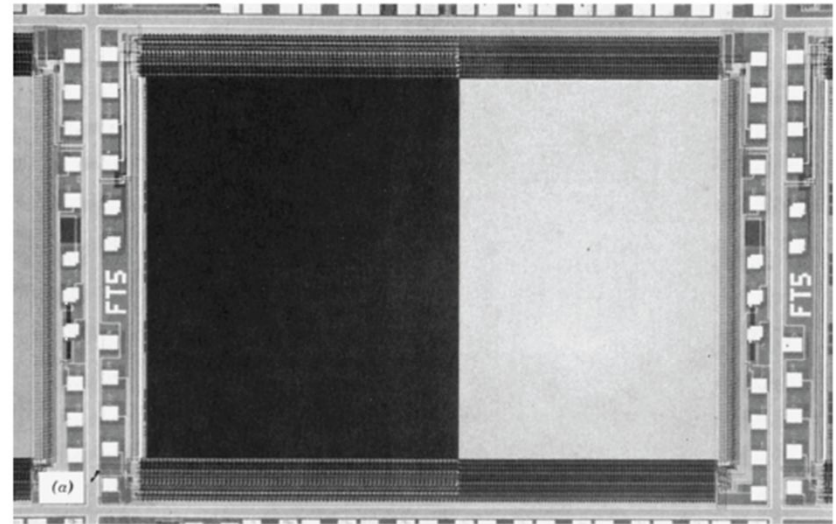
CCD Imager Chip

CCDs are extensively used in imaging applications, such as, digital camera, scanner, etc. An example is shown here:

a) A high-density frame-transfer CCD. The sensor array is on the left side of the chip and the light-shielded storage area is on the right. This CCD has 588 lines of 604 pixels each. The chip area is 38.22 mm^2 .

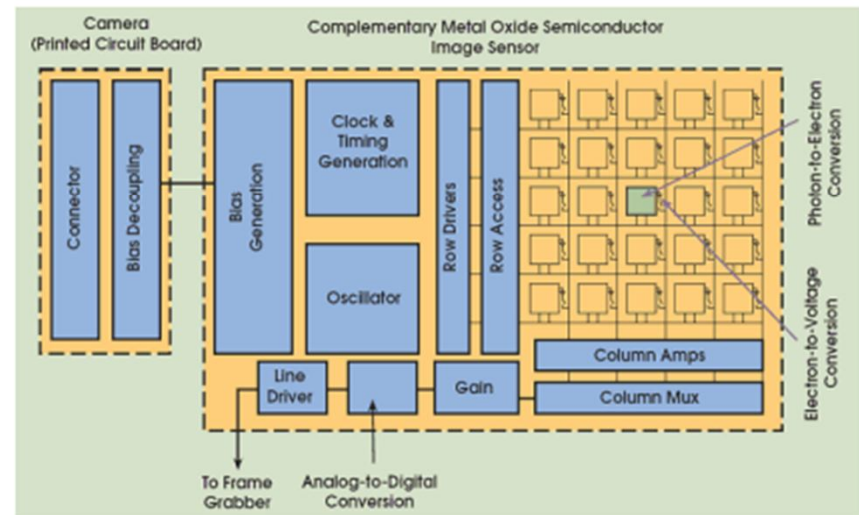
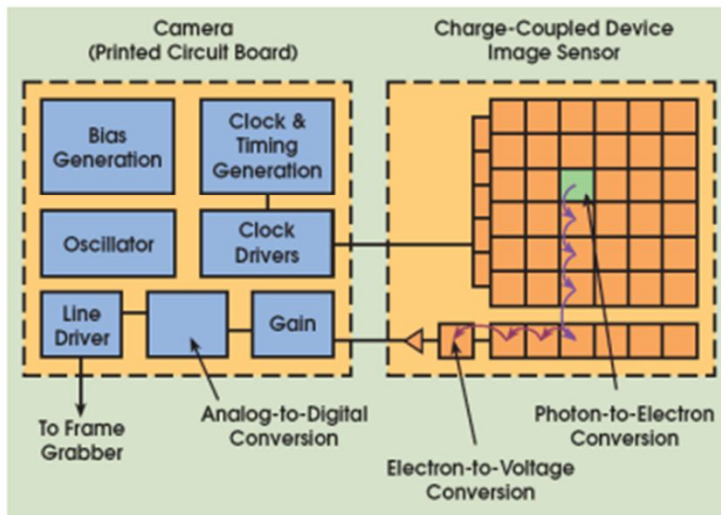
b) Peripheral CMOS circuitry for the CCD.

CCD with millions of pixels are commonly used in digital cameras today



CCD Vs CMOS

- ❑ *CMOS: Complementary Metal Oxide Semiconductor*
- ❑ *CCD: Charge Coupling Device*
- ❑ *Both are pixelated MOS and have the same process in charge generation and charge collection*
- ❑ *However, charge transfer and charge detection are different*
- ❑ *CCD: all pixels share one charge-to-voltage converter*
- ❑ *CMOS: every pixel has its own charge-to-voltage converter*



CCD Vs CMOS

CCD

- ❑ *offer high-quality, low-noise images*
- ❑ *greater sensitivity and fidelity*
- ❑ *100 times more power*
- ❑ *acceptable speed*
- ❑ *complicated anti-blooming technique*
- ❑ *require specialized assembly lines*
- ❑ *equal reliability*
- ❑ *older and more developed technology*

CMOS

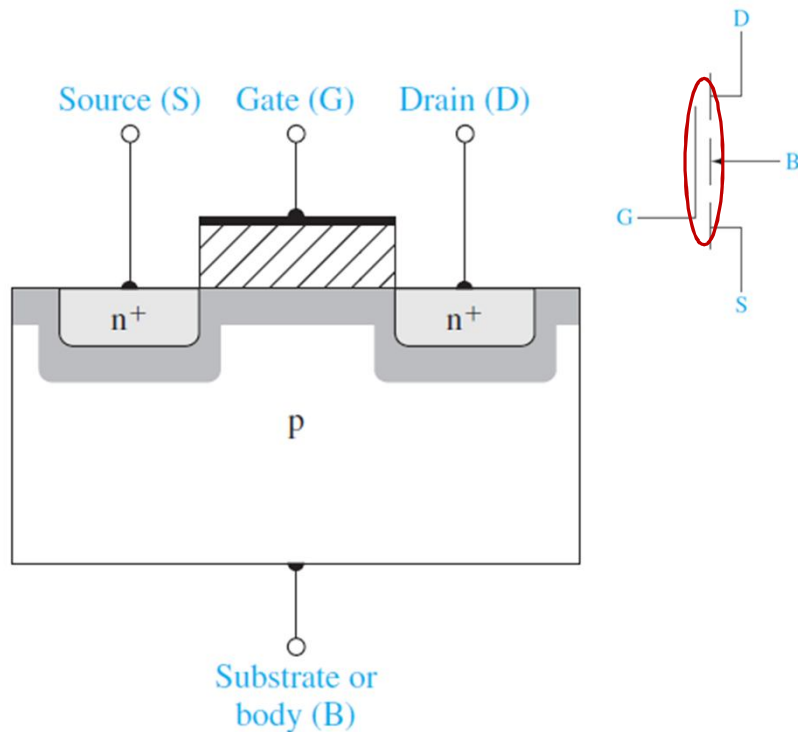
- ❑ *more susceptible to noise*
- ❑ *light sensitivity is lower*
- ❑ *consume little power*
- ❑ *much fast*
- ❑ *natural blooming immunity*
- ❑ *easy to manufacture*
- ❑ *highly integrated*
- ❑ *less expensive*

- ❑ *CMOS imagers offer superior integration, power dissipation and system size at the expense of image quality (particularly in low light) and flexibility. They are the technology of choice for high-volume, space-constrained applications where image quality requirements are low.*
- ❑ *CCDs offer superior image quality and flexibility at the expense of system size. They remain the most suitable technology for high-end imaging applications.*

10.3.1 MOSFET Structures

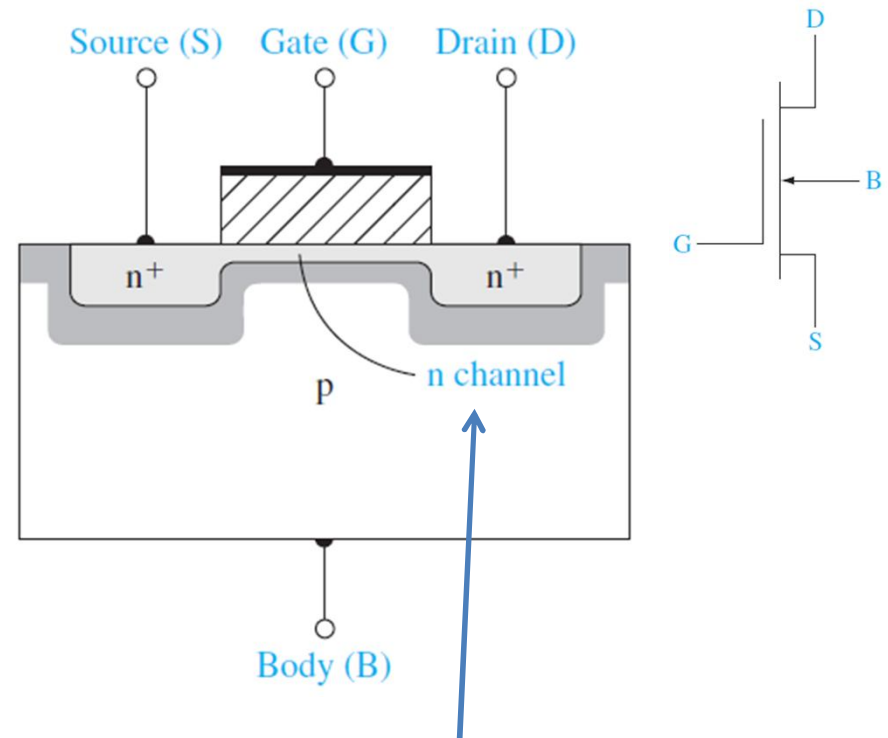
n-channel enhancement mode:

substrate is not inverted directly under the oxide with zero gate voltage



n-channel depletion mode:

An n-channel region exists under the oxide with 0 V applied to the gate

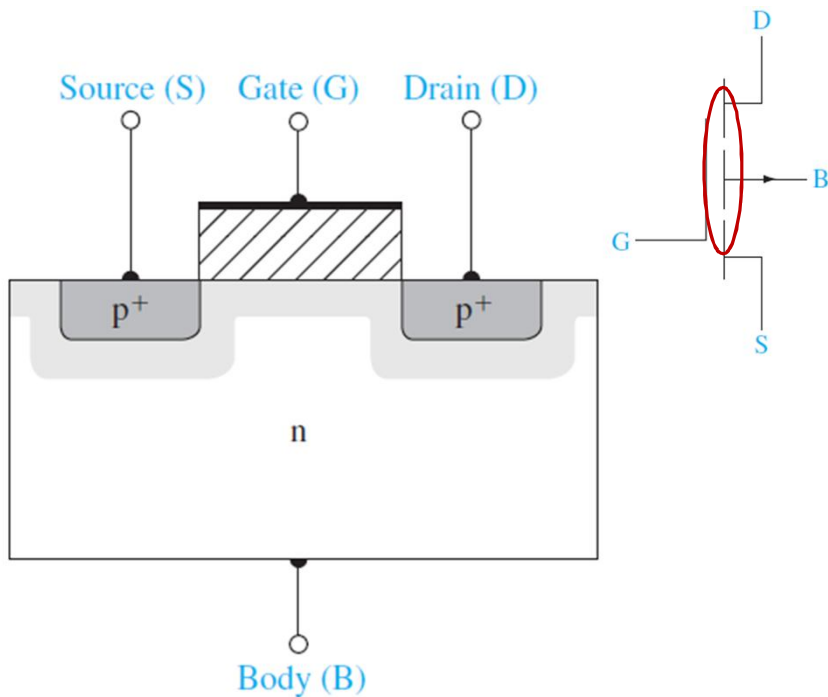


can be an electron inversion layer
or an intentionally doped n region

10.3.1 MOSFET Structures

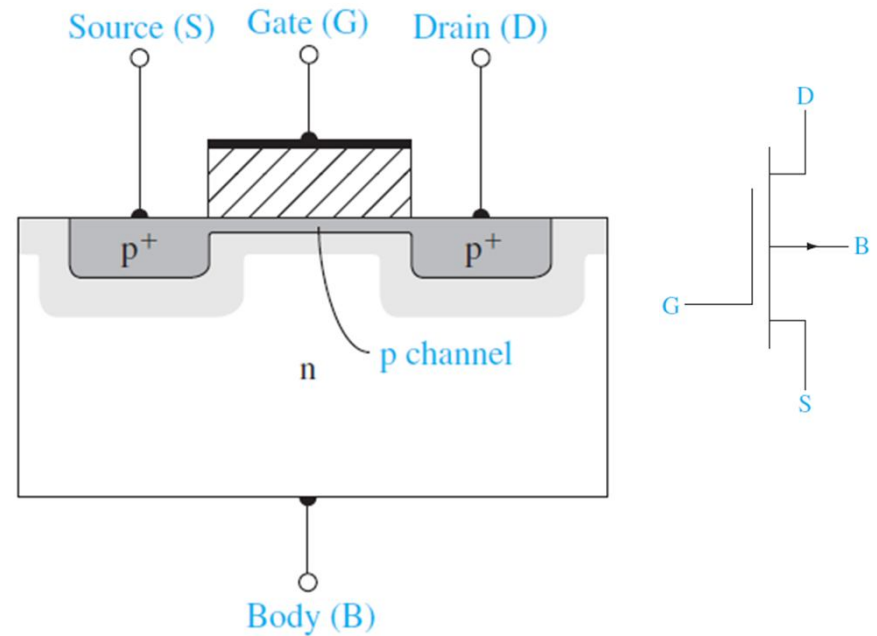
p-channel enhancement mode:

substrate is not inverted directly under the oxide with zero gate voltage

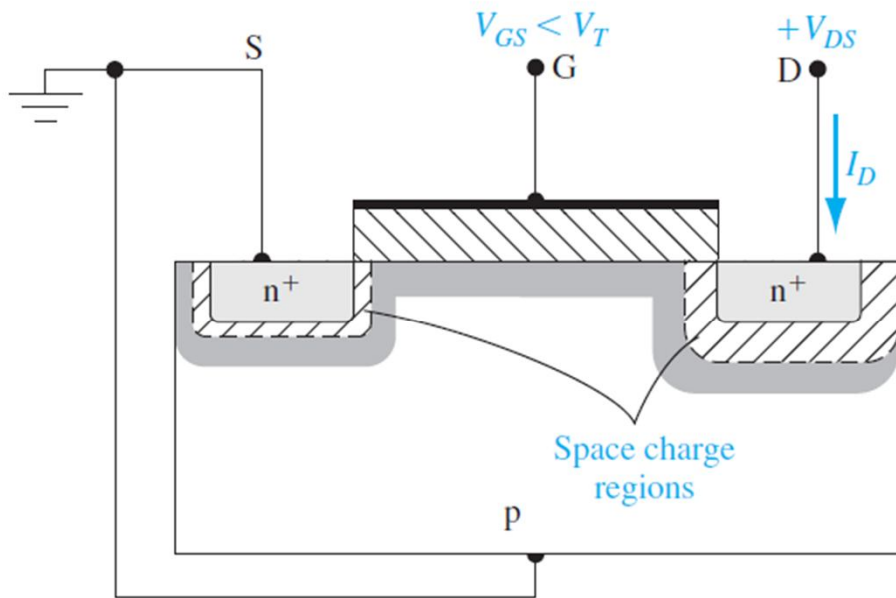


p-channel depletion mode:

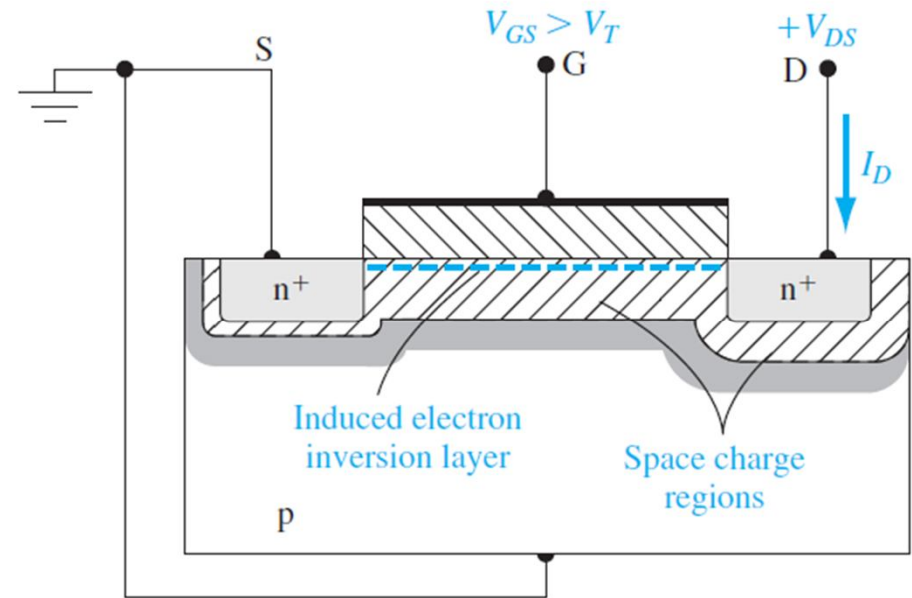
An p-channel region exists under the oxide with 0 V applied to the gate



Current–Voltage Relationship



(a)



(b)

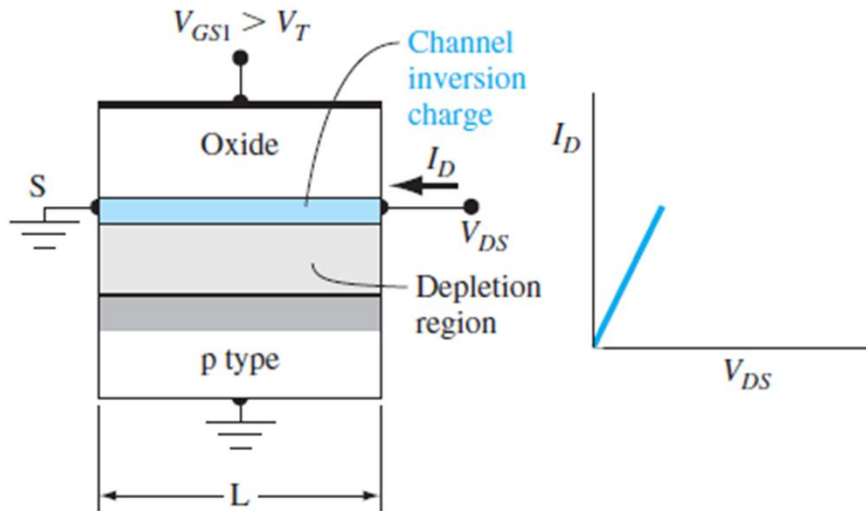
For small V_{DS} values, the channel region has the characteristics of a resistor, so we can write

$$I_D = g_d V_{DS} \quad (10.41)$$

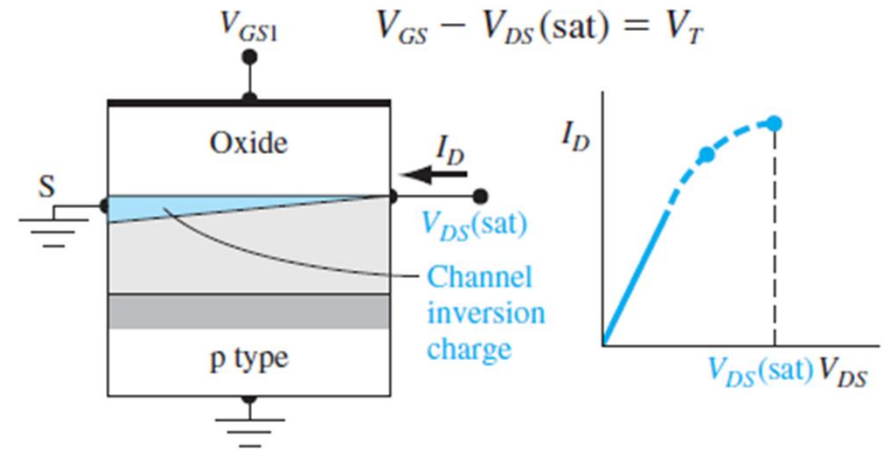
where g_d is defined as the channel conductance in the limit as $V_{DS} \rightarrow 0$. The channel conductance is given by

$$g_d = \frac{W}{L} \cdot \mu_n |Q'_n| \quad (10.42)$$

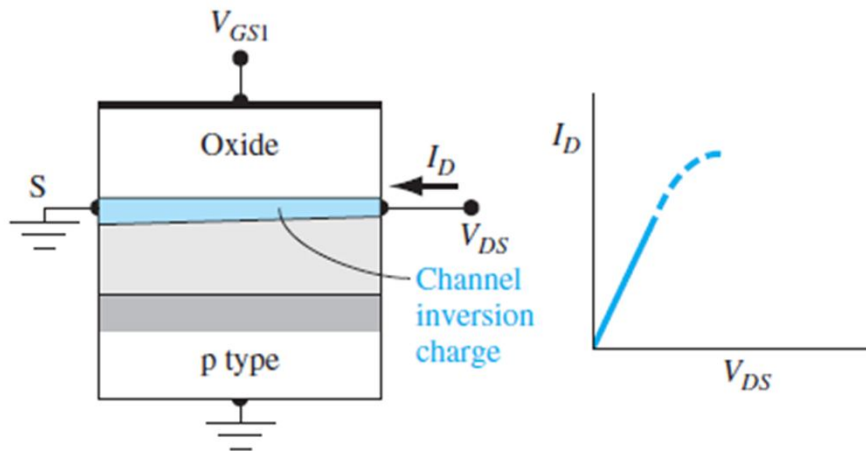
Current–Voltage Relationship



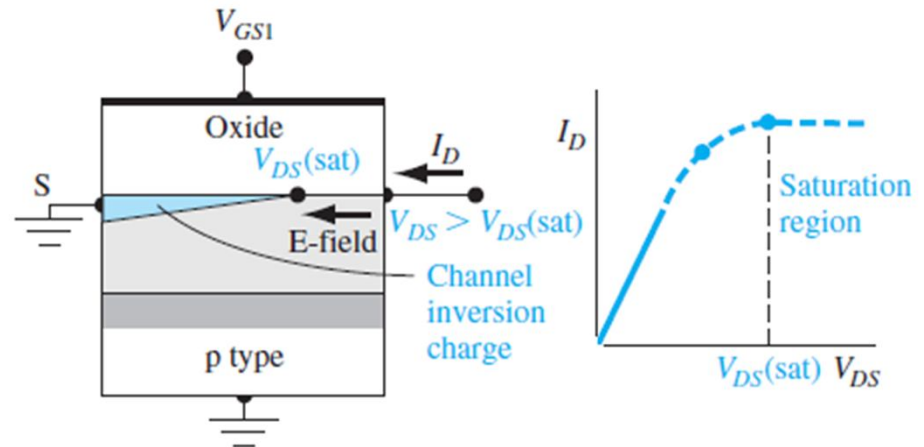
(a)



(c)

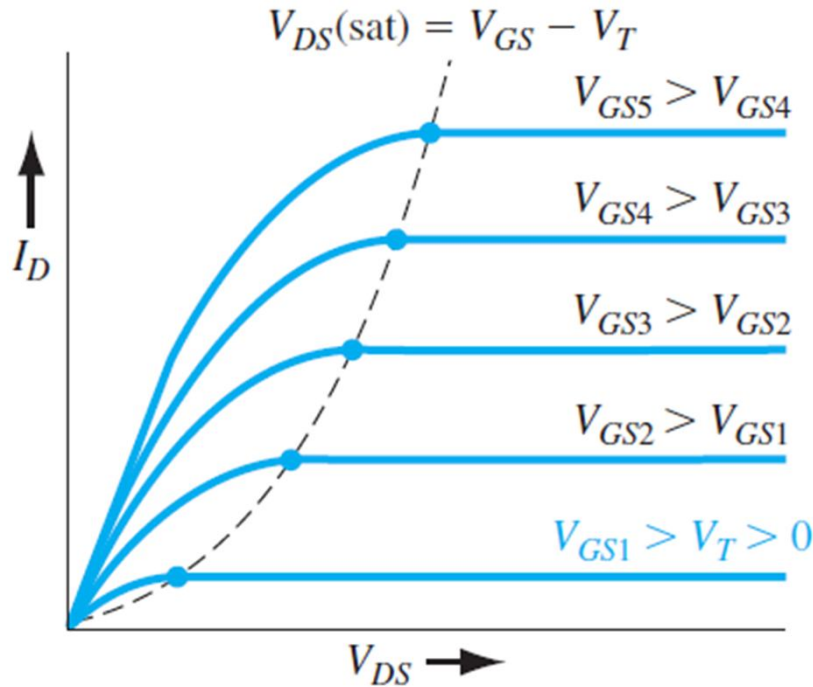


(b)



(d)

Current–Voltage Relationship



See later :

$$\begin{aligned}
 I_D &= \frac{W\mu_n C_{ox}}{2L} \left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2 \right] \\
 &= \frac{k'_n}{2} \frac{W}{L} \left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2 \right] \\
 &= K_n \left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2 \right]
 \end{aligned}$$

where $k'_n = \mu_n C_{ox}$ is the process conduction parameter

and $K_n = \frac{k'_n}{2} \frac{W}{L} = \frac{W\mu_n C_{ox}}{2L}$ is the conduction parameter

In saturation :

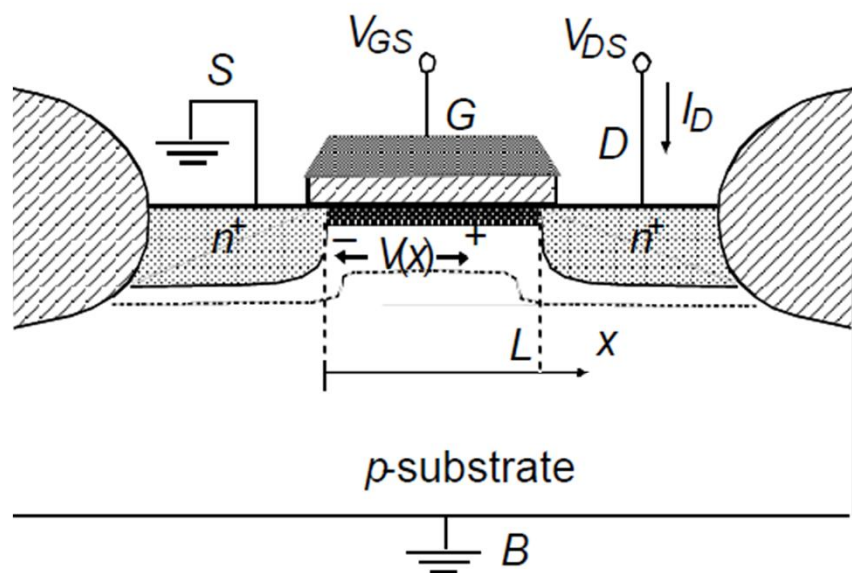
$$I_D = \frac{W\mu_n C_{ox}}{2L} (V_{GS} - V_T)^2$$

Mathematical Derivation of I-V relationship

Assumptions:

1. The current in the channel is due to drift rather than diffusion.
2. There is no current through the gate oxide.
3. A gradual channel approximation is used in which $dE_y/dy \gg dE_x/dx$. *This approximation means that E_x is essentially a constant.*
4. Any fixed oxide charge is an equivalent charge density at the oxide–semiconductor interface.
5. The carrier mobility in the channel is constant.

Current–Voltage Relationship



- When $V_{gs} < V_T$, no channel is formed.
- When $V_{gs} > V_T$, a voltage difference V_{ds} causes a current I_{ds} to flow.
- Current per unit area can be computed by expressing the amount of the charges in the channel.

□ Assume at a point x along the channel, the voltage is $V(x)$.
 □ The gate-to-channel voltage at that point equals $V_{gs} - V(x)$.
 □ Under the assumption that this voltage exceeds the threshold voltage all along the channel (so that an inversion layer is formed), the induced channel charge:

$$Q_i(x) = -C_{ox}[V_{gs} - V(x) - V_T]$$

Current–Voltage Relationship (linear region)

- ❑ The current is given as the product of the drift velocity of the carriers v_n and the the available charge.
- ❑ Due to charge conservation, the charge is constant over the length of the channel. W is width of the channel in a direction perpendicular to the current flow. We then have:

$$I_{ds} = -v_n(x)Q_i(x)W$$

- ❑ The electron velocity is related to the electric field through a parameter called *mobility* μ_n . The mobility is a complex function of crystal structure, local fields, and so on. We have:

$$v_n = -\mu_n E(x) = \mu_n \frac{dV}{dx}$$

- ❑ Combining the above two we have

$$I_{ds} dx = \mu_n C_{ox} W [V_{gs} - V(x) - V_T] dV$$

Current–Voltage Relationship (linear region)

□ Integrating the equation over the length of the channel L yields the voltage-current relation of the NMOS transistor:

$$I_{ds} = k'_n \frac{W}{L} \left[(V_{gs} - V_T) V_{ds} - \frac{V_{ds}^2}{2} \right] \quad I_{ds} = \beta_n \left[(V_{gs} - V_T) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

where k'_n is the process transconductance parameter and equal to:

$$\mu_n C_{ox} = \mu_n \frac{\epsilon_{ox}}{t_{ox}}$$

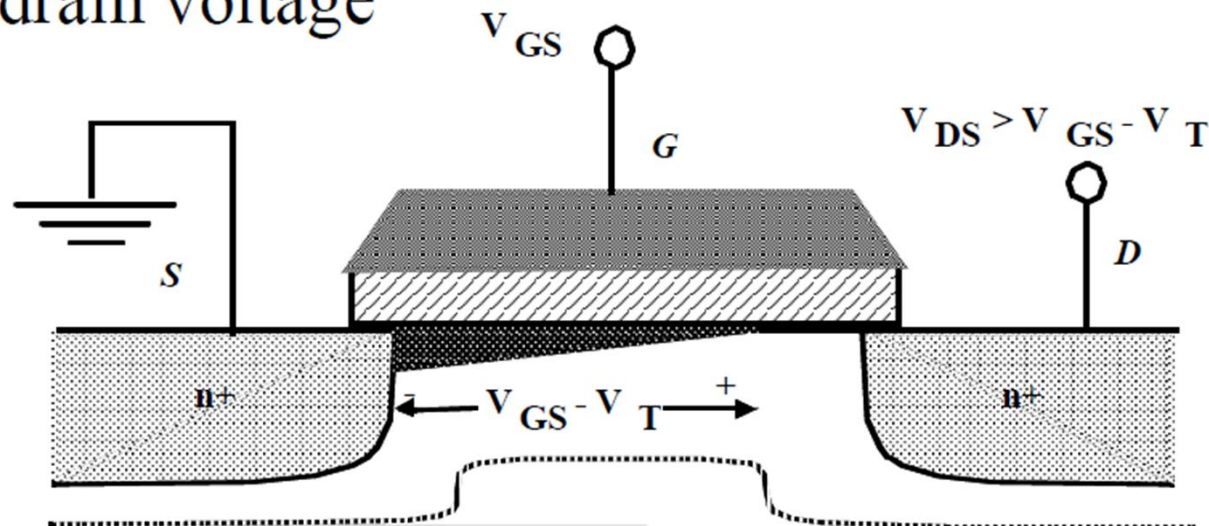
and β is the MOS transistor gain factor and is dependent on both the process parameters and the device geometry and is given by:

$$\beta_n = K_n = \mu_n \frac{\epsilon_{ox}}{t_{ox}} \left(\frac{W}{L} \right)$$

□ The above is the current equation at the linear region.

Current–Voltage Relationship (saturation region)

- As V_{ds} is further increased, the assumption that the channel voltage is larger than the threshold all along the channel ceases to hold. This happens when $V_{gs} - V(x) < V_T$.
- At this point the induced charge is zero, and the conducting channel disappears or is pinched off. **No channel exists in the vicinity of the drain region.** Conduction is brought about by a drift mechanism of electrons under the influence of the positive drain voltage



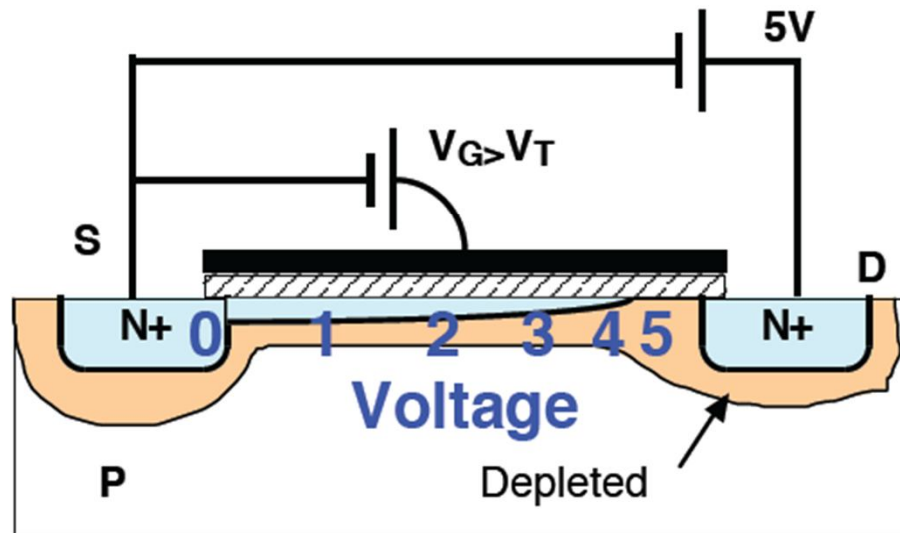
Current–Voltage Relationship (saturation region)

- As the electrons leave the channel, they are injected into the drain depletion region and are subsequently accelerated toward the drain.
- The voltage over the induced pinched-off channel (from the pinch-off point to the source) tends to remain fixed at $V_{gs} - V_T$ and consequently, the current remains constant (or saturates).
- Replacing V_{ds} by $V_{gs} - V_T$ yields:

$$I_{ds} = \frac{k_n}{2} \frac{W}{L} (V_{gs} - V_T)^2$$

Current–Voltage Relationship (saturation region)

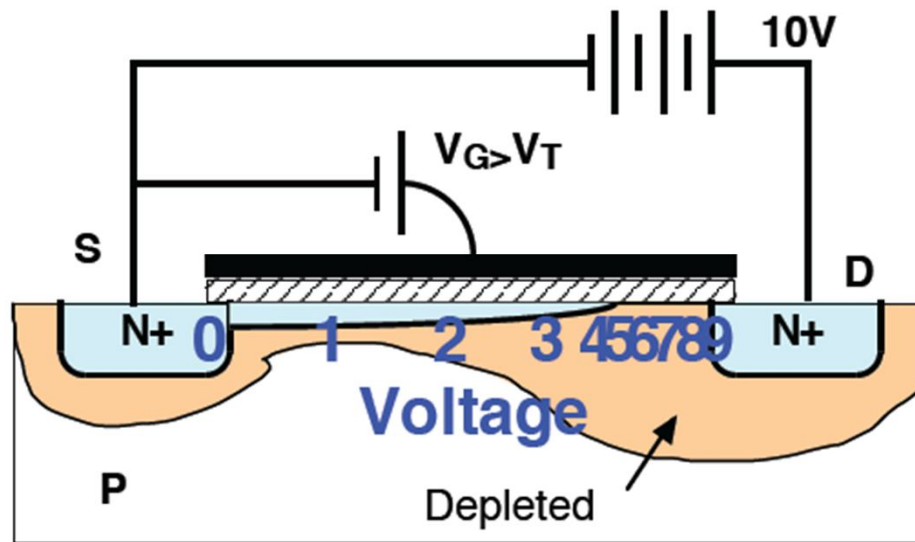
Why the current remains constant past pinchoff (saturation):



Near pinchoff, the voltage is decreasing approximately linearly, hence the E field is “relatively” constant throughout.

Current–Voltage Relationship (saturation region)

Why the current remains constant past pinchoff (saturation):



- At higher V_d , there is a larger depleted region, hence, greater voltage drop.
- E is still linear in the channel but very large in depletion region.
- Carriers rapidly get swept out of depletion region

Current–Voltage Relationship

- Cut-off region

$$I_{ds} = 0 \quad V_{gs} \leq V_t$$

- Linear region

$$I_{ds} = \beta \left[(V_{gs} - V_t) V_{ds} - \frac{V_{ds}^2}{2} \right] \quad 0 < V_{ds} < V_{gs} - V_t$$

- It is called linear region as I_{ds} varies linearly with V_{gs} and V_{ds} when the quadratic term $V_{ds}^2/2$ is very small.

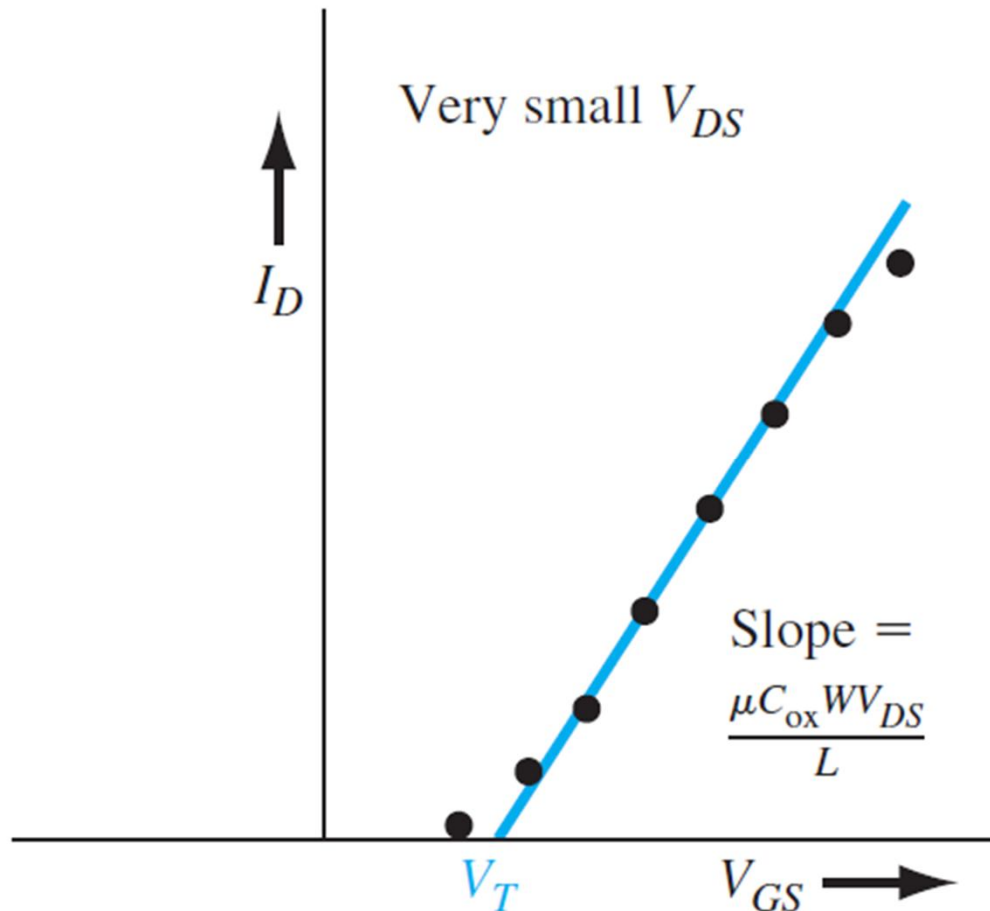
- Saturation region

$$I_{ds} = \beta \frac{(V_{gs} - V_t)^2}{2} \quad 0 < V_{gs} - V_t < V_{ds}$$

Parameters extraction (linear)

We can use the I – V relations to experimentally determine the mobility and threshold voltage parameters. For very small values of V_{DS} :

$$I_D = \frac{W\mu_n C_{ox}}{L} (V_{GS} - V_T) V_{DS}$$

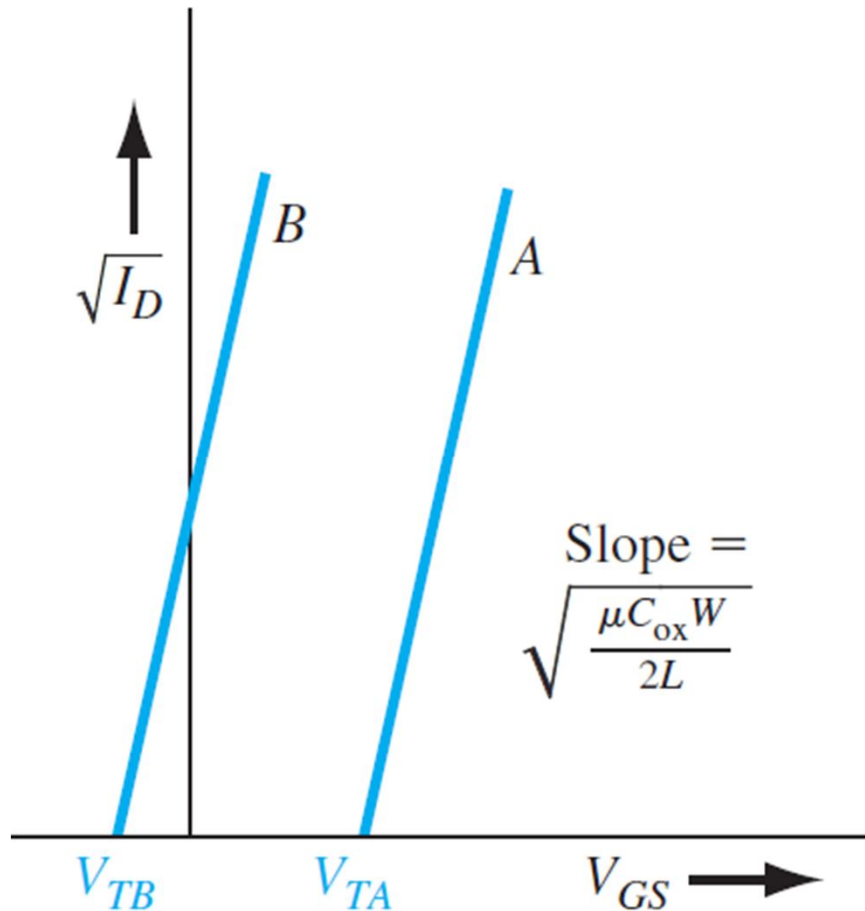


Parameters extraction (saturation)

For large V_{DS} and saturation mode:

$$I_D(\text{sat}) = \frac{W\mu_n C_{ox}}{2L} (V_{GS} - V_T)^2$$

$$\sqrt{I_D(\text{sat})} = \sqrt{\frac{W\mu_n C_{ox}}{2L}} (V_{GS} - V_T)$$



- the threshold voltage may be a function of V_{DS} in *short-channel devices*
- the V_T may differ from that extrapolated from the linear region
- The nonsaturation current-voltage characteristics will produce the more reliable data.

Transconductance

The MOSFET transconductance is defined as the change in drain current with respect to the corresponding change in gate voltage, which is sometimes referred to as the transistor gain.

$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

Linear region $g_{mL} = \frac{\partial I_D}{\partial V_{GS}} = \frac{W\mu_n C_{ox}}{L} \cdot V_{DS}$

Saturation region $g_{ms} = \frac{\partial I_D(\text{sat})}{\partial V_{GS}} = \frac{W\mu_n C_{ox}}{L} (V_{GS} - V_T)$

- The transconductance is a function of the geometry of the device as well as of carrier mobility and threshold voltage.
- In the design of MOSFET circuits, the size of the transistor, in particular the channel width W , is an important engineering design parameter.

Substrate Bias Effects

The source-to-substrate pn junction must always be zero or reverse biased, so V_{SB} *must always be greater than or equal to zero*. The newly created electrons will move laterally and flow out of the source terminal.

When $V_{SB} = 0$, we had

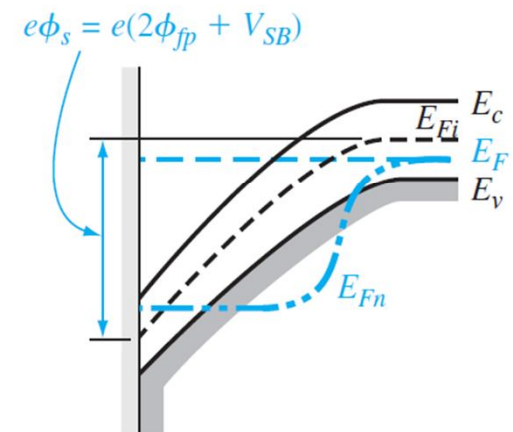
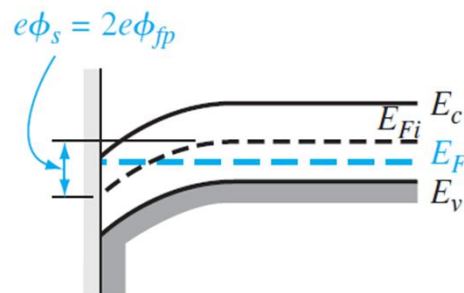
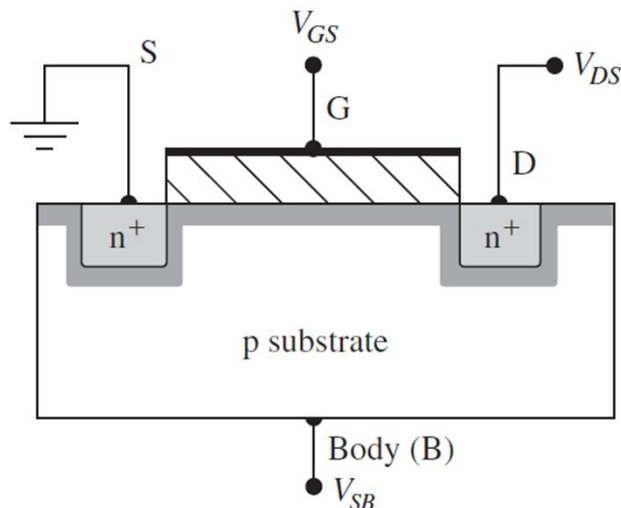
$$Q'_{SD} (\text{max}) = -eN_a x_{dT} = -\sqrt{2e\epsilon_s N_a (2\phi_{fp})}$$

When $V_{SB} > 0$, the space charge width increases and we now have

$$Q'_{SD} = -eN_a x_d = -\sqrt{2e\epsilon_s N_a (2\phi_{fp} + V_{SB})}$$

The change in the space charge density is then

$$\Delta Q'_{SD} = -\sqrt{2e\epsilon_s N_a} [\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}}]$$



Substrate Bias Effects

To reach the threshold condition, the applied gate voltage must be increased. The change in threshold voltage can be written as

$$\Delta V_T = -\frac{\Delta Q'_{SD}}{C_{ox}} = \frac{\sqrt{2e\epsilon_s N_a}}{C_{ox}} \left[\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}} \right] \quad (10.81)$$

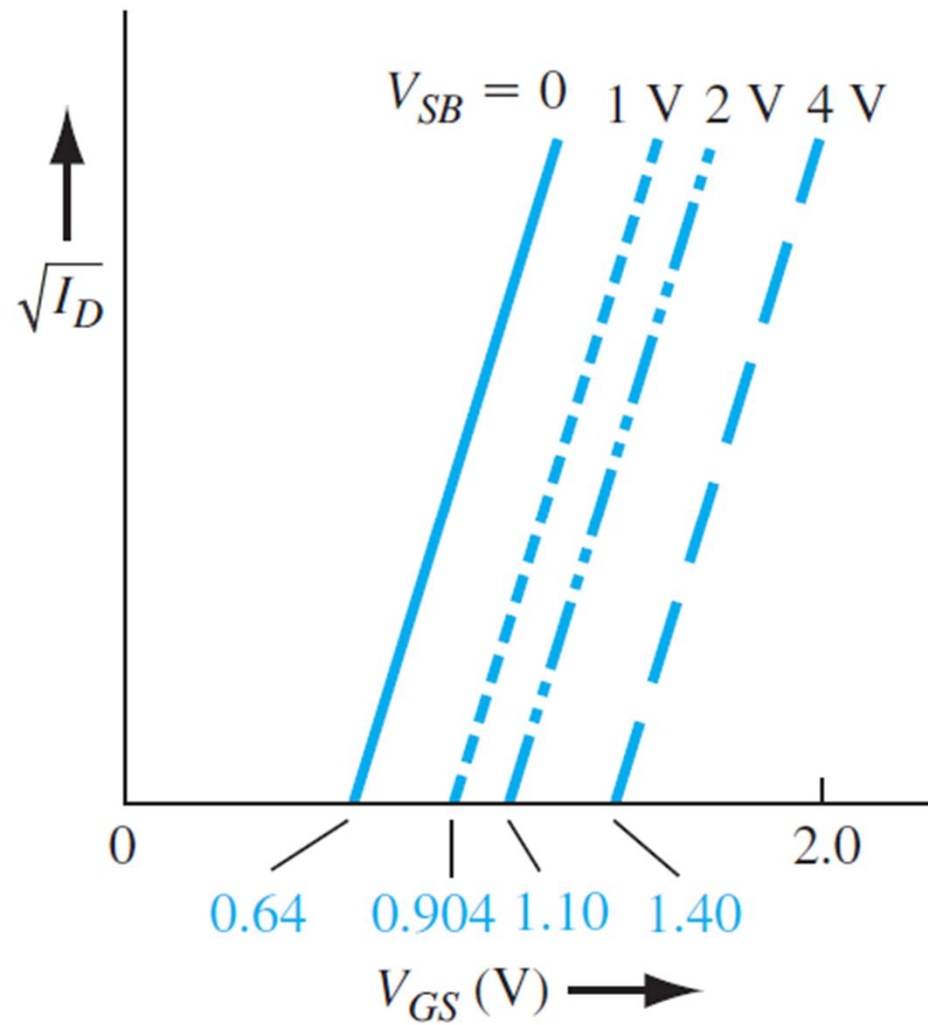
where $\Delta V_T = V_T(V_{SB} > 0) - V_T(V_{SB} = 0)$. We may note that V_{SB} must always be positive so that, for the n-channel device, ΔV_T is always positive. The threshold voltage of the n-channel MOSFET will increase as a function of the source–substrate junction voltage.

$$\Delta V_T = \gamma \left[\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}} \right]$$

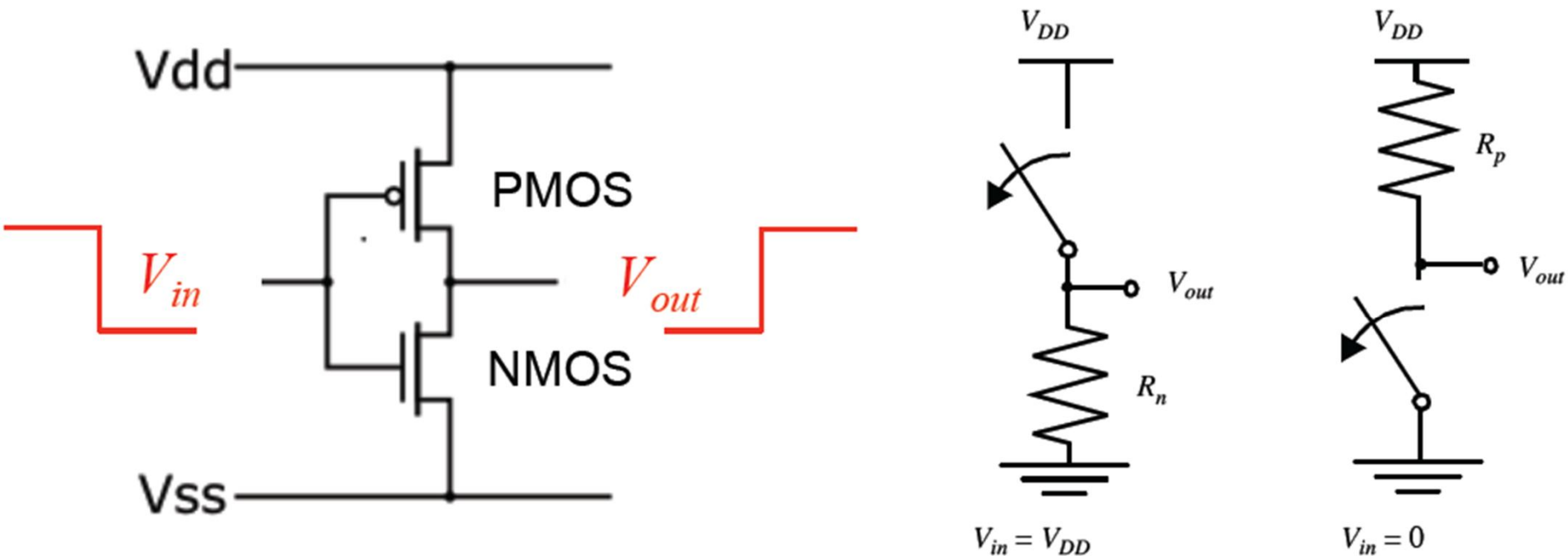
$$\gamma = \frac{\sqrt{2e\epsilon_s N_a}}{C_{ox}}$$

where γ is defined as the *body-effect coefficient*.

Substrate Bias Effects

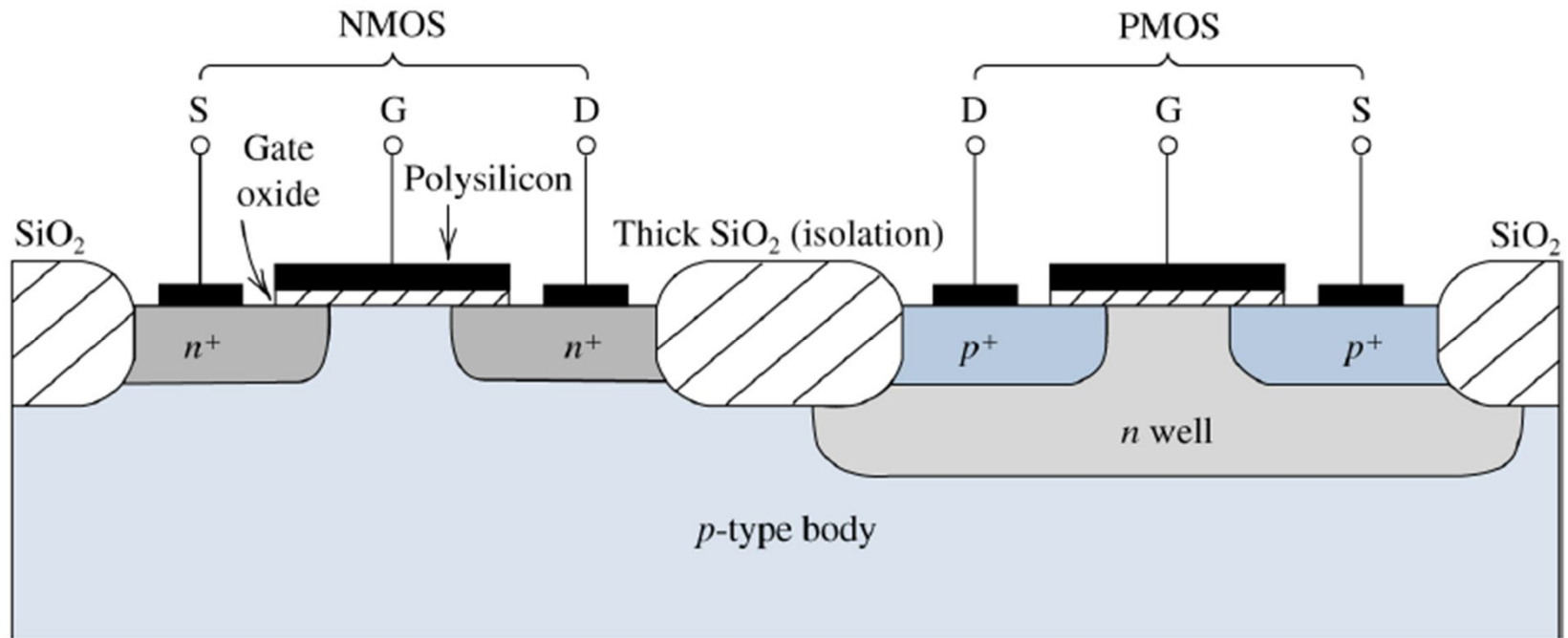


10.5 | THE CMOS TECHNOLOGY



- For $|V_g| < |V_t|$ the transistor is off represented by open circuit
- For $|V_g| > |V_t|$ the transistor is on represented by a resistor
- Output is an inverted form of input waveform
- CMOS inverter is the most important building block of modern logic circuits
- What is the power dissipation in this circuit?

10.5 | THE CMOS TECHNOLOGY



Cross section of a CMOS integrated circuit. Note that the PMOS transistor is formed in a separate n -type region, known as an n well.

Reading Materials

D. A. Neamen, “Semiconductor Physics and Devices: Basic Principles”, Fourth edition, Chapter 10, “Fundamentals of the Metal–Oxide–Semiconductor Field-Effect Transistor”.